

AIAA 3225

Learning and Optimization
for Artificial Intelligence

**Foundations, Convex Optimization, and Semidefinite
Applications**

Part I

Weeks 1–7

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Contents

Part I Guide and Notation	v
1 From AI Learning Problems to Optimization	1
1.1 The learning-and-optimization map	2
1.2 Optimization theory: a map of the field	3
1.3 Learning-theory preview: from training loss to population risk	4
1.4 The whole convex-optimization loop in one ridge problem	5
1.5 The objects in a supervised learning problem	7
1.6 Probability bridge: what the empirical average means	7
1.6.1 Random variables and distributions	7
1.6.2 Independent and identically distributed samples	8
1.7 Population, empirical, and regularized risk	9
1.7.1 Regularized empirical-risk minimization	9
1.7.2 Logistic regression as an ERM example	10
1.7.3 Matrix factorization for recommendation	11
1.8 From data separation to problem routing	12
1.8.1 Training objective versus evaluation criterion	12
1.8.2 Classify the optimization structure	13
1.9 A schematic foundation-model lifecycle	13
1.10 Chapter synthesis	15
1.11 Chapter practice set	15
1.12 Complete solutions	17
2 Convex Geometry and Structured First-Order Methods	20
2.1 The conceptual map	20
2.2 Anchor problem: sparse linear regression	21
2.3 Convex sets	21
2.4 Convex functions	22
2.4.1 A smooth maximum: log-sum-exp and softmax	22

2.4.2	The first-order characterization	23
2.4.3	Subgradients for nonsmooth convex functions	25
2.5	One regularized local model, four familiar updates	25
2.5.1	Gradient descent retrieval	26
2.5.2	The subgradient method	27
2.6	Constraints: projection and projected gradient	27
2.7	Composite objectives: proximal gradient	29
2.8	How to choose the first algorithm	31
2.9	Projection-free optimization: Frank–Wolfe	31
2.10	Computational mini-lab	32
2.11	Mastery checklist	34
2.12	Chapter practice set	34
2.13	Complete solutions	36
3	Separation, Certificates, and Conic Duality	41
3.1	The conceptual map	41
3.2	Anchor problem: projection onto a resource halfspace	42
3.3	Canonical constrained convex programs	42
3.3.1	Three logical questions	43
3.4	Separation turns geometry into a certificate	43
3.5	Farkas’ lemma: either a feasible vector or a refutation	44
3.6	Linear programming: the first complete certificate	45
3.6.1	Why the dual is a certificate	45
3.6.2	Vertices, basic feasible solutions, and the simplex idea	46
3.7	Lagrangian duality and constraint qualifications	48
3.7.1	The anchor problem in the dual	49
3.7.2	When the lower bound is exact	50
3.8	KKT conditions: a local-looking global certificate	50
3.8.1	Two compact KKT checks	51
3.9	The generic conic primal–dual pair	52
3.10	One problem, three certificate views	53
3.11	Chapter practice set	54
3.12	Complete solutions	56
4	From Newton and Barriers to Semidefinite Optimization	61
4.1	The conceptual map	61
4.2	Newton’s method: follow a local quadratic model	62

4.3	From a log barrier to the central path	63
4.4	Symmetric matrices and the PSD order	64
4.5	PSD-cone geometry and linear matrix inequalities	66
4.5.1	PSD matrices have nonnegative trace products	66
4.5.2	Linear matrix inequalities and spectrahedra	67
4.6	The SDP primal–dual pair and a complete certificate	68
4.6.1	Checking the SDP signs from the conic pair	69
4.6.2	Anchor problem: a complete 2×2 certificate	69
4.7	Spectral optimization: the largest eigenvalue as an SDP	70
4.8	SDP Slater and KKT as Week 3 specializations	71
4.9	From scalar barriers to the log-determinant barrier	72
4.10	Quantum representation bridge: binary state discrimination	73
4.11	Misconception checks	74
4.12	Chapter practice set	75
4.13	Complete solutions	77
5	Max-Cut, SDP Relaxation, and Randomized Rounding	83
5.1	The approximation process	84
5.2	Weighted Max-Cut	85
5.2.1	The graph Laplacian	85
5.2.2	When the edge weights are learned	86
5.3	Rank-one lifting	87
5.4	The SDP relaxation	88
5.4.1	The dual certificate	88
5.5	Gram-vector representation	89
5.6	Anchor example: K_4 and a tetrahedron	89
5.6.1	A matching dual certificate	90
5.7	Random-hyperplane rounding	90
5.8	The Goemans–Williamson guarantee	92
5.8.1	What the theorem does and does not say	93
5.9	Rounding the tetrahedron	93
5.10	Chapter practice set	95
5.11	Complete solutions	96
6	Shannon Capacity and the Lovász Theta Function	103
6.1	The chapter’s representation map	104
6.2	From a noisy channel to an independent set	105

6.2.1	Discrete memoryless channels	105
6.3	Repeated uses and the strong graph product	106
6.4	Lovász theta: fix the convention first	108
6.4.1	Orthonormal representations	108
6.5	The semidefinite program	109
6.5.1	Deriving the dual	110
6.6	Why theta bounds Shannon capacity	111
6.7	Anchor example: exact capacity of the pentagon	112
6.7.1	The primal certificate	113
6.7.2	The dual certificate	113
6.7.3	Representation switch: the pentagon umbrella	114
6.8	Chapter synthesis	114
6.9	Chapter practice set	115
6.10	Complete solutions	117
A	Formative Checks and Exit Tickets	122
A.1	Week 1: risk, first optimization loop, and attainment	122
A.2	Week 2: convexity and first-order operators	122
A.3	Week 3: separation, Farkas certificates, and conic duality	123
A.4	Week 4: Newton, barriers, and semidefinite certificates	123
A.5	Week 5: relaxation and randomized rounding	124
A.6	Week 6: zero-error capacity and theta	124
A.7	Week 7: cumulative certificate discipline	124
A.8	Answer key and diagnostic rationale	125
B	Normal Cones and the Unified Fermat Rule	126
B.1	Constraint and normal-cone language	126
B.2	Unified convex Fermat rule	127
B.3	Specializations to reuse	127

Part I Guide and Notation

Chapters 1–6 supply the new material for Weeks 1–6. Week 7 is a synthesis week: revisit the chapter exercises and use Appendix A for cumulative checks. The main route is learning objectives, convex updates, duality, PSD certificates, Max-Cut rounding, and zero-error capacity.

Notation	Meaning and convention
$\theta \in \mathcal{W}, \mathcal{S}, \mathcal{D}$	Trainable parameter and parameter space, observed sample, and data distribution.
$\hat{R}_{\mathcal{S}}(\theta), R(\theta)$	Empirical and population risk; neither includes a regularizer unless stated.
C or $\mathcal{X}$; $X \in \mathbb{S}^n$	Feasible sets use C or $\mathcal{X}$; an SDP matrix uses capital X . In Chapter 1, $\mathcal{X}$ instead denotes the input space.
$A^{\top}, A^{\dagger}; \text{Tr}(A)$	Real transpose, complex conjugate transpose, and trace. $\langle A, B \rangle = \text{Tr}(A^{\top} B)$ for real matrices.
$\text{diag}(X); \text{Diag}(v)$	Extract the diagonal as a vector; construct a diagonal matrix from a vector or a displayed list of entries.
$x \succeq 0; X \succeq 0$	Componentwise nonnegativity for vectors; positive semidefiniteness for symmetric matrices. The type determines the order.
$\mathbb{P}, \mathbb{E}; \log$	Probability and expectation over the stated randomness; natural logarithm unless a base is displayed. Rates in bits use $\log_2$.
x^*, p^*, d^*	Optimizer, primal optimal value, and dual optimal value. A star requires an attained optimum when attached to a point.
$\text{OPT}(G), \text{SDP}(G)$	Exact Max-Cut optimum and relaxed optimum. Feasible matrix and dual values are $v(X) = \langle L, X \rangle / 4$ and $U(y) = \sum_i y_i$.
$\alpha(G), \Theta(G), \vartheta(G)$	Independence number, Shannon capacity, and Lovász theta. Graphs are nonempty and edges mean confusability in Chapter 6.

Read bounds by problem direction. For a minimization primal, $d^* \leq p^*$; for a maximization primal, $p^* \leq d^*$. In Chapter 5,

$$\text{Cut}(x) \leq \text{OPT}(G) \leq \text{SDP}(G) \leq U(y), \quad v(X) \leq \text{SDP}(G),$$

for feasible x, X, y . A rounded sample, its expectation, and its guaranteed lower bound are three different quantities. In Chapter 6, $\alpha(G) \leq \Theta(G) \leq \vartheta(G)$.

Check a solution in four steps. Verify the objects and assumptions, the derivation, the numerical or symbolic certificate, and its interpretation and limits. Appendix B collects the convex optimality conditions and their assumptions.

Chapter 1

From AI Learning Problems to Optimization

Why this chapter

What should come next? Open a short-video app and the system must choose the next clip. Open a shopping app and it must rank the next products on the screen. Search, advertising, music, news, and social feeds repeatedly face the same mathematical task: choose one action, or an ordered list of actions, from many possibilities.

A prediction score alone does not make that choice. The score estimates what might happen; a decision rule uses those estimates to select and rank items under objectives and constraints. One recommendation therefore exposes two coupled optimization loops:

$$\begin{aligned} \text{learn:} \quad & \hat{R}_{\mathcal{S}}(\hat{\theta}) + \lambda \Omega(\hat{\theta}) \leq \inf_{\theta \in \mathcal{W}} \left[\frac{1}{n} \sum_{i=1}^n \ell(h_{\theta}(x_i), y_i) + \lambda \Omega(\theta) \right] + \varepsilon_{\text{opt}}, \\ \text{act:} \quad & s_t^* \in \arg \max_{s \in \mathcal{F}_t(c_t)} \hat{U}_{\hat{\theta}}(s \mid c_t). \end{aligned}$$

In the learning loop, the data (x_i, y_i) are fixed observations and $\theta \in \mathcal{W}$ is the trainable parameter. The pointwise loss is ℓ , the regularizer is Ω , and $\lambda \geq 0$ controls its weight. In the decision loop, c_t is the current context, s is an ordered recommendation slate, $\mathcal{F}_t(c_t)$ is the set of feasible slates, and $\hat{U}_{\hat{\theta}}$ is the utility predicted using the trained model. The nonnegative ε_{opt} records training suboptimality. For the serving display, assume $\mathcal{F}_t(c_t)$ is a nonempty finite candidate set, so the displayed maximizer exists; a fixed rule breaks ties. A short-video feed may have a one-item slate; a shopping page usually ranks several items at once.

Neither loop chooses its own goal. Should a video feed optimize the next click, total watch time, or long-term satisfaction? Should a shopping system optimize clicks, purchases, net value after returns, or continued trust? Must the slate also satisfy diversity, availability, safety, and latency requirements? A system can solve its stated optimization problem accurately and still behave badly when the objective or constraints do not represent the real task.

Optimization is not only about solving an objective;
it is also about choosing and checking the right objective.

The same language describes route planning, energy dispatch, network flow, resource allocation, experimental design, and engineering control. The optimization toolkit did

not grow along one simple timeline. Instead, several intertwined fronts answer different sources of difficulty:

1. **Continuous and convex optimization** uses geometry to turn local information into global certificates.
2. **Combinatorial optimization** handles discrete choices whose number can grow exponentially.
3. **Relaxation and approximation** seek tractable bounds or approximation guarantees when exact solution of an NP-hard problem is impractical; what can be guaranteed depends on the problem.
4. **Modern AI optimization** confronts enormous, stochastic, and often nonconvex objectives under limits on data, memory, energy, and computation.

This opening chapter establishes the language used throughout the course. An AI system is not specified by saying only “train a model.” We must state:

what data are observed, what is random, what the model predicts,
which variables are optimized, what loss is minimized,
and how performance is evaluated on unseen data.

Across applications, the recurring sequence is

data-generating process $\longrightarrow$ sample $\longrightarrow$ model and loss
 $\longrightarrow$ optimization $\longrightarrow$ held-out evaluation.

This week already asks when the resulting objective is convex and executes the first gradient-descent step. Week 2 then extends the same local-model idea to nonsmooth, constrained, and composite problems. Later weeks revisit this sequence for large-scale pretraining, low-rank adaptation, and generalization. Before specializing, we also build a field-level map of what optimization theory asks, how problems are classified, and which part of a reported AI conclusion optimization can—and cannot—verify.

1.1 The learning-and-optimization map

$$\underbrace{(X, Y) \sim \mathcal{D}}_{\text{unknown population}} \implies \underbrace{\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n}_{\text{observed sample}} \implies \underbrace{(h_\theta, \ell)}_{\text{model and loss}},$$

$$\hat{R}_\mathcal{S}(\hat{\theta}) + \lambda \Omega(\hat{\theta}) \leq \inf_{\theta} [\hat{R}_\mathcal{S}(\theta) + \lambda \Omega(\theta)] + \varepsilon_{\text{opt}}, \quad \hat{R}_\mathcal{S}(\hat{\theta}) \implies R(\hat{\theta}).$$

This display is a map, so its symbols are declared before the detailed definitions below. The distribution $\mathcal{D}$ generates the observed sample $\mathcal{S}$; h_θ is a predictor indexed by the trainable parameter θ ; ℓ is a pointwise loss; and $\hat{R}_\mathcal{S}$ is the sample-average loss. The nonnegative scalar λ weights a regularizer Ω . The displayed inequality allows an optimization error $\varepsilon_{\text{opt}} \geq 0$ and does not assume that the infimum is attained. If $\varepsilon_{\text{opt}} = 0$ and the minimum is attained, it is equivalent to $\hat{\theta} \in \arg \min_{\theta} [\hat{R}_\mathcal{S}(\theta) + \lambda \Omega(\theta)]$. Finally, $R(\hat{\theta})$ denotes population risk on a fresh draw, not another value of the training objective.

The upper row moves from the unknown data-generating process to quantities we can compute. The lower row separates two questions:

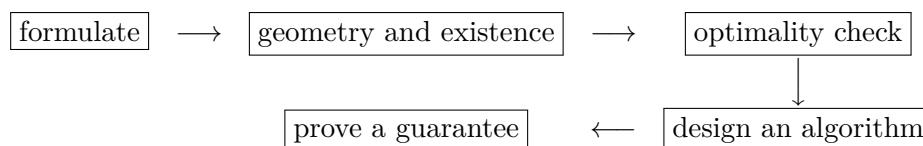
1. **Optimization question:** can we find a parameter with small training objective?
2. **Learning question:** does the returned parameter perform well on new data from the population?

Optimizing the empirical objective does not by itself answer the learning question. This distinction is the conceptual foundation of the course. In the opening recommender example, $h_{\hat{\theta}}$ supplies predictions to a separate serving-time optimizer. Small training loss does not by itself choose the right serving utility, define the feasible recommendation slate, or guarantee useful deployment behavior. The learning and decision loops must be formulated and checked separately. A concrete rating model appears in [section 1.7.3](#).

1.2 Optimization theory: a map of the field

Optimization is the disciplined study of choosing a decision under an objective and constraints. It is broader than any particular algorithm: gradient descent is one method, while optimization theory explains which problem has been posed, whether a solution exists, how an answer can be verified, what information an algorithm needs, and how its cost and error scale.

The recurring sequence of questions is



Each box asks a different question.

1. **Formulation: what is being chosen?** Name the decision variable, objective, feasible set, data, and hyperparameters. A badly aligned objective can be optimized perfectly and still produce the wrong engineering behavior.
2. **Existence and geometry: what shape does the problem have?** Compactness or coercivity may give attainment; convexity makes local lower models globally valid; smoothness, curvature, and constraint geometry determine which moves are meaningful.
3. **Optimality: how would we know we are done?** A zero gradient, a subgradient inclusion, KKT conditions, a dual bound, or a primal–dual gap can turn a candidate into a checkable conclusion. This check is often more informative than the final decimal value.
4. **Algorithms and information: what can we compute cheaply?** A method may access function values, gradients, stochastic gradients, projections, proximal maps, matrix–vector products, or a linear optimization oracle. The cheapest available oracle often determines the algorithm.
5. **Guarantees and tradeoffs: what does computation buy?** Convergence theory relates iterations, data passes, memory, precision, conditioning, and randomness to an output guarantee. A rate is only meaningful after its assumptions, cost unit, and reported output have been specified.

An optimization problem rarely has only one label. The same objective may be smooth, convex, constrained, and finite-sum at the same time. A useful first classification therefore uses several independent axes:

Axis	Typical categories	Why the distinction matters
geometry	convex / nonconvex; Euclidean / manifold	whether local information can verify global behavior and which directions are feasible
regularity	smooth / nonsmooth; weak / strong curvature	whether gradients exist and which rates or accelerated methods are available
constraints	unconstrained / simple-set / general constrained / conic	whether a step uses projection, a proximal map, multipliers, duality, or a specialized solver
information	exact / stochastic / derivative-free; first- / second-order	the accuracy and cost of the local information available to the method
scale and structure	finite sum, sparse, low rank, matrix, graph, saddle point, discrete	which decompositions, relaxations, and computational primitives can be exploited

These ideas recur far beyond model fitting. Resource allocation chooses how to spend limited budgets; control and robotics choose trajectories under dynamics and safety constraints; inverse problems infer hidden causes from measurements; network optimization routes flow; experimental design chooses informative measurements; robust and game-theoretic optimization protects against an adversary or uncertainty. AI training is one especially large family of such decision problems, not the definition of optimization itself.

1.3 Learning-theory preview: from training loss to population risk

Suppose a hypothesis family $\mathcal{H}$, meaning a set of candidate predictors, has been chosen and a training algorithm returns $\hat{h} \in \mathcal{H}$. Write $R(h)$ for the expected loss of predictor h on a fresh draw from the population. Optimization studies how well $\hat{h}$ minimizes the empirical objective. The scientific goal is usually different: small population risk and, ultimately, reliable behavior after deployment.

Under the fixed-distribution assumptions developed in Part 2 of this lecture note, a population guarantee has the following schematic form:

$$\underbrace{R(\hat{h}) - R_{\text{target}}^*}_{\text{excess population risk}} \lesssim \underbrace{\varepsilon_{\text{app}}}_{\text{approximation}} + \underbrace{\varepsilon_{\text{est}}}_{\text{estimation / selection}} + \underbrace{\varepsilon_{\text{opt}}}_{\text{optimization}} . \quad (1.1)$$

Here R_{target}^* is the best population risk in a named target class. Equation (1.1) is a roadmap, not yet a theorem: Week 13 states the assumptions and constants that make such a bound valid. The symbol $\lesssim$ means an upper bound up to the explicit constants and qualification terms supplied by that theorem; it is not an equality and carries no universal numerical constant here.

Term or gap	Question	What can help
approximation	Can the chosen model family represent a sufficiently good predictor?	a better representation, features, architecture, or hypothesis class
estimation / selection	Does performance measured on finite development data transfer to fresh data from the same population?	more independent data, controlled capacity, uniform guarantees, and honest validation
optimization	Did the algorithm make the empirical objective sufficiently small?	a better method, conditioning, compute, or stopping rule
evaluation / shift	Does the evaluation population, metric, and protocol match the deployment question?	fresh tests, provenance and contamination checks, stress tests, and an explicit evaluation scope

The last row is not automatically absorbed into a classical same-distribution generalization bound. A narrow confidence interval on the wrong benchmark still answers the wrong question.

There is also a subtle selection issue. For one predictor fixed before the sample is observed, an empirical average can concentrate around its population mean. Training, however, selects $\hat{h}$ after inspecting the same sample. Week 13 explains why a statement that holds for one fixed candidate cannot simply be substituted at a data-dependent candidate; one needs simultaneous control over the candidates, algorithm-dependent analysis, or a fresh evaluation sample. Capacity measures such as growth functions and VC dimension quantify one classical route to this simultaneous control.

Worked example: lower training loss need not mean better prediction

Suppose model A has training loss 0 and held-out loss 0.25, while model B has training loss 0.05 and held-out loss 0.08 under the same fixed protocol. The optimizer's training comparison favors A; the held-out result favors B on that protocol. Neither pair of numbers alone proves deployment performance. The example separates the optimization question from the estimation and evaluation questions before any concentration theorem is used.

This decomposition gives practical diagnostic rules. More optimization steps mainly target ε_{opt} ; a richer model may reduce approximation error while increasing selection difficulty; more suitable independent data mainly improve estimation; and none of these alone repairs distribution shift or an invalid metric. Week 1 keeps these distinctions qualitative. Week 13 turns them into concentration, uniform-convergence, capacity, and evaluation statements.

1.4 The whole convex-optimization loop in one ridge problem

Before building more language, inspect the complete loop on the three data points

$$(1, 2), \quad (2, 3), \quad (3, 5).$$

With $h_w(x) = wx$, squared loss, and $\lambda \geq 0$, regularized ERM is

$$\min_{w \in \mathbb{R}} f_\lambda(w) := \frac{1}{3} \sum_{i=1}^3 (wx_i - y_i)^2 + \lambda w^2. \quad (1.2)$$

The data are fixed; w is the decision variable; λ is a hyperparameter. Expanding derivatives gives

$$f'_\lambda(w) = \frac{2}{3}(14w - 23) + 2\lambda w, \quad f''_\lambda(w) = \frac{28}{3} + 2\lambda > 0. \quad (1.3)$$

Thus f_λ is a strictly convex coercive quadratic. It has the unique global minimizer

$$w_\lambda^* = \frac{23}{14 + 3\lambda}. \quad (1.4)$$

This one example already exposes the three central convex-optimization ideas.

1. **Geometry.** A differentiable convex function lies above every tangent line:

$$f(v) \geq f(w) + f'(w)(v - w).$$

Therefore $f'(w^*) = 0$ certifies global optimality.

2. **Existence.** Strict convexity gives uniqueness, while coercivity $f_\lambda(w) \rightarrow +\infty$ as $|w| \rightarrow \infty$ supplies existence.
3. **Algorithm.** If the exact solution is unavailable, minimize a linear local model stabilized by a quadratic movement cost:

$$w_{t+1} = \arg \min_w \left\{ f_\lambda(w_t) + f'_\lambda(w_t)(w - w_t) + \frac{1}{2\eta_t}(w - w_t)^2 \right\} = w_t - \eta_t f'_\lambda(w_t).$$

For example, with $\lambda = 0.1$, $w_0 = 0$, and $\eta_0 = 0.1$,

$$f'_{0.1}(0) = -\frac{46}{3}, \quad w_1 = 0 - 0.1 \left(-\frac{46}{3} \right) = \frac{23}{15} \approx 1.5333.$$

The exact minimizer is $w_{0.1}^* \approx 1.6084$, so the first step moves in the correct direction but does not need to solve the problem exactly. Week 9 will determine stable step sizes and rates.

Key result: the existence and attainment checklist

Before writing $x^* \in \arg \min_{x \in C} f(x)$, ask in this order:

1. Is C nonempty?
2. Is $\inf_{x \in C} f(x)$ finite rather than $-\infty$?
3. Is the infimum attained by some $x^* \in C$?

Two reusable sufficient routes in finite-dimensional spaces are:

- f continuous and C nonempty and compact;
- C is nonempty and closed, $f : C \rightarrow \mathbb{R}$ is continuous, and $f(x) \rightarrow +\infty$ along every feasible sequence with $\|x\|_2 \rightarrow \infty$.

The second route says that the objective grows along every unbounded feasible sequence. The more general extended-value formulation uses proper, lower semicontinuous, coercive objectives; these terms are defined in [definition 2.1](#).

Closed convex data alone are not enough. For example,

$$\inf_{x \in \mathbb{R}} e^x = 0,$$

but $e^x > 0$ for every finite x , so no minimum exists. Likewise, $\inf_{x > 0} x = 0$ is not a minimum because the domain excludes zero.

When a theorem displays $\min$ or $\arg \min$, locating the existence argument is part of reading the theorem, not an optional technicality.

1.5 The objects in a supervised learning problem

The notation can be organized in one chain:

$$(X, Y) \sim \mathcal{D} \text{ on } \mathcal{X} \times \mathcal{Y}, \quad h_\theta : \mathcal{X} \rightarrow \hat{\mathcal{Y}}, \quad L_\theta(x, y) = \ell(h_\theta(x), y).$$

Here $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$ is the observed sample, $\theta \in \mathcal{W} \subseteq \mathbb{R}^d$ is trainable, and the prediction space may differ from the target space. For example:

Task	Target	Prediction	Pointwise loss
scalar regression	$y \in \mathbb{R}$	$h_\theta(x) \in \mathbb{R}$	$(h_\theta(x) - y)^2$
binary classification	$y \in \{0, 1\}$	$p_\theta(x) \in (0, 1)$	$-y \log p_\theta(x) - (1 - y) \log(1 - p_\theta(x))$
sequence prediction	target tokens	token probabilities	average negative log-probability

Keep four objects distinct: a model produces a prediction, a pointwise loss scores one prediction, an empirical objective aggregates training losses, and a held-out metric evaluates the returned model. Before observation the examples are random; after observation their values are fixed inputs to training.

1.6 Probability bridge: what the empirical average means

The training sample is one realized dataset from an unknown population. We need a small amount of probability to express that uncertainty.

1.6.1 Random variables and distributions

Definition 1.1 (Discrete random variable). A discrete random variable Z takes values in a finite or countable set. Its probability mass function is

$$p_Z(z) = \mathbb{P}(Z = z), \quad p_Z(z) \geq 0, \quad \sum_z p_Z(z) = 1.$$

Definition 1.2 (Expectation and variance). For a discrete real-valued random variable Z , its expectation and variance are

$$\mathbb{E}[Z] = \sum_z z p_Z(z), \tag{1.5}$$

$$\text{Var}(Z) = \mathbb{E}[(Z - \mathbb{E}[Z])^2] = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2. \tag{1.6}$$

Expectation is a probability-weighted average. Variance measures squared spread around that average. Expectation is linear:

$$\mathbb{E}[aZ + bU] = a\mathbb{E}[Z] + b\mathbb{E}[U],$$

whether or not Z and U are independent.

Worked example: a random prediction loss

Suppose the loss L is 0 with probability $3/4$ and 2 with probability $1/4$. Then

$$\mathbb{E}[L] = 0 \cdot \frac{3}{4} + 2 \cdot \frac{1}{4} = \frac{1}{2}.$$

Also,

$$\mathbb{E}[L^2] = 0^2 \cdot \frac{3}{4} + 2^2 \cdot \frac{1}{4} = 1,$$

so

$$\text{Var}(L) = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}.$$

The expected loss is not itself a loss value observed on every example. It is the long-run probability-weighted average.

1.6.2 Independent and identically distributed samples

Definition 1.3 (Independent and identically distributed). Random pairs

$$(X_1, Y_1), \dots, (X_n, Y_n)$$

are i.i.d. from $\mathcal{D}$ when every pair has distribution $\mathcal{D}$ and joint event probabilities factor across examples. We write

$$(X_i, Y_i) \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}.$$

Identical distribution connects every example to the target population; independence allows variances to add without covariance terms. It does not mean the observed examples have equal values.

Definition 1.4 (Empirical mean). For observations $Z_1, \dots, Z_n$, the empirical mean is

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i.$$

Key result: mean and variance of an empirical average

Problem solved. How does averaging independent observations reduce uncertainty?

Assumptions. Let $Z_1, \dots, Z_n$ be i.i.d. with

$$\mathbb{E}[Z_i] = \mu, \quad \text{Var}(Z_i) = \sigma^2 < \infty.$$

Statement.

$$\mathbb{E}[\bar{Z}_n] = \mu, \quad \text{Var}(\bar{Z}_n) = \frac{\sigma^2}{n}. \quad (1.7)$$

Main idea. Averaging preserves the mean. Independent fluctuations partly cancel, so the standard deviation decreases as $1/\sqrt{n}$.

Why it matters. Empirical risk is an average of losses. The result explains the basic statistical value of having more independent examples, while later chapters explain why optimizing over many possible models requires additional generalization analysis.

Proof note. Use [equation \(1.7\)](#); the covariance expansion and its independent-sample specialization are worked out in [Solution 1.2](#). The main Week 1 route does not require reproducing this probability proof.

The empirical-average mean and variance are the probability objects later concentrated in Week 13: Week 1 names the objects, while Week 13 turns their fluctuations into high-probability bounds.

1.7 Population, empirical, and regularized risk

Training replaces an unknown population expectation by a sample average and then often adds a penalty. This section keeps those three objects separate before showing how they specialize to classification and rating prediction.

Definition 1.5 (Population risk). For a parameter $\theta \in \mathcal{W}$, the population risk is

$$R(\theta) = \mathbb{E}_{(X,Y) \sim \mathcal{D}} [\ell(h_\theta(X), Y)]. \quad (1.8)$$

It measures expected loss on a fresh draw from the target population.

The distribution $\mathcal{D}$ is normally unknown, so [equation \(1.8\)](#) cannot be evaluated exactly.

Definition 1.6 (Empirical risk). Given an observed sample $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^n$, the empirical risk is

$$\hat{R}_{\mathcal{S}}(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(h_\theta(x_i), y_i). \quad (1.9)$$

For a *fixed* parameter θ , the losses

$$Z_i = \ell(h_\theta(X_i), Y_i)$$

are i.i.d. when the examples are i.i.d. If their variance is finite, then [equation \(1.7\)](#) gives

$$\mathbb{E}[\hat{R}_{\mathcal{S}}(\theta)] = R(\theta), \quad \text{Var}(\hat{R}_{\mathcal{S}}(\theta)) = \frac{\text{Var}(Z_1)}{n}.$$

1.7.1 Regularized empirical-risk minimization

Definition 1.7 (Regularized ERM). Let $\Omega : \mathcal{W} \rightarrow \mathbb{R}_{\geq 0}$ measure a preferred notion of model complexity, and let $\lambda \geq 0$ be a regularization hyperparameter. Regularized empirical-risk minimization solves

$$\hat{F}_\lambda^* := \inf_{\theta \in \mathcal{W}} [\hat{R}_{\mathcal{S}}(\theta) + \lambda \Omega(\theta)]. \quad (1.10)$$

A training procedure is ε_{opt} -accurate when it returns $\hat{\theta}_\lambda \in \mathcal{W}$ satisfying

$$\hat{R}_{\mathcal{S}}(\hat{\theta}_\lambda) + \lambda \Omega(\hat{\theta}_\lambda) \leq \hat{F}_\lambda^* + \varepsilon_{\text{opt}}.$$

If the infimum is attained and $\varepsilon_{\text{opt}} = 0$, this reduces to exact regularized ERM.

The three ingredients have different roles:

- $\hat{R}_{\mathcal{S}}(\theta)$ measures data fit;

- $\Omega(\theta)$ expresses a preference such as small norm, smoothness, or sparsity;
- λ controls the tradeoff.

A *parameter*, such as θ , is selected by training. A *hyperparameter*, such as λ , model depth, or a step size, configures the training problem or algorithm and is normally selected using prior knowledge or validation data. The ridge anchor in equations (1.2) and (1.4) already shows the main effect: increasing λ shrinks the fitted coefficient, but does not by itself guarantee better test performance.

1.7.2 Logistic regression as an ERM example

Let $x_i \in \mathbb{R}^d$ and $y_i \in \{0, 1\}$. A linear score

$$z_\theta(x) = \theta^\top x$$

is converted into a probability using the logistic function

$$\sigma(z) = \frac{1}{1 + e^{-z}}, \quad p_\theta(x) = \sigma(\theta^\top x).$$

Using binary cross-entropy, the empirical risk is

$$\hat{R}_S(\theta) = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_\theta(x_i) + (1 - y_i) \log(1 - p_\theta(x_i))]. \quad (1.11)$$

For $\lambda > 0$, regularized logistic regression solves the attained problem

$$\min_{\theta \in \mathbb{R}^d} \hat{R}_S(\theta) + \lambda \|\theta\|_2^2.$$

Worked example: separable logistic loss and attainment

For one positive example $(x, y) = (1, 1)$, the scalar logistic loss is

$$\ell(w) = -\log \sigma(w) = \log(1 + e^{-w}). \quad (1.12)$$

It satisfies

$$\ell'(w) = -\frac{1}{1 + e^w} < 0, \quad \ell''(w) = \frac{e^w}{(1 + e^w)^2} > 0.$$

Hence the loss is strictly convex but decreases toward zero without attaining it:

$$\inf_{w \in \mathbb{R}} \ell(w) = 0, \quad \arg \min_{w \in \mathbb{R}} \ell(w) = \emptyset.$$

Strict convexity gives uniqueness *if* a minimizer exists; it does not guarantee existence. For every $\lambda > 0$, adding quadratic regularization (with the $\lambda/2$ normalization in this scalar example),

$$\ell_\lambda(w) = \log(1 + e^{-w}) + \frac{\lambda}{2} w^2. \quad (1.13)$$

gives a coercive, strictly convex objective. It therefore attains one unique minimum, characterized by

$$\lambda w^\star = \frac{1}{1 + e^{w^\star}}.$$

Here regularization changes both the statistical preference and the well-posedness of the optimization problem.

Common trap

“The training loss approaches its infimum” does not imply that the parameter approaches a finite optimizer. In [equation \(1.12\)](#), no optimizer exists. Before discussing convergence of iterates to w^* , first verify that w^* is a legitimate object.

1.7.3 Matrix factorization for recommendation

Return to the opening recommender, now with explicit numerical ratings. There are m users and q items. The training data are ratings $a_{ij} \in \mathbb{R}$ for pairs in a nonempty set $\mathcal{I}_{\text{tr}} \subseteq \{1, \dots, m\} \times \{1, \dots, q\}$. An unobserved rating is unknown, not zero. The symbol $\mathcal{I}_{\text{tr}}$ denotes observed training indices; Ω continues to denote a regularizer.

Choose a latent dimension $1 \leq r \leq \min\{m, q\}$, usually small. Represent user i and item j by trainable vectors $u_i, v_j \in \mathbb{R}^r$. Stack their transposes as rows of $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{q \times r}$, respectively. Predict

$$\hat{a}_{ij} = h_{(U,V)}(i, j) = u_i^\top v_j, \quad \hat{A} = UV^\top, \quad \text{rank}(\hat{A}) \leq r.$$

Shared factors let ratings from different users and items inform one another; this is a form of *collaborative filtering* [KBV09]. Low rank is a modeling assumption, not a property guaranteed for real preferences.

With $\lambda > 0$, train by minimizing the observed-entry objective

$$F_\lambda(U, V) := \frac{1}{2|\mathcal{I}_{\text{tr}}|} \sum_{(i,j) \in \mathcal{I}_{\text{tr}}} (u_i^\top v_j - a_{ij})^2 + \frac{\lambda}{2} (\|U\|_F^2 + \|V\|_F^2). \quad (1.14)$$

Here $\|U\|_F^2 = \sum_{i=1}^m \sum_{k=1}^r U_{ik}^2$, and similarly for V . Thus $\theta = (U, V)$, the loss is half the squared error, and $\Omega(U, V) = (\|U\|_F^2 + \|V\|_F^2)/2$. The objective is continuous; the positive penalty makes it coercive, so a minimum exists. The rank budget r and penalty weight λ are hyperparameters.

Worked example: predicting two missing ratings

For two users and three items, let question marks denote unobserved ratings. The illustrative rank-one factors give

$$A_{\text{obs}} = \begin{pmatrix} 1 & 2 & ? \\ 2 & ? & 5 \end{pmatrix}, \quad U = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad V = \begin{pmatrix} 1 \\ 2 \\ 5/2 \end{pmatrix}, \quad UV^\top = \begin{pmatrix} 1 & 2 & 5/2 \\ 2 & 4 & 5 \end{pmatrix}.$$

Hence $\hat{a}_{13} = 5/2$ and $\hat{a}_{22} = 4$. The observed-entry loss is zero, but $F_\lambda(U, V) = 65\lambda/8$; these factors illustrate prediction and are not claimed to minimize the regularized objective.

Fixing V leaves a convex ridge least-squares problem in each u_i ; fixing U gives the analogous item problems. Alternating these solves is *alternating least squares*. Joint optimization is generally nonconvex because both factors vary, so exact block solves alone do not certify a global optimum. After fitting, inner products score unrated items; held-out ratings assess prediction, while the opening decision loop determines which items to recommend. Week 12 develops the low-rank geometry further.

Key result: one ERM template, different prediction tasks

Linear regression, logistic regression, and matrix factorization share the architecture

parameter $\longrightarrow$ prediction $\longrightarrow$ pointwise loss $\longrightarrow$ sample average.

Linear regression and matrix factorization predict real values with squared loss; logistic regression predicts a probability with cross-entropy. The prediction task informs the loss, while the parameterization determines the optimization geometry: the same squared loss can yield a convex linear model or a nonconvex factorized model.

1.8 From data separation to problem routing

Good optimization starts before an algorithm is selected: decide which data may influence each choice, specify the objective, and then classify the mathematical structure.

Definition 1.8 (Three-way data split). A dataset is commonly divided into:

1. **training data**, used to fit model parameters;
2. **validation data**, used to choose hyperparameters and compare candidate models;
3. **test data**, opened only after the model, hyperparameters, prompts, and stopping rule are fixed, to estimate final performance.

Let Λ be a nonempty finite set of candidates, each defining a trained parameter $\hat{\theta}_\lambda$. Validation data can select

$$\hat{\lambda} \in \arg \min_{\lambda \in \Lambda} \hat{R}_{\text{val}}(\hat{\theta}_\lambda).$$

using a fixed deterministic rule for ties. Only after this choice is fixed is the final test score computed for $\hat{\theta}_{\hat{\lambda}}$.

Repeatedly changing the model after viewing the test score makes that score adaptive validation evidence, not a final independent estimate; a new untouched test set is then required. Cross-validation can improve estimation when data are scarce, but it does not license repeated inspection of the final test set.

1.8.1 Training objective versus evaluation criterion

The training objective must be optimizable. The deployment criterion must be meaningful. They may differ:

training objective	possible evaluation criterion
cross-entropy	accuracy, calibration, worst-group accuracy
squared loss	mean absolute error, domain-specific cost
next-token loss	task success, factuality, safety evaluation.

Using a surrogate loss is legitimate only when its relationship to the actual evaluation goal is understood and checked on held-out data.

1.8.2 Classify the optimization structure

After specifying the data and loss, write the decision problem in the canonical form

$$\underset{\theta \in \mathcal{W}}{\text{minimize}} \quad f(\theta). \quad (1.15)$$

Here θ is the decision variable, $\mathcal{W}$ is the feasible set, and f is the objective. Then classify the relevant axes independently: convexity alone does not determine smoothness, constraints, stochastic access, or exploitable matrix structure.

Structure	Example	Where the course develops it
smooth convex	ridge or linear logistic regression	gradient methods and convergence
nonsmooth or composite convex	sparse estimation and ℓ_1 penalties	subgradient and proximal methods
constrained, conic, or discrete	budgets, PSD constraints, routing, and Max-Cut	projection, KKT, SDP, relaxation, and rounding
large finite sum or expectation	supervised learning and pretraining	stochastic and variance-reduced methods
nonconvex, saddle-point, or structured	deep networks, robust learning, and low-rank models	local stationarity plus problem-specific geometry

The table is a navigation device, not a substitute for assumptions. After a dataset is observed, the full empirical objective is deterministic; a minibatch gradient is random because the algorithm samples only part of that fixed dataset. Generalization is a separate population question.

Example 1.9 (Routing a two-layer neural-network objective). Let $\{(x_i, y_i)\}_{i=1}^n$ be a data set with inputs $x_i \in \mathbb{R}^d$ and (scalar) targets $y_i \in \mathbb{R}$. For a componentwise nonlinear activation σ , consider

$$h_{w_1, W_2}(x) = w_1^\top \sigma(W_2^\top x),$$

where $W_2 \in \mathbb{R}^{d \times h}$ and $w_1 \in \mathbb{R}^h$. Least-squares training is

$$\inf_{w_1, W_2} \frac{1}{n} \sum_{i=1}^n \left[y_i - w_1^\top \sigma(W_2^\top x_i) \right]^2. \quad (1.16)$$

Jointly, the two parameter blocks interact through a nonlinear composition, so the problem is generally nonconvex. The block structure matters: if W_2 is fixed, the hidden features are fixed and the objective is a convex quadratic in w_1 ; if w_1 is fixed, the nonlinear dependence on W_2 is generally nonconvex. A useful classification should expose such structure rather than attach only the label “nonconvex.”

1.9 A schematic foundation-model lifecycle

LLM foundation models are trained on broad data and then adapted to many downstream tasks [B⁺21]. The systems are much larger than the anchor regression example, but the optimization grammar remains recognizable. The table is schematic: a real system may omit, combine, or reorder stages, and preference optimization is only one post-training route [RSM⁺23].

Stage	Data	Trainable object	Optimization view
pretraining	broad unlabeled or weakly labeled corpora	most or all model parameters	large finite-sum or expected prediction loss
adaptation	task- or domain-specific examples	all parameters or a structured update	new empirical risk, often with parameter or rank constraints
post-training	instructions, preferences, feedback, or verifier signals	model or policy parameters	supervised, pairwise, or reward-shaped objectives
evaluation	held-out tasks and stress tests	no parameter should be fit on the final test set	estimate capability, robustness, and failure modes

Low-rank adaptation is one example of changing the trainable object: instead of updating an entire pretrained weight matrix, LoRA parameterizes a low-rank update while keeping the base weight fixed [HSW⁺22]. Week 12 derives this geometry. Week 10 studies the stochastic optimization and scaling side of pretraining. Week 13 returns to the gap between training objectives and population performance.

Connection to AI practice: the lifecycle is a sequence of optimization problems

Modern AI development is not one monolithic training run. Each lifecycle stage changes at least one of:

data,
 trainable variables,
 loss,
constraints,
evaluation distribution.

The first diagnostic question for a new technique is therefore not “which model name does it use?” but “which mathematical object changed, and what new optimization problem resulted?”

1.10 Chapter synthesis

Summary and connections

- Specify the data, trainable variables, predictor, loss, and evaluation criterion. Regularized ERM combines empirical risk with a penalty $\lambda\Omega(\theta)$.
- Empirical risk measures training fit; population risk concerns fresh data. Training, validation, and test data serve different roles, and low training loss alone does not establish deployment performance.
- Classify geometry, regularity, constraints, and available information before choosing an algorithm or claiming a guarantee.
- An infimum need not be attained, even for strictly convex logistic loss. Positive quadratic regularization restores a unique finite minimizer in the chapter's linear logistic model.
- For smooth unconstrained convex problems, zero gradient certifies global optimality. Gradient descent uses a linear local model plus a quadratic movement penalty; ridge regression supplies the worked example.
- Matrix factorization predicts ratings by $u_i^\top v_j$ and fits observed entries. Fixing either factor gives convex ridge subproblems; joint optimization is generally nonconvex.
- A recommender couples learned predictions with a serving-time decision under a separate objective and constraints. Optimizing one loop does not settle the other.

1.11 Chapter practice set

Exercise 1.1 (Compute: expectation and variance). Let Z take values $-1, 0, 2$ with probabilities $1/4, 1/2, 1/4$. Compute $\mathbb{E}[Z]$, $\mathbb{E}[Z^2]$, and $\text{Var}(Z)$. If $Z_1, \dots, Z_{100}$ are i.i.d., compute the mean and variance of their empirical average.

Exercise 1.2 (Derive: empirical-risk variance). Redo the proof of [equation \(1.7\)](#) without independence. Show that

$$\text{Var}(\bar{Z}_n) = \frac{1}{n^2} \left[\sum_i \text{Var}(Z_i) + 2 \sum_{i < j} \text{Cov}(Z_i, Z_j) \right].$$

Explain how positive correlation changes the benefit of more data.

Exercise 1.3 (Derive: ridge regression). For general scalar observations $(x_i, y_i)_{i=1}^n$ and $\lambda \geq 0$, derive the minimizer of

$$\frac{1}{n} \sum_{i=1}^n (wx_i - y_i)^2 + \lambda w^2.$$

State the condition under which the unregularized minimizer is unique.

Exercise 1.4 (Check: infimum versus minimum). For each problem below, state whether the optimal value is finite and whether it is attained. Use inf rather than min whenever no optimizer exists:

$$\inf_{x>0} x, \quad \inf_{x \in \mathbb{R}} \log(1 + e^{-x}), \quad \min_{x \in [-1, 1]} e^x, \quad \min_{x \in \mathbb{R}} (x^2 + e^x).$$

Name the compactness or coercivity route when it applies.

Exercise 1.5 (Formulate: logistic regression). Let $A \in \mathbb{R}^{n \times d}$ contain examples and $y \in \{0, 1\}^n$ contain labels. Write a dimensionally consistent regularized logistic-regression objective. Identify the decision variable, sample-dependent terms, and hyperparameter.

1.12 Complete solutions

Solution conventions. Throughout this chapter, the mean squared error (MSE) of a predictor on a finite dataset is the arithmetic mean of its squared residuals; it does not include a regularization penalty.

Solution to Exercise 1.1

Direct calculation gives

$$\mathbb{E}[Z] = (-1)\frac{1}{4} + 0\frac{1}{2} + 2\frac{1}{4} = \frac{1}{4}, \quad \mathbb{E}[Z^2] = 1\frac{1}{4} + 0\frac{1}{2} + 4\frac{1}{4} = \frac{5}{4}.$$

Consequently,

$$\text{Var}(Z) = \mathbb{E}[Z^2] - \mathbb{E}[Z]^2 = \frac{5}{4} - \frac{1}{16} = \frac{19}{16}.$$

For $\bar{Z}_{100} = 100^{-1} \sum_{i=1}^{100} Z_i$, linearity of expectation and independence yield

$$\mathbb{E}[\bar{Z}_{100}] = \frac{1}{4}, \quad \text{Var}(\bar{Z}_{100}) = \frac{1}{100^2} \sum_{i=1}^{100} \text{Var}(Z_i) = \frac{19}{1600}.$$

$V(ax) = a^2 V(x)$
 $V(x+y) = V(x) + V(y)$

Solution to Exercise 1.2

Assume that all Z_i have finite second moments, and put $\mu_i = \mathbb{E}[Z_i]$. No independence or identical-distribution assumption is needed for the algebra below. Since

$$\bar{Z}_n - \mathbb{E}[\bar{Z}_n] = \frac{1}{n} \sum_{i=1}^n (Z_i - \mu_i),$$

expanding the square gives

$\text{Var}(x) = \mathbb{E}[(x - \mathbb{E}[x])^2]$

$$\begin{aligned} \text{Var}(\bar{Z}_n) &= \mathbb{E} \left[\left(\frac{1}{n} \sum_{i=1}^n (Z_i - \mu_i) \right)^2 \right] \\ &= \frac{1}{n^2} \left[\sum_{i=1}^n \mathbb{E}[(Z_i - \mu_i)^2] + 2 \sum_{i < j} \mathbb{E}[(Z_i - \mu_i)(Z_j - \mu_j)] \right] \\ &= \frac{1}{n^2} \left[\sum_{i=1}^n \text{Var}(Z_i) + 2 \sum_{i < j} \text{Cov}(Z_i, Z_j) \right]. \end{aligned}$$

Thus positive pairwise covariances make the variance larger than the independent sample benchmark. For example, if every Z_i has variance $\sigma^2 > 0$ and every distinct pair has correlation $\rho \geq 0$, then

$$\text{Var}(\bar{Z}_n) = \frac{\sigma^2}{n} (1 + (n-1)\rho).$$

For a fixed $\rho > 0$, this converges to $\rho\sigma^2$, not to zero. In the extreme case $\rho = 1$, averaging does not reduce variance at all. Positive correlation therefore lowers the effective amount of independent information and can replace the usual $1/n$ variance decay by a nonzero variance floor.

Solution to Exercise 1.3

Assume, as in the chapter, that $\lambda \geq 0$, and define

$$F_\lambda(w) = \frac{1}{n} \sum_{i=1}^n (wx_i - y_i)^2 + \lambda w^2.$$

Its first two derivatives are

$$F'_\lambda(w) = 2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 + \lambda \right) w - \frac{2}{n} \sum_{i=1}^n x_i y_i, \quad F''_\lambda(w) = 2 \left(\frac{1}{n} \sum_{i=1}^n x_i^2 + \lambda \right).$$

Whenever $\sum_i x_i^2 + n\lambda > 0$, the objective is strictly convex, so its unique stationary point is its unique global minimizer:

$$w_\lambda^* = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2 + n\lambda}.$$

For the unregularized problem ($\lambda = 0$), uniqueness holds if and only if

$$\sum_{i=1}^n x_i^2 > 0,$$

equivalently, at least one input is nonzero. If all $x_i = 0$, the objective is $n^{-1} \sum_i y_i^2$, independent of w , and every $w \in \mathbb{R}$ is a minimizer.

Solution to Exercise 1.4

1. Since $x > 0$ for every feasible point and $x \downarrow 0$ is possible,

$$\inf_{x>0} x = 0.$$

The value is finite but is not attained. The feasible set is not closed (and hence is not compact), so the compactness route does not apply.

2. The function $q(x) = \log(1 + e^{-x})$ is strictly positive for every finite x , while $q(x) \rightarrow 0$ as $x \rightarrow +\infty$. Hence

$$\inf_{x \in \mathbb{R}} \log(1 + e^{-x}) = 0,$$

and the finite infimum is not attained. The domain is not compact and q is not coercive.

3. Because e^x is increasing,

$$\min_{x \in [-1, 1]} e^x = e^{-1},$$

attained uniquely at $x = -1$. Existence also follows from continuity on the nonempty compact set $[-1, 1]$.

4. Let $g(x) = x^2 + e^x$. It is continuous and coercive because $g(x) \rightarrow +\infty$ both as $x \rightarrow -\infty$ and as $x \rightarrow +\infty$. Thus it attains a finite minimum. Moreover,

$$g''(x) = 2 + e^x > 0,$$

so the minimizer is unique. The equation $g'(x) = 2x + e^x = 0$ gives

$$x^* = -W_0(1/2) \approx -0.351733711,$$

where W_0 is the principal Lambert function, defined by $W_0(t)e^{W_0(t)} = t$. Consequently,

$$\min_{x \in \mathbb{R}} (x^2 + e^x) = W_0(1/2)^2 + 2W_0(1/2) \approx 0.827184026.$$

Solution to Exercise 1.5

Let the i th row of A be $a_i^\top \in \mathbb{R}^{1 \times d}$ and let $\theta \in \mathbb{R}^d$. Then $A\theta \in \mathbb{R}^n$, and the predicted probabilities are $p_i = \sigma(a_i^\top \theta)$, where $\sigma(t) = (1 + e^{-t})^{-1}$. Following the regularization convention of the chapter, a dimensionally consistent objective is

$$\inf_{\theta \in \mathbb{R}^d} F_\lambda(\theta) := \frac{1}{n} \sum_{i=1}^n \left[\log(1 + e^{a_i^\top \theta}) - y_i a_i^\top \theta \right] + \lambda \|\theta\|_2^2, \quad \lambda \geq 0.$$

Indeed, $\log(1 + e^z) - yz = -y \log \sigma(z) - (1 - y) \log(1 - \sigma(z))$, so the finite-sum term is precisely the empirical binary cross-entropy. The decision variable is θ . The rows $a_i^\top$ and labels y_i are observed, sample-dependent quantities and are fixed during training. The scalar λ is a hyperparameter; the penalty itself is sample-independent. An intercept can be included by appending a constant column to A , with the choice of whether to penalize its coefficient stated separately.

When $\lambda > 0$, the quadratic penalty makes this convex objective coercive and strongly convex, so the infimum is attained at a unique parameter. When $\lambda = 0$, separable data can make the infimum finite but unattained, as in the chapter's scalar logistic example.

Suggested reading. The learning-theory notation follows [SSBD14]. Broader deep-learning context is available in [GBC16]. Matrix-factorization recommenders are introduced in [KBV09]. The foundation-model framing and its downstream inheritance of capabilities and failures are discussed in [B⁺21]; low-rank adaptation is introduced in [HSW⁺22]. The optimization notation is consistent with [BV04].

Historical note

Cauchy's 1847 paper described an iterative descent construction based on derivatives, an early ancestor of the gradient methods used today [Cau47]. Modern machine learning replaces a small analytic objective by empirical risk over data and vastly increases the scale, but the basic loop—differentiate, update, evaluate, and repeat—remains recognizable.

Chapter 2

Convex Geometry and Structured First-Order Methods

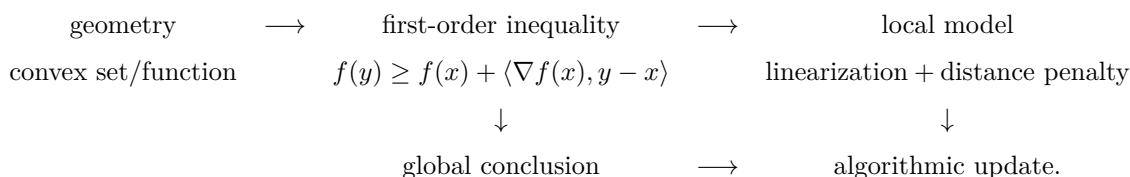
Learning objectives

By the end of this chapter, you should be able to:

1. verify convexity and use a subgradient inequality to build a valid global affine lower model;
2. derive subgradient, projected-gradient, and proximal-gradient updates from one regularized local model;
3. compute Euclidean projections and proximal maps, including soft-thresholding, with their existence conditions;
4. select a method by identifying whether the available oracle returns a subgradient, projection, proximal map, or linear minimizer.

2.1 The conceptual map

The same optimization problem can be viewed through several compatible representations:



Convexity makes a local linear model globally informative. A distance penalty then converts this model into a finite step. Changing the feasible set or the distance model produces different first-order algorithms.

Definition 2.1 (Local notation for norms and effective domains). For $x, y \in \mathbb{R}^d$, the Euclidean inner product and norm are

$$\langle x, y \rangle = x^\top y, \quad \|x\|_2 = \sqrt{\langle x, x \rangle},$$

while $\|x\|_1 = \sum_{j=1}^d |x_j|$. Convex functions in this chapter may take the value $+\infty$ to encode impossible choices. Their effective domain is

$$\text{dom } f := \{x \mid f(x) < +\infty\}.$$

Thus a condition such as $C \cap \text{dom } f \neq \emptyset$ says that at least one feasible point has finite objective value.

An extended-value function is *proper* when its effective domain is nonempty. It is *closed* when its epigraph is closed—equivalently, in finite dimensions, when it is lower semicontinuous. It is *coercive* when $f(x) \rightarrow +\infty$ whenever $\|x\|_2 \rightarrow \infty$. For a convex set C , $\text{relint } C$ is its interior relative to its affine hull. Its indicator function is $\delta_C(x) = 0$ on C and $+\infty$ outside C ; thus $f + \delta_C$ encodes the constraint $x \in C$ in the objective.

2.2 Anchor problem: sparse linear regression

Suppose $A \in \mathbb{R}^{n \times d}$ contains n observations and $b \in \mathbb{R}^n$ contains their responses. Sparse linear regression seeks a vector $x \in \mathbb{R}^d$ that fits the data while using few nonzero coefficients. A standard convex formulation is the LASSO:

$$\min_{x \in \mathbb{R}^d} F(x) := \underbrace{\frac{1}{2n} \|Ax - b\|_2^2}_{f(x): \text{ smooth data fit}} + \underbrace{\lambda \|x\|_1}_{g(x): \text{ nonsmooth regularizer}}, \quad \lambda \geq 0. \quad (2.1)$$

This one model creates the main questions of the chapter.

- Why is F convex?
- What replaces the gradient when $\|x\|_1$ is nondifferentiable?
- Why can ordinary gradient descent not be applied directly to F ?
- How can an algorithm exploit the split $F = f + g$?
- Why do some coefficients become exactly zero?

We will answer all five questions and return to [equation \(2.1\)](#) in a complete worked example. The chapter follows the development in [BV04, Bub15, PB14].

2.3 Convex sets

Definition 2.2 (Convex set). A set $\mathcal{X} \subseteq \mathbb{R}^d$ is *convex* if, for every $x, y \in \mathcal{X}$ and every $\theta \in [0, 1]$,

$$(1 - \theta)x + \theta y \in \mathcal{X}.$$

The point $(1 - \theta)x + \theta y$ is a convex combination of x and y .

Thus a set is convex precisely when the line segment joining any two feasible points remains feasible.

Example 2.3 (Convex and nonconvex sets). The following sets are convex:

$$\mathbb{R}^d, \quad \{x \mid a^\top x \leq b\}, \quad \{x \mid \|x\|_2 \leq r\}, \quad [\ell_1, u_1] \times \cdots \times [\ell_d, u_d].$$

Here the closed ball uses $r \geq 0$. For $r > 0$, the sphere $\{x \mid \|x\|_2 = r\}$ is not convex (already when $d \geq 1$): the midpoint of two antipodal points is 0, which is not on the sphere. A finite set containing two distinct points is also nonconvex.

Proposition 2.4 (Intersections preserve convexity). *If $\{\mathcal{X}_i\}_{i \in I}$ is any family of convex sets, then $\bigcap_{i \in I} \mathcal{X}_i$ is convex.*

Proof. Take $x, y \in \bigcap_{i \in I} \mathcal{X}_i$ and $\theta \in [0, 1]$. Then $x, y \in \mathcal{X}_i$ for every i . Convexity of each $\mathcal{X}_i$ gives $(1 - \theta)x + \theta y \in \mathcal{X}_i$ for every i , hence the convex combination lies in their intersection. $\square$

Why it matters. Linear inequality constraints define halfspaces, so a system $Ax \leq b$ defines an intersection of convex sets. This simple observation is one reason linear programs and many constrained learning models are convex.

2.4 Convex functions

Definition 2.5 (Convex function). Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex. A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is *convex* if, for all $x, y \in \mathcal{X}$ and $\theta \in [0, 1]$,

$$f((1 - \theta)x + \theta y) \leq (1 - \theta)f(x) + \theta f(y). \quad (2.2)$$

The graph of a scalar convex function lies below every chord joining two points on the graph. Equivalently, the epigraph

$$\text{epi } f := \{(x, t) \in \mathcal{X} \times \mathbb{R} \mid f(x) \leq t\}$$

is a convex set.

Proposition 2.6 (Basic convexity-preserving operations). *Let $f_1, \dots, f_m$ be convex functions.*

1. *If $\alpha_i \geq 0$, then $\sum_{i=1}^m \alpha_i f_i$ is convex.*
2. *If f is convex and $x \mapsto Ax + b$ is affine, then $x \mapsto f(Ax + b)$ is convex on its domain.*
3. *The pointwise maximum $\max_i f_i(x)$ is convex.*

Proof. Each statement follows by applying equation (2.2). For example, for the affine composition $h(x) = f(Ax + b)$,

$$\begin{aligned} h((1 - \theta)x + \theta y) &= f((1 - \theta)(Ax + b) + \theta(Ay + b)) \\ &\leq (1 - \theta)f(Ax + b) + \theta f(Ay + b) \\ &= (1 - \theta)h(x) + \theta h(y). \end{aligned}$$

The weighted-sum and pointwise-maximum proofs use the same inequality term by term. $\square$

Example 2.7 (Why the LASSO objective is convex). The scalar map $u \mapsto u^2$ is convex, so each squared residual $(a_i^\top x - b_i)^2$ is convex by affine composition. Their nonnegative sum is convex. The ℓ_1 norm $\|x\|_1 = \sum_{j=1}^d |x_j|$ is convex because the absolute value is convex and sums preserve convexity. Therefore F in equation (2.1) is convex.

2.4.1 A smooth maximum: log-sum-exp and softmax

The pointwise maximum is convex but nonsmooth when two coordinates tie. A standard smooth replacement is the *log-sum-exp* function. For $z = (z_1, \dots, z_m)^\top \in \mathbb{R}^m$ and temperature $\tau > 0$, define

$$\text{LSE}_\tau(z) := \tau \log \left(\sum_{i=1}^m e^{z_i/\tau} \right). \quad (2.3)$$

Let $M = \max_i z_i$. Factoring out the largest exponential gives

$$\text{LSE}_\tau(z) = M + \tau \log \left(\sum_{i=1}^m e^{(z_i - M)/\tau} \right).$$

One term in the sum equals 1, and every term is at most 1. Therefore

$$\max_i z_i \leq \text{LSE}_\tau(z) \leq \max_i z_i + \tau \log m. \quad (2.4)$$

Thus LSE_τ approximates the maximum with a known additive error. Smaller τ makes the approximation sharper.

Differentiation reveals a familiar object:

$$\frac{\partial}{\partial z_i} \text{LSE}_\tau(z) = p_i(z) := \frac{e^{z_i/\tau}}{\sum_j e^{z_j/\tau}}. \quad (2.5)$$

The gradient is exactly the *softmax* probability vector. Its Hessian is

$$\nabla^2 \text{LSE}_\tau(z) = \frac{1}{\tau} \left(\text{Diag}(p) - pp^\top \right) \succeq 0. \quad (2.6)$$

Indeed, for any $v \in \mathbb{R}^m$,

$$v^\top \left(\text{Diag}(p) - pp^\top \right) v = \sum_i p_i v_i^2 - \left(\sum_i p_i v_i \right)^2 = \text{Var}_{i \sim p}(v_i) \geq 0.$$

This proves convexity and gives the matrix a statistical interpretation.

Worked example: two scores and one temperature

Take $z = (0, 1)^\top$ and $\tau = 1$. Then

$$\text{LSE}_1(z) = \log(1 + e) \approx 1.3133, \quad p \approx (0.2689, 0.7311)^\top.$$

The exact guarantee reads

$$1 \leq 1.3133 \leq 1 + \log 2 \approx 1.6931.$$

The larger score receives the larger softmax weight, but the smaller score is not discarded. As $\tau \downarrow 0$, the value approaches the hard maximum. With a unique maximum, its weight tends to one; with several tied maxima, the standard softmax divides the limiting mass equally among them. Larger temperatures spread the weights more evenly.

The factored expression above is also the numerically stable way to compute log-sum-exp: subtract M before exponentiating, then add it back. This avoids overflow when all scores are shifted by a large constant.

2.4.2 The first-order characterization

Key result: supporting-hyperplane inequality

Problem solved. How can we recognize and use differentiable convexity without checking every chord?

Assumptions. Let $\mathcal{X} \subseteq \mathbb{R}^d$ be open and convex, and let $f : \mathcal{X} \rightarrow \mathbb{R}$ be differentiable.

Statement. The function f is convex if and only if

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \quad \text{for every } x, y \in \mathcal{X}. \quad (2.7)$$

Main idea. A convex differentiable function lies above every tangent hyperplane. The tangent model is local, but the inequality is global.

Why it matters. If $\nabla f(x^*) = 0$, then $f(y) \geq f(x^*)$ for every y . Thus a stationary point is globally optimal.

Proof. Assume first that f is convex. Fix $x, y \in \mathcal{X}$, write $d = y - x$, and define $\phi(t) = f(x + td)$ for $t \in [0, 1]$. Convexity gives, for $t \in (0, 1]$,

$$\phi(t) \leq (1 - t)\phi(0) + t\phi(1).$$

Rearranging,

$$\phi(1) \geq \phi(0) + \frac{\phi(t) - \phi(0)}{t}.$$

Letting $t \downarrow 0$ and using differentiability,

$$f(y) = \phi(1) \geq \phi(0) + \phi'(0) = f(x) + \langle \nabla f(x), y - x \rangle.$$

Conversely, assume equation (2.7). For $z = (1 - \theta)x + \theta y$, apply the inequality twice, using z as the base point:

$$f(x) \geq f(z) + \langle \nabla f(z), x - z \rangle, \quad f(y) \geq f(z) + \langle \nabla f(z), y - z \rangle.$$

Multiply the first inequality by $1 - \theta$, the second by θ , and add. Since $(1 - \theta)(x - z) + \theta(y - z) = 0$, the inner-product terms cancel:

$$(1 - \theta)f(x) + \theta f(y) \geq f(z).$$

This is the definition of convexity. □

Corollary 2.8 (First-order global optimality). *If $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable and convex, then $\nabla f(x^*) = 0$ implies $x^* \in \arg \min_x f(x)$.*

Proof. Set $x = x^*$ in equation (2.7). □

Common trap

Mistake: “A stationary point is always a global minimizer.”

Why it is wrong: The conclusion uses convexity. For $f(t) = t^3$, $f'(0) = 0$, but 0 is not even a local minimum.

Self-check: Before using stationarity as a global condition, identify the exact convexity assumption and the domain on which it holds.

2.4.3 Subgradients for nonsmooth convex functions

Definition 2.9 (Subgradient and subdifferential). Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex and finite at x . A vector $g \in \mathbb{R}^d$ is a *subgradient* of f at x if

$$f(y) \geq f(x) + \langle g, y - x \rangle \quad \text{for every } y \in \mathbb{R}^d. \quad (2.8)$$

The set of all such vectors is the *subdifferential* $\partial f(x)$.

If f is differentiable and convex, then $\partial f(x) = \{\nabla f(x)\}$. At a kink, the subdifferential may contain an interval or a higher-dimensional convex set.

Example 2.10 (Absolute value). For $h(t) = |t|$,

$$\partial h(t) = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0. \end{cases}$$

At $t = 0$, the inequality $|y| \geq gy$ holds for every y exactly when $g \in [-1, 1]$.

Proposition 2.11 (Subgradient optimality). Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper and convex, and let $x^* \in \text{dom } f$. Then

$$0 \in \partial f(x^*) \iff x^* \in \arg \min_x f(x).$$

Proof. If $0 \in \partial f(x^*)$, then

$$f(y) \geq f(x^*) + \langle 0, y - x^* \rangle = f(x^*)$$

for every y . Thus x^* is globally optimal. Conversely, if x^* is a global minimizer, then $f(y) \geq f(x^*) = f(x^*) + \langle 0, y - x^* \rangle$ for every y . This is precisely $0 \in \partial f(x^*)$. No relative-interior assumption is needed for this equivalence under the stated subgradient definition. $\square$

2.5 One regularized local model, four familiar updates

Let $C \subseteq \mathbb{R}^d$ be a nonempty closed convex set, let $r : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex, and assume $C \cap \text{dom } r \neq \emptyset$. Given a first-order vector g_t and a stepsize $\eta_t > 0$, define

$$x_{t+1} = \arg \min_{x \in C} \left\{ \langle g_t, x - x_t \rangle + \frac{1}{2\eta_t} \|x - x_t\|_2^2 + r(x) \right\}. \quad (2.9)$$

The quadratic term dominates at infinity, so the closed convex subproblem is coercive and attains its minimum. It is also strongly convex, hence the minimizer is unique. The choices below recover the chapter's methods:

Method	First-order vector	Structure kept exact
GD	$g_t = \nabla f(x_t)$	$C = \mathbb{R}^d, r = 0$
subgradient	$g_t \in \partial f(x_t)$	$C = \mathbb{R}^d, r = 0$
projected GD	$g_t = \nabla f(x_t)$	C with $r = 0$
proximal GD	$g_t = \nabla f(x_t)$	$C = \mathbb{R}^d$ and nonsmooth r

The general first-order condition behind the subproblem is the normal-cone Fermat rule in [equation \(B.4\)](#). The body of this chapter uses its projection and proximal specializations; [Appendix B](#) records the general statement and its qualification. In [figure 2.1](#), the projection arrow is perpendicular to the rectangle's upper edge; the other panels show different local oracles.

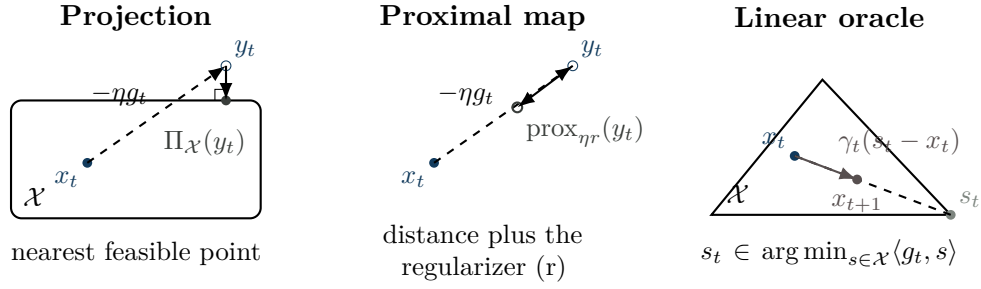


Figure 2.1: Three distinct first-order primitives. Projection and the proximal map first form $y_t = x_t - \eta g_t$ and then correct it. Frank–Wolfe instead calls a linear minimization oracle and moves only partway from x_t toward the returned atom s_t ; it does not form or project y_t .

2.5.1 Gradient descent retrieval

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable. At x_t , replace f by its first-order model and penalize a step that moves too far:

$$x_{t+1} = \arg \min_{x \in \mathbb{R}^d} \left\{ f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + \frac{1}{2\eta_t} \|x - x_t\|_2^2 \right\}. \quad (2.10)$$

The constant $f(x_t)$ does not affect the minimizer. Differentiating the local model with respect to x and setting the result to zero gives

$$\nabla f(x_t) + \frac{1}{\eta_t}(x - x_t) = 0,$$

so

$$x_{t+1} = x_t - \eta_t \nabla f(x_t).$$

Algorithm: Gradient descent (GD)

Input: differentiable f , starting point x_0 , step sizes $\eta_t > 0$, number of steps T .

1. For $t = 0, \dots, T - 1$, compute $g_t = \nabla f(x_t)$.
2. Update $x_{t+1} = x_t - \eta_t g_t$.
3. Return a specified output, usually x_T .

Scope: The algorithm is defined for any differentiable f , but descent and convergence require additional assumptions and suitable step sizes.

Worked example: one GD step on an anisotropic quadratic

Let

$$f(x_1, x_2) = \frac{1}{2}x_1^2 + 2x_2^2, \quad \nabla f(x_1, x_2) = (x_1, 4x_2)^\top.$$

Starting from $x_0 = (2, 1)^\top$ with $\eta = 1/4$,

$$x_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \frac{1}{4} \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 0 \end{pmatrix}.$$

The second coordinate reaches zero in one step, while the first moves only part of the way. Different curvature in different directions produces different time scales; Week 9 formalizes this through conditioning.

2.5.2 The subgradient method

For a convex but nondifferentiable objective, choose any $g_t \in \partial f(x_t)$ and update

$$x_{t+1} = x_t - \eta_t g_t.$$

Algorithm: Subgradient method

Input: convex f , subgradient oracle, x_0 , $\eta_t > 0$, T .

1. Choose $g_t \in \partial f(x_t)$.
2. Update $x_{t+1} = x_t - \eta_t g_t$.
3. Return an averaged or best iterate specified by the convergence theorem.

Limitation: A negative subgradient is not necessarily a descent direction for a finite step. Step-size control and averaging are essential; Week 8 proves the relevant guarantee.

Example 2.12 (Subgradient step for $|x|$). At $x_t = 2$, the only subgradient is $g_t = 1$, so a step with $\eta_t = 1/2$ gives $x_{t+1} = 3/2$. At $x_t = 0$, any $g_t \in [-1, 1]$ is valid. Choosing $g_t = 0$ correctly leaves the optimum fixed.

2.6 Constraints: projection and projected gradient

Definition 2.13 (Euclidean projection). Let $\mathcal{X} \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. The Euclidean projection of $y \in \mathbb{R}^d$ onto $\mathcal{X}$ is

$$\Pi_{\mathcal{X}}(y) := \arg \min_{x \in \mathcal{X}} \frac{1}{2} \|x - y\|_2^2.$$

In finite dimensions, the squared distance is continuous and coercive. Together with nonemptiness and closedness of $\mathcal{X}$, this ensures attainment; strict convexity over a convex feasible set ensures uniqueness. These are separate existence and uniqueness checks.

Key result: projection optimality

Problem solved. How can we recognize a projection without solving the distance problem again?

Assumptions. $\mathcal{X} \subseteq \mathbb{R}^d$ is nonempty, closed, and convex.

Statement. A point $p \in \mathcal{X}$ equals $\Pi_{\mathcal{X}}(y)$ if and only if

$$\langle y - p, x - p \rangle \leq 0 \quad \text{for every } x \in \mathcal{X}. \quad (2.11)$$

Main idea. The vector from p to y forms an obtuse angle with every feasible direction leaving p .

Why it matters. This inequality is the main geometric step in the convergence proof of projected gradient descent.

In the language of Appendix B, equation (2.11) says exactly

$$y - p \in N_{\mathcal{X}}(p),$$

where the normal cone is

$$N_{\mathcal{X}}(p) := \left\{ v \in \mathbb{R}^d \mid \langle v, x - p \rangle \leq 0 \text{ for every } x \in \mathcal{X} \right\}.$$

This is the projection specialization of equation (B.4).

Proof. Suppose first that $p = \Pi_{\mathcal{X}}(y)$. For any $x \in \mathcal{X}$, convexity implies $p + \theta(x - p) \in \mathcal{X}$ for $\theta \in [0, 1]$. Define

$$\phi(\theta) = \frac{1}{2} \|p + \theta(x - p) - y\|_2^2.$$

Because $\theta = 0$ minimizes ϕ over $[0, 1]$, the right derivative satisfies

$$\phi'(0) = \langle p - y, x - p \rangle \geq 0,$$

which is equivalent to equation (2.11).

Conversely, assume equation (2.11). For every $x \in \mathcal{X}$,

$$\begin{aligned} \|x - y\|_2^2 &= \|(x - p) + (p - y)\|_2^2 \\ &= \|x - p\|_2^2 + \|p - y\|_2^2 + 2 \langle x - p, p - y \rangle \\ &\geq \|p - y\|_2^2. \end{aligned}$$

Thus p minimizes the distance to y . □

Corollary 2.14 (Nonexpansiveness of projection). *If $\mathcal{X} \subseteq \mathbb{R}^d$ is nonempty, closed, and convex, then for every $u, v \in \mathbb{R}^d$,*

$$\|\Pi_{\mathcal{X}}(u) - \Pi_{\mathcal{X}}(v)\|_2 \leq \|u - v\|_2.$$

Proof. Let $p = \Pi_{\mathcal{X}}(u)$ and $q = \Pi_{\mathcal{X}}(v)$. Applying equation (2.11) with $x = q$ and then with $x = p$ gives

$$\langle u - p, q - p \rangle \leq 0, \quad \langle v - q, p - q \rangle \leq 0.$$

After adding and rearranging,

$$\|p - q\|_2^2 \leq \langle u - v, p - q \rangle.$$

Cauchy–Schwarz gives $\|p - q\|_2^2 \leq \|u - v\|_2 \|p - q\|_2$. If $p = q$, the result is immediate; otherwise divide by $\|p - q\|_2$. □

Algorithm: Projected gradient descent (PGD)

Problem:

$$\min_{x \in \mathcal{X}} f(x),$$

where f is differentiable and $\mathcal{X}$ is nonempty, closed, and convex.

1. Start from $x_0 \in \mathcal{X}$.
2. Compute the unconstrained proposal $y_{t+1} = x_t - \eta_t \nabla f(x_t)$.
3. Restore feasibility: $x_{t+1} = \Pi_{\mathcal{X}}(y_{t+1})$.

Equivalently, the PGD step solves the constrained local model

$$x_{t+1} = \arg \min_{x \in \mathcal{X}} \left\{ \langle \nabla f(x_t), x - x_t \rangle + \frac{1}{2\eta_t} \|x - x_t\|_2^2 \right\}. \quad (2.12)$$

Worked example: projection onto an ℓ_2 ball

Let $\mathcal{X} = \{x \in \mathbb{R}^d \mid \|x\|_2 \leq r\}$ with $r > 0$. Then

$$\Pi_{\mathcal{X}}(y) = \begin{cases} y, & \|y\|_2 \leq r, \\ r \frac{y}{\|y\|_2}, & \|y\|_2 > r. \end{cases}$$

For $y = (3, 4)^\top$ and $r = 2$, $\|y\|_2 = 5$, so

$$\Pi_{\mathcal{X}}(y) = 2 \frac{(3, 4)^\top}{5} = (6/5, 8/5)^\top.$$

The projected point has norm 2 and lies on the ray from 0 through y .

2.7 Composite objectives: proximal gradient

Projection treats a constraint by minimizing squared distance over a set. A proximal operator generalizes this idea by adding a convex function instead of a hard constraint.

Definition 2.15 (Proximal operator). Let $g : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex, and let $\eta > 0$. The proximal operator is

$$\text{prox}_{\eta g}(v) := \arg \min_{x \in \mathbb{R}^d} \left\{ g(x) + \frac{1}{2\eta} \|x - v\|_2^2 \right\}. \quad (2.13)$$

Because a proper closed convex function has an affine lower support on its domain, the quadratic term makes the subproblem coercive and closed; a minimizer therefore exists. The same term makes the objective strongly convex, so that minimizer is unique. Equivalently, the proximal point is characterized by equation (B.6). If g is the indicator of a closed convex set,

$$g(x) = \begin{cases} 0, & x \in \mathcal{X}, \\ +\infty, & x \notin \mathcal{X}, \end{cases}$$

then $\text{prox}_{\eta g}(v) = \Pi_{\mathcal{X}}(v)$. Projection is therefore a special proximal operator.

Algorithm: Proximal gradient**Problem:**

$$\min_{x \in \mathbb{R}^d} F(x) := f(x) + g(x),$$

where f is convex and differentiable and g is proper, closed, convex, and has an efficiently computable proximal operator.

1. Form the gradient proposal $v_{t+1} = x_t - \eta_t \nabla f(x_t)$.
2. Apply the structured correction $x_{t+1} = \text{prox}_{\eta_t g}(v_{t+1})$.

Key result: soft-thresholding

For $g(t) = \lambda |t|$, $\lambda \geq 0$,

$$\text{prox}_{\eta g}(v) = \text{soft}(v, \eta\lambda) := \begin{cases} v - \eta\lambda, & v > \eta\lambda, \\ 0, & |v| \leq \eta\lambda, \\ v + \eta\lambda, & v < -\eta\lambda. \end{cases} \quad (2.14)$$

Proof. We minimize

$$\lambda |t| + \frac{1}{2\eta}(t - v)^2.$$

Optimality is equivalent to

$$0 \in \lambda \partial |t| + \frac{1}{\eta}(t - v).$$

If $t > 0$, then $\partial |t| = \{1\}$, so $t = v - \eta\lambda$, consistent with $t > 0$ precisely when $v > \eta\lambda$. If $t < 0$, then $t = v + \eta\lambda$, consistent precisely when $v < -\eta\lambda$. If $t = 0$, the condition becomes

$$\frac{v}{\eta} \in \lambda[-1, 1],$$

or $|v| \leq \eta\lambda$. These three cases give equation (2.14). $\square$

Worked example: the anchor LASSO in one dimension

Consider

$$\min_{x \in \mathbb{R}} F(x) = \frac{1}{2}(x - 3)^2 + |x|.$$

Here $f(x) = \frac{1}{2}(x - 3)^2$, $g(x) = |x|$, and $f'(x) = x - 3$. With $\eta = 1$, the gradient proposal is

$$v_{t+1} = x_t - (x_t - 3) = 3,$$

independent of x_t . The proximal step gives

$$x_{t+1} = \text{soft}(3, 1) = 2.$$

Thus proximal gradient reaches $x^* = 2$ in one step.

We can verify optimality directly:

$$0 \in (x^* - 3) + \partial |x^*| = -1 + \{1\} = \{0\}.$$

The coefficient is reduced from the unregularized value 3 to 2. If the data-fit center were 1/2 instead of 3, soft-thresholding would set the solution exactly to zero.

Connection to AI practice: structured parameter updates

Many learning objectives combine a differentiable data-fit term with a nonsmooth regularizer or simple constraint. Proximal and projected methods make the structure explicit rather than hiding it inside a generic solver. The same design principle will reappear in low-rank adaptation: choose a parameterization or update that expresses the structure one wants the algorithm to exploit.

2.8 How to choose the first algorithm

Problem structure	First candidate	Diagnostic question
Smooth, unconstrained f	Gradient descent	Can $\nabla f(x)$ be evaluated, and is a safe step size known or estimated?
Convex, nonsmooth f	Subgradient method	Can a bounded subgradient be obtained, and is slow convergence acceptable?
Smooth f , simple convex $\mathcal{X}$	PGD	Can $\Pi_{\mathcal{X}}(y)$ be computed substantially more easily than the full problem?
$f + g$, smooth f , simple convex g	Proximal gradient	Is $\text{prox}_{\eta g}(v)$ available in closed form or by a cheap subroutine?

This table selects a plausible starting point, not a universal winner. The convergence rate, cost per step, data access pattern, and numerical stability must still be analyzed.

2.9 Projection-free optimization: Frank–Wolfe

Projected gradient is attractive when projection onto the feasible set is cheap. Some important convex sets have the opposite profile: projection is expensive, but minimizing a linear function over the set is easy. Frank–Wolfe, also called the conditional-gradient method, is designed for this case.

Let f be differentiable and convex on a nonempty compact convex set $\mathcal{X}$. At x_t , first call a linear minimization oracle:

$$s_t \in \arg \min_{s \in \mathcal{X}} \langle \nabla f(x_t), s \rangle.$$

Then move toward its answer:

$$x_{t+1} = (1 - \gamma_t)x_t + \gamma_t s_t, \quad \gamma_t \in [0, 1]. \quad (2.15)$$

Both endpoints are feasible, so convexity of $\mathcal{X}$ keeps every iterate feasible. Unlike projected gradient, the method never asks for the nearest feasible point.

The same oracle produces a computable gap:

$$g_{\text{FW}}(x_t) := \langle \nabla f(x_t), x_t - s_t \rangle. \quad (2.16)$$

If x^* minimizes f over $\mathcal{X}$, convexity gives

$$f(x_t) - f(x^*) \leq \langle \nabla f(x_t), x_t - x^* \rangle.$$

The oracle ensures $\langle \nabla f(x_t), s_t \rangle \leq \langle \nabla f(x_t), x^* \rangle$, hence

$$0 \leq f(x_t) - f(x^*) \leq g_{\text{FW}}(x_t).$$

The Frank–Wolfe gap is therefore both a direction-selection quantity and an upper bound on current suboptimality.

Worked example: one Frank–Wolfe step on a simplex

Let

$$\mathcal{X} = \Delta_3 := \left\{ x \in \mathbb{R}^3 \mid x_i \geq 0, \sum_i x_i = 1 \right\}, \quad f(x) = \frac{1}{2} \|x - e_3\|_2^2,$$

and start from $x_0 = (1/3, 1/3, 1/3)^\top$. The gradient is $\nabla f(x_0) = (1/3, 1/3, -2/3)^\top$. A linear function over a simplex is minimized at a vertex, so the smallest gradient coordinate gives $s_0 = e_3$. With $\gamma_0 = 1/2$,

$$x_1 = \frac{1}{2}x_0 + \frac{1}{2}e_3 = (1/6, 1/6, 2/3)^\top.$$

Moreover,

$$g_{\text{FW}}(x_0) = \|x_0 - e_3\|_2^2 = \frac{2}{3}, \quad f(x_0) - f^* = \frac{1}{3}.$$

Thus the computable Frank–Wolfe gap certifies the current suboptimality, but the bound need not be tight.

Frank–Wolfe is useful only when its linear oracle is materially cheaper than projection. When initialized at a simplex vertex, after t updates its iterate is a mixture of at most $t + 1$ selected vertices; its rate and step-size theory are deferred.

2.10 Computational mini-lab

Computational mini-lab: GD, PGD, and proximal gradient in inspectable NumPy

Goal. Verify that the three updates enforce different forms of structure.

Setup. Python 3 and NumPy; no optimizer library or randomness.

```

1 import numpy as np
2
3 def project_box(x, lower=-1.0, upper=1.0):
4     return np.clip(x, lower, upper)
5
6 def soft_threshold(v, tau):
7     return np.sign(v) * np.maximum(np.abs(v) - tau, 0.0)
8
9 def grad_quadratic(x):
10     # f(x) = 0.5 * (x_1^2 + 4 x_2^2)
11     return np.array([x[0], 4.0 * x[1]])
12
13 def run_gd(x0, eta, steps):
14     xs = [np.array(x0, dtype=float)]
15     for _ in range(steps):
16         xs.append(xs[-1] - eta * grad_quadratic(xs[-1]))
17     return np.array(xs)
18
19 def run_pgd(x0, eta, steps):
20     # Minimize 0.5 ||x - (2,-2)||^2 over [-1,1]^2.
21     target = np.array([2.0, -2.0])
22     xs = [np.array(x0, dtype=float)]
23     proposals = []

```

```

24     for _ in range(steps):
25         proposal = xs[-1] - eta * (xs[-1] - target)
26         proposals.append(proposal)
27         xs.append(project_box(proposal))
28     return np.array(xs), np.array(proposals)
29
30 def run_prox_lasso(x0, eta, lam, steps):
31     # F(x) = 0.5 * (x - 3)^2 + lam * |x|
32     xs = [float(x0)]
33     for _ in range(steps):
34         proposal = xs[-1] - eta * (xs[-1] - 3.0)
35         xs.append(float(soft_threshold(proposal, eta * lam)))
36     return np.asarray(xs)
37
38 gd = run_gd([2.0, 1.0], eta=0.25, steps=8)
39 pgd, proposals = run_pgd([0.0, 0.0], eta=0.75, steps=4)
40 prox = run_prox_lasso(0.0, eta=0.5, lam=1.0, steps=6)
41
42 assert np.allclose(gd[1], [1.5, 0.0])
43 assert np.allclose(proposals[0], [1.5, -1.5])
44 assert np.allclose(pgd[1], [1.0, -1.0])
45 assert np.all(np.abs(pgd) <= 1.0 + 1e-12)
46 assert np.allclose(prox[:4], [0.0, 1.0, 1.5, 1.75])
47
48 print("GD first step:", gd[1])
49 print("PGD proposal then projection:", proposals[0], pgd[1])
50 print("Proximal iterates:", prox)

```

Expected checkpoints.

- The first GD step is $(1.5, 0)^T$.
- PGD maps the infeasible proposal $(1.5, -1.5)^T$ to $(1, -1)^T$, and every stored iterate is feasible.
- With $\eta = 1/2$ and $\lambda = 1$, the proximal iterates begin 0, 1, 1.5, 1.75 and approach the minimizer 2.

Interpretation. GD follows a gradient model, PGD additionally enforces a hard feasible set, and proximal gradient applies a soft structural correction.

Further exploration. Change the GD stepsize to $\eta = 0.6$. For the two coordinates, compute the update multipliers $1 - \eta$ and $1 - 4\eta$, then explain why one coordinate contracts while the other oscillates and diverges.

2.11 Mastery checklist

Summary and connections

- Convexity makes gradient and subgradient supporting inequalities global; subgradients also cover nonsmooth convex functions.
- Log-sum-exp smoothly approximates a maximum, with softmax as its gradient.
- One regularized local model yields GD, subgradient, PGD, or proximal GD by changing the first-order vector and the structure kept exact.
- Projection and proximal subproblems are strongly convex: check attainment, then uniqueness follows. Appendix B unifies their optimality conditions through the normal-cone Fermat rule.
- Frank–Wolfe replaces projection by linear optimization and supplies a computable convex suboptimality bound.

2.12 Chapter practice set

Exercise 2.1 (Recall: convex sets). Let $a \in \mathbb{R}^d$, $\beta \in \mathbb{R}$, $A \in \mathbb{R}^{m \times d}$, $b \in \mathbb{R}^m$, and $r \geq 0$. Determine which sets are convex and justify the answer:

$$\{x \in \mathbb{R}^d \mid a^\top x = \beta\}, \quad \{x \mid \|x\|_2 \geq 1\}, \quad \{x \mid \|Ax - b\|_2 \leq r\}.$$

Exercise 2.2 (Derivation: convexity from the Hessian). Let $f(x) = \frac{1}{2}x^\top Qx + c^\top x$, where $Q = Q^\top$. Show that f is convex if $Q \succeq 0$. Give a two-dimensional counterexample when Q has a negative eigenvalue.

Exercise 2.3 (Computation: subdifferential). Compute $\partial \|x\|_1$ at $x = (2, 0, -3)^\top$. Use the answer to test whether $x = 0$ minimizes the following objective for a given $b \in \mathbb{R}^d$ and $\lambda \geq 0$:

$$F(x) = \frac{1}{2} \|x - b\|_2^2 + \lambda \|x\|_1$$

Exercise 2.4 (Proof: local minima are global). Let f be convex on a convex set $\mathcal{X}$. Prove directly from the definition that every local minimizer is global. State where convexity of the set is used.

Exercise 2.5 (Projection onto a box). Let $y \in \mathbb{R}^d$ and $\ell_i \leq u_i$ for every i . Derive the coordinate-wise projection of y onto $\mathcal{X} = [\ell_1, u_1] \times \cdots \times [\ell_d, u_d]$. Verify the projection-angle condition.

Exercise 2.6 (Projection onto an ℓ_2 ball). Prove the formula used in the worked example for a radius- r Euclidean ball with $r > 0$, and state separately what happens when $r = 0$. Then compute the projection of $(3, 4, 5)^\top$ for $r = 4$.

Exercise 2.7 (Proximal operator of a quadratic). For $g(x) = \frac{\lambda}{2} \|x\|_2^2$, with $\lambda \geq 0$ and $\eta > 0$, derive $\text{prox}_{\eta g}(v)$. Interpret the result as shrinkage.

Exercise 2.8 (Transfer: simplex constraint). The probability simplex is

$$\Delta_d = \left\{ x \in \mathbb{R}^d \mid x_i \geq 0, \sum_{i=1}^d x_i = 1 \right\}.$$

Explain why it is convex. Formulate a PGD method for minimizing a differentiable function over Δ_d . You may use a simplex-projection subroutine without deriving it.

Exercise 2.9 (The Frank–Wolfe gap). For the simplex problem in the Frank–Wolfe worked example:

- (a) compute s_0 , x_1 , and $g_{\text{FW}}(x_0)$ when $\gamma_0 = 1/2$;
- (b) compute the true initial objective gap and verify that it is no larger than the Frank–Wolfe gap;
- (c) explain why a linear minimization oracle over a simplex needs only the smallest coordinate of the gradient.

2.13 Complete solutions

Solution conventions. Throughout this section, all vector spaces are finite-dimensional Euclidean spaces. For a nonempty closed convex set $\mathcal{X}$, its Euclidean projection is

$$\Pi_{\mathcal{X}}(y) := \arg \min_{x \in \mathcal{X}} \frac{1}{2} \|x - y\|_2^2.$$

The projection is unique and is characterized by

$$p = \Pi_{\mathcal{X}}(y) \iff p \in \mathcal{X} \text{ and } \langle y - p, x - p \rangle \leq 0 \text{ for every } x \in \mathcal{X}. \quad (2.17)$$

Unless stated otherwise, radii and regularization parameters are nonnegative, and every stepsize is strictly positive.

Solution to Exercise 2.1

The affine level set S_1 is convex. Indeed, if $x, y \in S_1$ and $\theta \in [0, 1]$, then

$$a^\top((1 - \theta)x + \theta y) = (1 - \theta)a^\top x + \theta a^\top y = \beta.$$

This conclusion also covers the degenerate cases: if $a = 0$ and $\beta = 0$, then $S_1 = \mathbb{R}^d$; if $a = 0$ and $\beta \neq 0$, then $S_1 = \emptyset$, which is convex vacuously.

For $d \geq 1$, the set S_2 is not convex. Let e_1 denote the first standard basis vector. Both e_1 and $-e_1$ belong to S_2 , but their midpoint is 0, whose norm is strictly less than 1.

The set S_3 is convex for $r \geq 0$. If $x, y \in S_3$ and $\theta \in [0, 1]$, the triangle inequality gives

$$\begin{aligned} \|A((1 - \theta)x + \theta y) - b\|_2 &= \|(1 - \theta)(Ax - b) + \theta(Ay - b)\|_2 \\ &\leq (1 - \theta)\|Ax - b\|_2 + \theta\|Ay - b\|_2 \\ &\leq r. \end{aligned}$$

Equivalently, S_3 is the inverse image of a Euclidean ball under an affine map. If one allows the nonstandard case $r < 0$, then S_3 is empty and hence still convex.

Solution to Exercise 2.2

Symmetry of Q yields

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q.$$

More directly, for arbitrary $x, y \in \mathbb{R}^d$ and $h = y - x$,

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle = \frac{1}{2} h^\top Q h.$$

If $Q \succeq 0$, the right-hand side is nonnegative. Thus

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \text{ for all } x, y,$$

which is the first-order characterization of convexity.

For a two-dimensional counterexample, take

$$Q = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad c = 0, \quad f(x_1, x_2) = \frac{1}{2}(x_1^2 - x_2^2).$$

With $u = (0, 1)^\top$ and $v = (0, -1)^\top$,

$$f\left(\frac{u+v}{2}\right) = f(0) = 0 > -\frac{1}{2} = \frac{f(u) + f(v)}{2},$$

so the convexity inequality fails. In fact, this argument generalizes: if Q has a negative eigenvalue with eigenvector $w \neq 0$, the two points w and $-w$ violate midpoint convexity (the linear terms cancel). Hence this quadratic is convex on $\mathbb{R}^d$ if and only if $Q \succeq 0$.

Solution to Exercise 2.3

For a scalar t ,

$$\partial|t| = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0. \end{cases}$$

Since the ℓ_1 norm is a separable sum, its subdifferential is the Cartesian product of the coordinate subdifferentials. Therefore

$$\partial\|(2, 0, -3)^\top\|_1 = \{(1, s, -1)^\top \mid s \in [-1, 1]\}.$$

Now assume $b \in \mathbb{R}^d$ and $\lambda \geq 0$. The smooth term has gradient $x - b$, and

$$\partial\|0\|_1 = [-1, 1]^d.$$

Because F is convex, the subgradient optimality condition is necessary and sufficient:

$$\begin{aligned} 0 \in \partial F(0) &\iff 0 \in -b + \lambda[-1, 1]^d \\ &\iff |b_i| \leq \lambda \quad \text{for every } i \\ &\iff \|b\|_\infty \leq \lambda. \end{aligned}$$

Thus $x = 0$ minimizes F exactly when every coordinate of b lies in $[-\lambda, \lambda]$. The quadratic term is strongly convex, so whenever this condition holds, 0 is in fact the unique minimizer. In the edge case $\lambda = 0$, the condition reduces to $b = 0$.

Source-data note. The exercise supplies no numerical values for b and λ ; consequently, the complete answer is the criterion above rather than a numerical yes/no decision.

Solution to Exercise 2.4

Let $x^* \in \mathcal{X}$ be a local minimizer relative to $\mathcal{X}$. Thus there is an $\varepsilon > 0$ such that

$$f(x^*) \leq f(z) \quad \text{whenever } z \in \mathcal{X} \text{ and } \|z - x^*\|_2 < \varepsilon.$$

Suppose, toward a contradiction, that x^* is not globally minimizing. Then some $y \in \mathcal{X}$ satisfies $f(y) < f(x^*)$. Necessarily $y \neq x^*$. Choose

$$0 < \theta < \min \left\{ 1, \frac{\varepsilon}{\|y - x^*\|_2} \right\} \quad \text{and set} \quad z_\theta = (1 - \theta)x^* + \theta y.$$

Convexity of $\mathcal{X}$ is used precisely here: it ensures $z_\theta \in \mathcal{X}$. Moreover, $\|z_\theta - x^*\|_2 = \theta\|y - x^*\|_2 < \varepsilon$, whereas convexity of f gives

$$f(z_\theta) \leq (1 - \theta)f(x^*) + \theta f(y) < f(x^*).$$

This contradicts local minimality. Hence no such y exists, and x^* is a global minimizer.

Solution to Exercise 2.5

For $y \in \mathbb{R}^d$, the projection problem separates by coordinate:

$$\underset{x \in \mathcal{X}}{\text{minimize}} \frac{1}{2} \|x - y\|_2^2 = \underset{\substack{x \in \mathbb{R}^d \\ \ell_i \leq x_i \leq u_i, i=1, \dots, d}}{\text{minimize}} \frac{1}{2} \sum_{i=1}^d (x_i - y_i)^2.$$

Each scalar squared distance is minimized by leaving y_i unchanged when it is in the interval and otherwise moving it to the nearest endpoint. Hence, for $p = \Pi_{\mathcal{X}}(y)$,

$$p_i = \min\{u_i, \max\{\ell_i, y_i\}\} = \begin{cases} \ell_i, & y_i < \ell_i, \\ y_i, & \ell_i \leq y_i \leq u_i, \\ u_i, & y_i > u_i. \end{cases} \quad (2.18)$$

To verify the angle condition, fix any $x \in \mathcal{X}$ and inspect each coordinate. If $y_i < \ell_i$, then $y_i - p_i < 0$ and $x_i - p_i = x_i - \ell_i \geq 0$. If $y_i \in [\ell_i, u_i]$, then $y_i - p_i = 0$. If $y_i > u_i$, then $y_i - p_i > 0$ and $x_i - p_i = x_i - u_i \leq 0$. Thus $(y_i - p_i)(x_i - p_i) \leq 0$ in every case, and summing gives

$$\langle y - p, x - p \rangle = \sum_{i=1}^d (y_i - p_i)(x_i - p_i) \leq 0.$$

Equation (2.17) therefore confirms that (2.18) is the Euclidean projection.

Solution to Exercise 2.6

Let

$$B_r = \{x \in \mathbb{R}^d \mid \|x\|_2 \leq r\}, \quad r > 0.$$

We claim that

$$\Pi_{B_r}(y) = \begin{cases} y, & \|y\|_2 \leq r, \\ r \frac{y}{\|y\|_2}, & \|y\|_2 > r. \end{cases} \quad (2.19)$$

If $\|y\|_2 \leq r$, then y is feasible and achieves the smallest possible distance, namely zero. Now suppose $\|y\|_2 > r$, set $u = y/\|y\|_2$, and let $p = ru$. Then $p \in B_r$ and, for every $x \in B_r$,

$$\begin{aligned} \langle y - p, x - p \rangle &= (\|y\|_2 - r) \langle u, x - ru \rangle \\ &= (\|y\|_2 - r) (\langle u, x \rangle - r) \\ &\leq (\|y\|_2 - r)(\|x\|_2 - r) \leq 0, \end{aligned}$$

where Cauchy–Schwarz gives $\langle u, x \rangle \leq \|x\|_2$. The projection-angle characterization proves (2.19). For $r = 0$, the ball is $\{0\}$ and its projection is 0, consistent with the natural extension of the formula.

For $y = (3, 4, 5)^\top$,

$$\|y\|_2 = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50} = 5\sqrt{2} > 4.$$

Therefore

$$\begin{aligned} \Pi_{B_4}(3, 4, 5)^\top &= \frac{4}{\sqrt{50}}(3, 4, 5)^\top \\ &= \left(\frac{6\sqrt{2}}{5}, \frac{8\sqrt{2}}{5}, 2\sqrt{2} \right)^\top. \end{aligned}$$

Its Euclidean norm is 4, as required.

Solution to Exercise 2.7

Under the standard proximal-operator assumptions $\eta > 0$ and $\lambda \geq 0$,

$$\text{prox}_{\eta g}(v) = \arg \min_{x \in \mathbb{R}^d} \left\{ \frac{\lambda}{2} \|x\|_2^2 + \frac{1}{2\eta} \|x - v\|_2^2 \right\}.$$

The objective is differentiable and strongly convex. Its unique stationary point satisfies

$$\lambda x + \frac{1}{\eta}(x - v) = 0,$$

and hence

$$\boxed{\text{prox}_{\eta g}(v) = \frac{1}{1 + \eta\lambda} v.} \quad (2.20)$$

The factor $1/(1 + \eta\lambda)$ lies in $(0, 1]$: it equals 1 when $\lambda = 0$ and is strictly smaller than 1 when $\lambda > 0$. Thus every coordinate is scaled uniformly toward the origin. Unlike soft-thresholding, finite quadratic regularization does not send a nonzero coordinate exactly to zero.

Solution to Exercise 2.8

Let $x, y \in \Delta_d$ and $\theta \in [0, 1]$. Each coordinate of $z = (1 - \theta)x + \theta y$ is nonnegative, and

$$\sum_{i=1}^d z_i = (1 - \theta) \sum_{i=1}^d x_i + \theta \sum_{i=1}^d y_i = (1 - \theta) + \theta = 1.$$

Therefore $z \in \Delta_d$, proving convexity. Equivalently, Δ_d is the intersection of the halfspaces $x_i \geq 0$ with the affine hyperplane $\mathbf{1}^\top x = 1$.

For a differentiable f (defined on an open set containing Δ_d), choose $x_0 \in \Delta_d$ and stepsizes $\eta_t > 0$, and iterate

$$y_{t+1} = x_t - \eta_t \nabla f(x_t), \quad x_{t+1} = \Pi_{\Delta_d}(y_{t+1}). \quad (2.21)$$

The projection subproblem is

$$\Pi_{\Delta_d}(y) = \arg \min_{x_i \geq 0, \sum_i x_i = 1} \frac{1}{2} \|x - y\|_2^2.$$

It has a unique solution because Δ_d is nonempty, closed, and convex. In particular, every iterate in (2.21) is feasible. Suitable stepsize and smoothness assumptions are additionally needed for a convergence-rate claim, but not for the definition of the PGD update.

Solution to Exercise 2.9

The gradient at the initial point is

$$\nabla f(x_0) = x_0 - e_3 = \left(\frac{1}{3}, \frac{1}{3}, -\frac{2}{3} \right)^\top.$$

(a) The smallest gradient coordinate is the third, so the linear oracle returns $s_0 = e_3$. With the Frank–Wolfe update $x_1 = (1 - \gamma_0)x_0 + \gamma_0 s_0$,

$$x_1 = \frac{1}{2}x_0 + \frac{1}{2}e_3 = \left(\frac{1}{6}, \frac{1}{6}, \frac{2}{3} \right)^\top.$$

Furthermore, $x_0 - s_0 = x_0 - e_3 = \nabla f(x_0)$, so

$$\begin{aligned} g_{\text{FW}}(x_0) &= \|x_0 - e_3\|_2^2 \\ &= \frac{1}{9} + \frac{1}{9} + \frac{4}{9} = \frac{2}{3}. \end{aligned}$$

(b) Since $e_3 \in \Delta_3$ and $f(x) \geq 0$, the exact minimizer is $x^* = e_3$ and $f^* = 0$. Hence

$$f(x_0) - f^* = \frac{1}{2} \|x_0 - e_3\|_2^2 = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3} \leq \frac{2}{3} = g_{\text{FW}}(x_0).$$

This verifies the Frank–Wolfe upper bound, which need not be tight.

(c) Let $c \in \mathbb{R}^d$ be any gradient vector and let $j \in \arg \min_i c_i$. For every $s \in \Delta_d$,

$$c^\top s = \sum_{i=1}^d c_i s_i \geq \left(\min_i c_i \right) \sum_{i=1}^d s_i = \min_i c_i.$$

Equality is attained at the vertex $s = e_j$. Thus the oracle only needs to identify a smallest gradient coordinate and return its simplex vertex. If several coordinates tie for the minimum, any corresponding vertex—or any convex combination of those tied vertices—is optimal.

Historical note

Moreau’s work in the 1960s made proximal mappings and convex duality into systematic tools for nonsmooth analysis [Mor65]. Projection, soft-thresholding, and proximal gradient are therefore not unrelated tricks: they are modern algorithmic forms of one geometric idea—replace a difficult objective locally by a simpler problem with a quadratic movement penalty.

Chapter 3

Separation, Certificates, and Conic Duality

Why this chapter

Week 2 used convex geometry to build primal algorithms. Constraints create a second question: can feasibility or optimality be certified by a short object that another person can check? We begin with geometric separation, turn it into Farkas' linear infeasibility alternative, and then develop Lagrangian bounds, KKT certificates, and a generic conic primal–dual pair. This chapter focuses on what a valid certificate contains; Chapter 4 introduces Newton and barrier methods as one route for computing such certificates. Dual ascent moves to Week 11, where prices and constrained dynamics fit the non-Euclidean/min–max block.

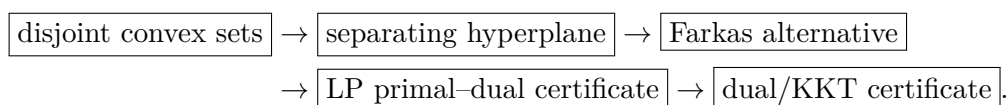
Learning objectives

By the end of this chapter, you should be able to:

1. verify linear infeasibility using a separation argument and a Farkas certificate;
2. formulate a small linear program, solve it geometrically, and verify optimality with its dual and complementary slackness;
3. derive a Lagrangian dual bound and verify a convex candidate with the KKT conditions;
4. derive the generic conic primal–dual gap and interpret primal feasibility, dual feasibility, and complementarity.

3.1 The conceptual map

For a constrained problem, the main logical route is



The generic conic language then packages the same logic for the nonnegative orthant, second-order cones, and the PSD cone used in Week 4.

Throughout this chapter, $u \preceq v$ and $u \succeq v$ for vectors mean componentwise inequalities. Chapter 4 reuses the glyphs for symmetric matrices, where they denote the PSD order instead. Before drawing any inequality, first identify whether the primal problem is a minimization or a maximization:

Primal sense	Feasible quantity	Bound on the optimum	Role
minimization	primal-feasible $f(x)$	upper bound on $p^\star$	candidate quality
minimization	dual-feasible $g(\lambda, \nu)$	lower bound on $p^\star$	checkable bound
maximization LP	primal-feasible $p^\top x$	lower bound on $v^\star$	candidate reward
maximization LP	dual-feasible $d^\top y$	upper bound on $v^\star$	resource-price bound
zero gap	matching feasible values	exact optimum	both sides agree

First identify whether the primal is a minimization or maximization; only then draw the bound arrows.

3.2 Anchor problem: projection onto a resource halfspace

Throughout the chapter, consider

$$\begin{aligned} & \underset{x \in \mathbb{R}^2}{\text{minimize}} && f_0(x) = \frac{1}{2} \|x - c\|_2^2 \\ & \text{subject to} && a^\top x \leq b, \end{aligned} \quad c = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad a = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad b = 1. \quad (3.1)$$

This model chooses the point closest to c while enforcing the total-resource limit $x_1 + x_2 \leq 1$. It is deliberately small enough to solve by geometry, KKT conditions, and dual maximization. Chapter 4 revisits the same point along a barrier path, giving a fourth view that must agree with these certificates.

The unconstrained minimizer c violates the constraint because $a^\top c = 3 > 1$. Hence the constraint should be active at the solution. The geometric projection of c onto $a^\top x = b$ is

$$x^\star = c - \frac{a^\top c - b}{\|a\|_2^2} a = \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \frac{2}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The rest of the chapter explains how a general-purpose solver discovers and certifies this result without relying on a drawing.

3.3 Canonical constrained convex programs

We use the standard form

$$\begin{aligned} & \underset{x \in \mathbb{R}^d}{\text{minimize}} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m, \\ & && Ax = b, \end{aligned} \quad (3.2)$$

where $f_0, f_1, \dots, f_m : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ are proper convex functions and the equality constraints are affine. Define their common effective domain

$$\mathcal{D} := \bigcap_{i=0}^m \text{dom } f_i.$$

Its feasible set is

$$\mathcal{X} = \{x \in \mathcal{D} \mid f_i(x) \leq 0 \text{ for all } i, Ax = b\}.$$

The optimal primal value is denoted by p^* , with $p^* = +\infty$ when the problem is infeasible and $p^* = -\infty$ when feasible objective values are unbounded below. A finite value alone does not assert that an optimizer is attained.

Common trap

An equality $h(x) = 0$ is compatible with convex optimization only when h is affine in the standard formulation. Replacing it by the two inequalities $h(x) \leq 0$ and $-h(x) \leq 0$ does not fix a nonlinear equality: both functions would need to be convex, which normally forces h to be affine.

3.3.1 Three logical questions

Keep the following questions separate.

1. **Feasibility:** does any x satisfy all constraints?
2. **Optimality:** if a candidate x is feasible, can we certify that no other feasible point has smaller objective value?
3. **Computation:** which iterative procedure finds such a candidate?

Separation, Farkas, duality, and KKT primarily answer the first two questions. Interior-point methods answer the third. A certificate is not automatically an algorithm.

3.4 Separation turns geometry into a certificate

Key result: strictly separate a point from a closed convex set

Problem solved. How can we prove that a proposed point does not belong to a convex feasible region without testing every point in the region?

Assumptions. Let $C \subseteq \mathbb{R}^d$ be nonempty, closed, and convex, and let $z \notin C$.

Statement. There exist $a \neq 0$ and $\beta \in \mathbb{R}$ such that

$$\langle a, x \rangle \leq \beta < \langle a, z \rangle \quad \text{for every } x \in C. \quad (3.3)$$

Main idea. Project z onto C . The residual from the projection to z is normal to every feasible displacement and therefore defines the separator.

Why it matters. The vector a and threshold β are a finite, checkable nonmembership certificate. Farkas' lemma and strong duality are algebraic descendants of this geometry.

Figure 3.1 shows the residual direction and the strict inequality that excludes the external point.

Proof note. Know the projection proof roadmap and use the theorem. A more general two-set separation proof is not required in the main assessment.

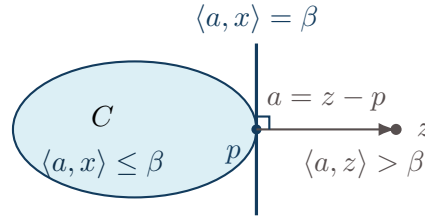


Figure 3.1: With $p = \Pi_C(z)$, the residual $a = z - p$ is normal to the supporting line and $\beta = \langle a, p \rangle$. Thus every $x \in C$ satisfies $\langle a, x \rangle \leq \beta$, while z satisfies the strict reverse inequality.

Proof. Because C is nonempty, closed, and convex, the projection $p = \Pi_C(z)$ exists and is unique. Set

$$a = z - p \neq 0, \quad \beta = \langle a, p \rangle.$$

Projection optimality from equation (2.11) gives $\langle z - p, x - p \rangle \leq 0$ for every $x \in C$, hence

$$\langle a, x \rangle \leq \langle a, p \rangle = \beta.$$

On the other hand,

$$\langle a, z \rangle = \langle a, p \rangle + \|z - p\|_2^2 > \beta.$$

This proves equation (3.3). $\square$

For two arbitrary disjoint convex sets, strict separation needs additional conditions; weak separation is the safer general statement. Closedness matters above because it guarantees a nearest point. The open-set example $C = \{x \in \mathbb{R} \mid x < 0\}$, $z = 0$ has no positive separation gap.

3.5 Farkas' lemma: either a feasible vector or a refutation

Key result: Farkas alternative

Problem solved. Can the linear system $Ax = b$, $x \succeq 0$, be feasible?

Assumptions. Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Statement. Exactly one of the following systems has a solution:

$$(P) \quad Ax = b, \quad x \succeq 0, \quad (3.4)$$

$$(F) \quad A^T y \succeq 0, \quad b^T y < 0. \quad (3.5)$$

The vector y in (F) is an infeasibility certificate for (P).

Main idea. The attainable right-hand sides form the polyhedral cone $K = \{Ax \mid x \succeq 0\}$. If $b \notin K$, a separating hyperplane produces y with the signs in (F).

Why it matters. A solver does not merely report “infeasible”: it can return a short vector whose matrix products independently verify infeasibility.

The two systems cannot both hold. If they did, then

$$0 \leq x^T A^T y = (Ax)^T y = b^T y < 0,$$

a contradiction. For the converse, the cone $K = \{Ax \mid x \succeq 0\}$ is a closed convex polyhedral cone. Separating $b \notin K$ from K gives precisely a vector satisfying (F). This is the separation-to-certificate step; a full polyhedral-closedness proof is not required here.

Worked example: a checkable infeasibility certificate

Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Solving $Ax = b$ gives $x_1 = 3/2$ and $x_2 = -1/2$, so nonnegativity fails. More importantly, the vector

$$y = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

is a certificate because

$$A^T y = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \succeq 0, \quad b^T y = 1 - 2 = -1 < 0.$$

Anyone can check these two products without resolving the primal system.

3.6 Linear programming: the first complete certificate

A *linear program* (LP) has a linear objective and affine equality or inequality constraints. The feasible set is therefore a polyhedron. LP is the simplest setting in which geometry, an algorithm, and a dual certificate can be seen together without nonlinear algebra.

For a resource-allocation interpretation, it is convenient to use the pair

$$\begin{aligned} (\text{P}_{\text{LP}}) \quad & \underset{x \in \mathbb{R}^n}{\text{maximize}} \quad p^T x \quad \text{subject to} \quad Bx \preceq d, \quad x \succeq 0, \\ (\text{D}_{\text{LP}}) \quad & \underset{y \in \mathbb{R}^m}{\text{minimize}} \quad d^T y \quad \text{subject to} \quad B^T y \succeq p, \quad y \succeq 0, \end{aligned} \tag{3.6}$$

where x contains activity levels, p contains rewards per unit, $Bx \preceq d$ encodes m resource limits, and y assigns a nonnegative price to each resource. Denote the common optimal value, when it exists, by v_{LP}^* .

3.6.1 Why the dual is a certificate

Suppose x and y are feasible for their respective problems. Because $x \succeq 0$, $B^T y \succeq p$, $y \succeq 0$, and $Bx \preceq d$,

$$p^T x \leq x^T B^T y = y^T Bx \leq y^T d. \tag{3.7}$$

Thus every dual-feasible y gives an upper bound on every primal-feasible reward. This is LP weak duality in one line. The dual constraints are exactly what is needed for a weighted sum of the resource inequalities to dominate the objective coefficient by coefficient.

The difference between the two bounds decomposes as

$$d^T y - p^T x = \underbrace{y^T (d - Bx)}_{\text{unused-resource terms}} + \underbrace{x^T (B^T y - p)}_{\text{reduced-reward terms}}. \tag{3.8}$$

Every term on the right is nonnegative. Hence a feasible pair has zero gap if and only if it satisfies the LP complementary-slackness equations

$$y_i(d_i - (Bx)_i) = 0, \quad x_j((B^T y)_j - p_j) = 0 \quad (3.9)$$

for every resource i and activity j . Economically: an unused resource has zero price, while an activity used at positive level has zero reduced reward.

Worked example: a resource LP from polygon to certificate

Two activities earn rewards 3 and 2 per unit and consume two limited resources:

$$\begin{aligned} & \underset{x_1, x_2}{\text{maximize}} && 3x_1 + 2x_2 \\ & \text{subject to} && x_1 + x_2 \leq 4, \\ & && 2x_1 + x_2 \leq 5, \\ & && x_1, x_2 \geq 0. \end{aligned} \quad (3.10)$$

The feasible polygon has vertices

$$(0, 0), \quad (0, 4), \quad (1, 3), \quad (5/2, 0).$$

Evaluating the objective gives, respectively,

$$0, \quad 8, \quad 9, \quad 15/2.$$

The best vertex is therefore $x^* = (1, 3)^T$, with value 9.

The dual problem is

$$\begin{aligned} & \underset{y_1, y_2}{\text{minimize}} && 4y_1 + 5y_2 \\ & \text{subject to} && y_1 + 2y_2 \geq 3, \\ & && y_1 + y_2 \geq 2, \\ & && y_1, y_2 \geq 0. \end{aligned} \quad (3.11)$$

The choice $y^* = (1, 1)^T$ is dual feasible and has value 9. Consequently,

$$\underbrace{9}_{\text{primal feasible}} \leq v_{\text{LP}}^* \leq \underbrace{9}_{\text{dual feasible}},$$

so both points are optimal. In fact, the certificate can be read without any duality terminology: adding the two resource inequalities gives

$$3x_1 + 2x_2 \leq 4 + 5 = 9,$$

and $(1, 3)$ attains the bound.

At this pair, both resources are fully used and both dual inequalities are tight. Therefore every product in [equation \(3.9\)](#) is zero. Geometry, dual bounds, and complementary slackness all identify the same solution.

3.6.2 Vertices, basic feasible solutions, and the simplex idea

Inequalities can be converted to equalities by adding slack variables. For [equation \(3.6\)](#),

$$Bx + s = d, \quad x \succeq 0, \quad s \succeq 0,$$

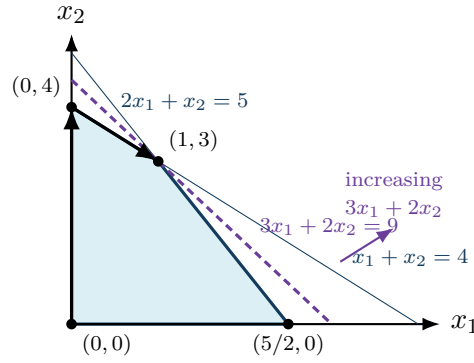


Figure 3.2: The feasible set of equation (3.10). Moving parallel objective lines in the indicated direction stops at $(1, 3)$. The exact vertex path $(0, 0) \rightarrow (0, 4) \rightarrow (1, 3)$ illustrates two improving simplex pivots.

where $s_i = d_i - (Bx)_i$ is the unused amount of resource i . This gives the standard equality form

$$\underset{z}{\text{minimize}} \quad c^T z \quad \text{subject to} \quad Az = b, \quad z \succeq 0, \quad (3.12)$$

after changing the sign of a maximization objective if necessary.

Assume $A \in \mathbb{R}^{m \times n}$ has full row rank. Choose m columns whose matrix A_B is nonsingular, set all remaining variables to zero, and solve

$$z_B = A_B^{-1}b.$$

If $z_B \succeq 0$, the resulting point is a *basic feasible solution* (BFS). Geometrically, BFSs are precisely the vertices of a standard-form feasible polyhedron, allowing for degeneracy: several bases may describe the same vertex.

Key result: the fundamental theorem of linear programming

Problem solved. Why is it enough for a vertex-based method to inspect a discrete set of candidates rather than every point in a polyhedron?

Assumptions. Consider equation (3.12) with a nonempty feasible set and finite optimal value; remove redundant equality rows so that A has full row rank.

Statement. There exists an optimal basic feasible solution, hence an optimal vertex.

Main idea. If a feasible point is not extreme, it lies on a segment between two distinct feasible points. Linearity makes its objective the same convex combination of the endpoint objectives, so moving toward a no-worse endpoint does not hurt. Repeating within a finite-dimensional face reaches an extreme point.

Why it matters. The theorem is the geometric foundation of the simplex method. It is also a special LP existence result: for a nonempty standard-form LP, a finite optimum is attained. One must still distinguish infeasible, unbounded, and finite cases before invoking it.

The *simplex idea* is now short to state: start from a BFS, pivot toward an adjacent basis selected by reduced costs, and stop when those costs certify that no improving adjacent direction remains. With degeneracy, a pivot can have zero step length, leave the geometric point and objective

unchanged, and even cycle without an anti-cycling rule. In figure 3.2, one possible path is

$$(0, 0) \longrightarrow (0, 4) \longrightarrow (1, 3).$$

This chapter does not develop tableau bookkeeping or pivot rules. The goal is to recognize what a pivot changes and why the stopping test is a dual certificate.

	Simplex viewpoint	Interior path (Chapter 4 preview)
Iterates	Vertices/basic feasible solutions	Strictly interior points
Characteristic step	Pivot to an adjacent basis	Newton step toward the central path
What certifies stopping?	Reduced-cost/dual-feasibility test	Small primal–dual residuals and gap
Why keep both?	Sparse bases, warm starts, and vertex solutions can be valuable	Smooth path-following extends naturally to large conic formulations

Connection to AI practice: why LP belongs in an AI optimization course

LP is not included only as historical background. Linear subproblems appear when choosing dataset-mixture weights, allocating compute or bandwidth, routing jobs among devices, matching agents to tasks, and constructing convex relaxations of discrete decisions. More importantly, LP is the first complete model of a reusable AI-systems pattern:

candidate decision + independent dual bound = checkable performance certificate.

Week 4 replaces nonnegative numbers by positive-semidefinite matrices, but the certificate logic remains the same.

Common trap

A problem is not an LP merely because its constraints are linear: its objective must also be linear. Also, “an optimum occurs at a vertex” means that at least one optimal vertex exists under the theorem’s assumptions. An entire edge can be optimal, and an arbitrary optimal point on that edge need not be a vertex.

3.7 Lagrangian duality and constraint qualifications

The LP dual above used nonnegative prices to combine resource inequalities into an objective bound. The Lagrangian now extends exactly that idea to nonlinear convex objectives and constraints.

Definition 3.1 (Lagrangian). For equation (3.2), introduce inequality multipliers $\lambda \in \mathbb{R}^m$ and equality multipliers $\nu \in \mathbb{R}^p$. The Lagrangian is

$$\mathcal{L}(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \nu^\top (Ax - b). \quad (3.13)$$

The inequality multipliers are restricted to $\lambda \succeq 0$; equality multipliers are unrestricted.

If x is primal feasible and $\lambda \succeq 0$, then

$$\sum_{i=1}^m \lambda_i f_i(x) \leq 0, \quad \nu^\top (Ax - b) = 0.$$

Therefore $\mathcal{L}(x, \lambda, \nu) \leq f_0(x)$. This simple sign observation is the engine behind weak duality.

Definition 3.2 (Dual function and dual problem). The dual function is the best lower bound produced by fixed multipliers:

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} \mathcal{L}(x, \lambda, \nu). \quad (3.14)$$

Restricting the infimum to the common effective domain keeps every Lagrangian term finite before the multipliers are applied and avoids undefined expressions such as $0 \cdot (+\infty)$. The Lagrange dual problem is

$$\begin{aligned} & \underset{\lambda, \nu}{\text{maximize}} && g(\lambda, \nu) \\ & \text{subject to} && \lambda \succeq 0. \end{aligned} \quad (3.15)$$

Its optimal value is denoted by d^* .

For fixed x , the Lagrangian is affine in (λ, ν) . Its pointwise infimum is therefore concave, so g is concave even when the primal problem is not convex.

Theorem 3.3 (Weak duality). *For every primal-feasible x , every $\lambda \succeq 0$, and every ν ,*

$$g(\lambda, \nu) \leq f_0(x).$$

Consequently,

$$d^* \leq p^*.$$

Proof. By the definition of an infimum,

$$g(\lambda, \nu) = \inf_z \mathcal{L}(z, \lambda, \nu) \leq \mathcal{L}(x, \lambda, \nu).$$

Primal feasibility, $\lambda \succeq 0$, and the sign argument above give $\mathcal{L}(x, \lambda, \nu) \leq f_0(x)$. Maximizing the left side over dual-feasible multipliers and minimizing the right side over primal-feasible points yields $d^* \leq p^*$. $\square$

Key result: a lower-bound certificate

Any dual-feasible pair (λ, ν) gives a certified lower bound $g(\lambda, \nu)$ on the primal optimum. Any primal-feasible x gives an upper bound $f_0(x)$. Their difference,

$$f_0(x) - g(\lambda, \nu) \geq 0,$$

is a computable primal–dual gap. A zero gap certifies that both points are optimal.

3.7.1 The anchor problem in the dual

Write the constraint in equation (3.1) as $a^\top x - b \leq 0$. With one multiplier $\lambda \geq 0$,

$$\mathcal{L}(x, \lambda) = \frac{1}{2} \|x - c\|_2^2 + \lambda(a^\top x - b).$$

Minimizing over x gives

$$\nabla_x \mathcal{L} = x - c + \lambda a = 0 \implies x(\lambda) = c - \lambda a.$$

Substitution yields

$$g(\lambda) = -\frac{1}{2} \lambda^2 \|a\|_2^2 + \lambda(a^\top c - b) = -\lambda^2 + 2\lambda, \quad \lambda \geq 0. \quad (3.16)$$

The maximum occurs at $\lambda^* = 1$, so

$$d^* = g(1) = 1, \quad x(\lambda^*) = c - a = (1, 0)^\top, \quad f_0(x^*) = 1.$$

The primal and dual values agree.

3.7.2 When the lower bound is exact

Weak duality is automatic, but equality $d^* = p^*$ needs additional conditions.

Definition 3.4 (Slater's condition: the form used here). Assume the objective and inequality functions are finite and convex on an open domain containing the feasible set. The problem in equation (3.2) satisfies Slater's condition if there exists $\tilde{x}$ in that domain such that

$$\begin{aligned} f_i(\tilde{x}) &< 0, & i = 1, \dots, m, \\ A\tilde{x} &= b. \end{aligned}$$

Such a point is strictly feasible. This easy-to-check version is slightly stronger than the most general relative-interior formulation, which is not needed in this chapter.

Theorem 3.5 (Strong duality under Slater's condition). *Suppose equation (3.2) is convex, its optimal value is finite, and Slater's condition holds. Then*

$$d^* = p^*,$$

and a dual optimum is attained.

A proof uses separation and is deferred. The reusable logical chain is

convexity + constraint qualification $\implies$ no duality gap and valid KKT multipliers.

Slater's condition is sufficient, not necessary. For the anchor problem, $\tilde{x} = (0, 0)^\top$ has $a^\top \tilde{x} - b = -1 < 0$; hence Slater explains the matching values $p^* = d^* = 1$.

3.8 KKT conditions: a local-looking global certificate

Assume for now that all f_i are differentiable. The Karush–Kuhn–Tucker (KKT) conditions for equation (3.2) are:

$$f_i(x^*) \leq 0, \quad Ax^* = b, \quad \text{primal feasibility,} \quad (3.17)$$

$$\lambda^* \succeq 0, \quad \text{dual feasibility,} \quad (3.18)$$

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m, \quad \text{complementary slackness,} \quad (3.19)$$

$$\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + A^\top \nu^* = 0, \quad \text{stationarity.} \quad (3.20)$$

Complementary slackness says that each inequality is in one of two regimes:

$$f_i(x^*) < 0 \implies \lambda_i^* = 0, \quad \lambda_i^* > 0 \implies f_i(x^*) = 0.$$

An inactive constraint has zero marginal price. A positive price can occur only when the corresponding resource is fully used.

KKT is the computable multiplier form of the normal-cone condition $0 \in \partial f_0(x^*) + N_{\mathcal{X}}(x^*)$ from Appendix B.

Key result: KKT certificate and characterization

Suppose the objective and inequality functions are convex and differentiable and the equalities are affine.

- Any triple satisfying the four KKT conditions is primal–dual optimal with zero duality gap.
- If Slater’s condition also holds, every primal optimum admits KKT multipliers. Consequently,

$$x^* \text{ is optimal} \iff \text{there exist } (\lambda^*, \nu^*) \text{ satisfying KKT.}$$

For sufficiency, stationarity minimizes the convex Lagrangian over x , while feasibility and complementary slackness reduce its value to $f_0(x^*)$. Weak duality then closes the certificate. Convexity supplies this direction; Slater supplies the reverse direction.

Common trap

Without convexity, a KKT point need not be globally optimal. Without a constraint qualification, an optimum may exist without valid multipliers. Always state which direction of the KKT claim is being used.

3.8.1 Two compact KKT checks

Anchor problem. For equation (3.1), the KKT conditions become

$$\begin{aligned} a^\top x^* - b &\leq 0, \\ \lambda^* &\geq 0, \\ \lambda^*(a^\top x^* - b) &= 0, \\ x^* - c + \lambda^* a &= 0. \end{aligned}$$

The already computed pair

$$x^* = (1, 0)^\top, \quad \lambda^* = 1$$

satisfies all four lines: the constraint is active and $x^* - c + \lambda^* a = 0$. KKT therefore certifies the same solution without repeating the geometric or dual derivation.

Hard-margin SVM. Consider labeled data

$$(x_i, y_i), \quad x_i \in \mathbb{R}^d, \quad y_i \in \{-1, +1\}.$$

Assume the data are strictly linearly separable. Rescaling a separator gives the hard-margin support-vector machine

$$\begin{aligned} &\underset{w, b}{\text{minimize}} \quad \frac{1}{2} \|w\|_2^2 \\ &\text{subject to} \quad 1 - y_i(w^\top x_i + b) \leq 0, \quad i = 1, \dots, n. \end{aligned} \tag{3.21}$$

The two margin planes are $2/\|w\|_2$ apart, so minimizing $\|w\|_2$ maximizes their separation.

Introduce one multiplier $\alpha_i \geq 0$ per training point. The Lagrangian is

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|_2^2 + \sum_{i=1}^n \alpha_i [1 - y_i(w^\top x_i + b)].$$

Stationarity with respect to w and b gives

$$w = \sum_i \alpha_i y_i x_i, \quad \sum_i \alpha_i y_i = 0.$$

Complementary slackness reads

$$\alpha_i [1 - y_i(w^\top x_i + b)] = 0.$$

Thus $\alpha_i > 0$ implies that point i lies on a margin plane, and stationarity shows that these support vectors assemble w .

Worked example: a compact SVM certificate

For $(-1, -1), (1, +1), (2, +1)$, take $w^* = 1$, $b^* = 0$, and $\alpha^* = (1/2, 1/2, 0)^\top$. The signed margins are 1, 1, 2, and

$$\sum_i \alpha_i^* y_i = 0, \quad \sum_i \alpha_i^* y_i x_i = 1 = w^*, \quad \frac{1}{2} \|w^*\|_2^2 = \frac{1}{2}.$$

Feasibility, stationarity, and complementary slackness all hold, so KKT certifies optimality. The third point is correctly classified but has zero multiplier and does not determine the boundary.

Scaling a strict separator until every margin exceeds 1 verifies Slater's condition, completing the chain

strict separation $\implies$ Slater $\implies$ KKT multipliers $\implies$ support-vector certificate.

Common trap

Hard-margin SVM asks for the maximum-margin separator, not merely any separator. A point strictly beyond the margin must have zero multiplier.

Classroom activity: KKT case analysis

Consider

$$\min_x (x - 3)^2 \quad \text{subject to} \quad x \leq u.$$

In pairs, derive x^* and λ^* for the two cases $u \geq 3$ and $u < 3$. Then explain the multiplier as the magnitude of the sensitivity of the optimal value to a small relaxation of u : wherever the value function $p^*(u)$ is differentiable, $dp^*(u)/du = -\lambda^*$.

Checkpoint.

$$x^* = \min\{3, u\}, \quad \lambda^* = \begin{cases} 0, & u \geq 3, \\ 2(3 - u), & u < 3. \end{cases}$$

3.9 The generic conic primal–dual pair

Let $K \subseteq \mathbb{R}^n$ be a nonempty closed convex cone. Its dual cone is

$$K^* := \{s \in \mathbb{R}^n \mid \langle s, x \rangle \geq 0 \text{ for every } x \in K\}. \quad (3.22)$$

The standard conic pair used in the rest of the first half is

$$\begin{aligned} \text{(P)} \quad & \underset{x}{\text{minimize}} \quad \langle c, x \rangle \quad \text{subject to} \quad Ax = b, \quad x \in K, \\ \text{(D)} \quad & \underset{y,s}{\text{maximize}} \quad \langle b, y \rangle \quad \text{subject to} \quad A^\top y + s = c, \quad s \in K^*. \end{aligned} \quad (3.23)$$

For any primal–dual feasible triple,

$$\langle c, x \rangle - \langle b, y \rangle = \langle A^\top y + s, x \rangle - \langle y, Ax \rangle = \langle s, x \rangle \geq 0. \quad (3.24)$$

This single identity gives conic weak duality. The condition

$$\langle s, x \rangle = 0 \quad (3.25)$$

is conic complementarity; when strong duality and attainment hold, it closes the certificate.

Cone K	Dual cone	Problem class
$\mathbb{R}_+^n$	$\mathbb{R}_+^n$	linear programming
second-order cone	itself	norm/conic quadratic constraints
$\mathbb{S}_+^n$	$\mathbb{S}_+^n$	semidefinite programming

Week 4 specializes [equation \(3.23\)](#) to the PSD cone. It will prove the matrix-specific facts—self-duality, trace inner products, and Gram representations—without rederiving generic weak duality or KKT.

Worked example: the nonnegative cone recovers an LP certificate

Take $K = \mathbb{R}_+^n$. Then $s \succeq 0$, $x \succeq 0$, and $\langle s, x \rangle = \sum_i s_i x_i$. If the sum is zero, every nonnegative term must be zero, so $s_i x_i = 0$ coordinatewise. Ordinary LP complementary slackness is therefore the nonnegative-cone form of [equation \(3.25\)](#).

3.10 One problem, three certificate views

View	Main computation	What it reveals
Geometry	Project c orthogonally onto the half-space.	The solution lies on the boundary and the residual is normal to it.
KKT	Solve feasibility, stationarity, and complementary slackness.	$x^* = (1, 0)^\top$ and the active constraint has price $\lambda^* = 1$.
Dual	Maximize $g(\lambda) = -\lambda^2 + 2\lambda$ for $\lambda \geq 0$.	The dual lower bound meets the primal value at 1.

Connection to AI practice: constraints as prices in AI systems

Lagrange multipliers turn requirements into adaptive prices. This viewpoint appears in constrained model training, resource-aware inference, distributed learning, and alignment objectives with explicit safety or behavior constraints. The mathematical lesson is durable: before adding a penalty by hand, ask whether the requirement should be a hard constraint, what its multiplier means, and whether a primal–dual gap can certify the result.

Geometry explains the candidate, KKT checks local-looking equations, and the dual supplies an independent bound. Chapter 4 adds the computational bridge: Newton steps follow local curvature, while a barrier keeps those steps inside the feasible region.

Summary and connections

- Projection gives a separating certificate for a point outside a nonempty closed convex set; Farkas' alternative certifies linear infeasibility by matrix and sign checks.
- LP geometry connects feasible vertices and simplex pivots; a zero primal–dual gap certifies optimality.
- The Lagrangian produces dual lower bounds. Weak duality gives $d^* \leq p^*$; strong duality needs additional conditions.
- For differentiable convex problems under Slater's condition, KKT conditions characterize optimality.
- In hard-margin SVM, positive multipliers identify margin points contributing to the maximum-margin separator.

3.11 Chapter practice set

Exercise 3.1 (Farkas certificate). For

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

determine whether there exists $x \succeq 0$ with $Ax = b$. If not, find y satisfying $A^\top y \succeq 0$ and $b^\top y < 0$, then verify the certificate by direct multiplication.

Exercise 3.2 (LP geometry, duality, and a basis). Consider

$$\begin{aligned} & \underset{u,v}{\text{maximize}} && 3u + 3v \\ & \text{subject to} && 2u + v \leq 8, \\ & && u + 2v \leq 8, \\ & && u, v \geq 0. \end{aligned}$$

- List the vertices of the feasible polygon and find a primal optimum.
- Derive the dual using [equation \(3.6\)](#) and find a dual-feasible point with the same value.
- Verify every equation in [equation \(3.9\)](#).
- Introduce slacks s_1, s_2 , write the equality-form matrix, and show that choosing u, v as the basic variables recovers the optimal vertex.

Exercise 3.3 (Conic weak duality). Starting from [equation \(3.23\)](#), derive [equation \(3.24\)](#) without invoking a memorized duality theorem. Name the assumption that makes the final inner product nonnegative, and state what a zero gap certifies when both optima are attained.

Exercise 3.4 (KKT certificate for a small SVM). Consider the one-dimensional labeled data

$$(-2, -1), \quad (1, +1), \quad (3, +1).$$

Use the hard-margin SVM formulation [equation \(3.21\)](#).

- (a) Verify primal feasibility for $w^* = 2/3$, $b^* = 1/3$, and identify the margin points.
- (b) Verify every KKT condition using $\alpha^* = (2/9, 2/9, 0)^\top$.
- (c) Compute the primal and dual values and use their equality to certify optimality.

Exercise 3.5 (KKT and water-filling). Allocate one unit of a resource between two channels:

$$\begin{aligned} & \underset{p_1, p_2}{\text{maximize}} && \log(1 + p_1) + \log(1 + 4p_2) \\ & \text{subject to} && p_1 \geq 0, \quad p_2 \geq 0, \quad p_1 + p_2 \leq 1. \end{aligned}$$

Rewrite the problem as convex minimization, introduce multipliers $\mu_1, \mu_2 \geq 0$ for the lower bounds and $\nu \geq 0$ for the budget, and derive the KKT conditions. Verify

$$p_1^* = \frac{1}{8}, \quad p_2^* = \frac{7}{8}, \quad \mu_1^* = \mu_2^* = 0, \quad \nu^* = \frac{8}{9}.$$

Explain why the common derivative ν^* is the marginal value of one more unit of budget.

Exercise 3.6 (A duality-gap certificate). Suppose a minimization algorithm returns a primal-feasible point with objective value 12.4, while a dual-feasible point has dual value 12.1.

- (a) Give an interval containing p^* .
- (b) Give a certified bound on the suboptimality of the primal point.
- (c) Can you conclude that either point is optimal? Explain.

3.12 Complete solutions

Solution conventions. Throughout this chapter, vector inequalities are componentwise. We use the course convention that a convex minimization constraint is written as $f_i(x) \leq 0$ and receives a multiplier $\lambda_i \geq 0$. Thus its Lagrangian contribution is $+\lambda_i f_i(x)$. For the maximization-form linear program, we use the primal–dual pair

$$\max\{p^\top x : Bx \leq d, x \geq 0\}, \quad \min\{d^\top y : B^\top y \geq p, y \geq 0\}.$$

Solution to Exercise 3.1

The matrix A is nonsingular because $\det A = 1 - 4 = -3$. Consequently, $Ax = b$ has the unique solution

$$x = A^{-1}b = -\frac{1}{3} \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

Its first component is negative, so no $x \geq 0$ can satisfy $Ax = b$.

A directly checkable Farkas witness is

$$y = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Indeed, since A is symmetric,

$$A^\top y = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \geq 0, \quad b^\top y = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = -1 < 0.$$

To see the contradiction without solving the linear system, suppose that $Ax = b$ and $x \geq 0$. Then

$$0 \leq x^\top A^\top y = (Ax)^\top y = b^\top y = -1,$$

which is impossible. This proves infeasibility.

Solution to Exercise 3.2

Write

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad d = \begin{bmatrix} 8 \\ 8 \end{bmatrix}, \quad p = \begin{bmatrix} 3 \\ 3 \end{bmatrix}.$$

(a) Geometry and the primal optimum. On the u -axis, feasibility gives $0 \leq u \leq 4$, and on the v -axis it gives $0 \leq v \leq 4$. The two non-axis boundary lines meet at

$$2u + v = 8, \quad u + 2v = 8,$$

whose solution is $u = v = 8/3$. The vertices and their objective values are

(u, v)	$(0, 0)$	$(4, 0)$	$(0, 4)$	$(8/3, 8/3)$
$3u + 3v$	0	12	12	16.

Therefore

$$x^* = \begin{bmatrix} u^* \\ v^* \end{bmatrix} = \begin{bmatrix} 8/3 \\ 8/3 \end{bmatrix}, \quad p^\top x^* = 16.$$

(b) Dual certificate. The dual associated with $Bx \leq d$, $x \geq 0$ is

$$\begin{aligned} & \underset{y_1, y_2}{\text{minimize}} && 8y_1 + 8y_2 \\ & \text{subject to} && 2y_1 + y_2 \geq 3, \\ & && y_1 + 2y_2 \geq 3, \\ & && y_1, y_2 \geq 0. \end{aligned}$$

The point $y^* = (1, 1)^\top$ is dual feasible and has value $d^\top y^* = 16$. Weak duality now yields

$$16 = p^\top x^* \leq v_{\text{LP}}^* \leq d^\top y^* = 16.$$

Thus both feasible points are optimal and the common optimal value is 16.

(c) Complementary slackness. At the proposed pair,

$$d - Bx^* = \begin{bmatrix} 8 \\ 8 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 8/3 \\ 8/3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

and

$$B^\top y^* - p = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Consequently, every scalar equation is verified explicitly:

$$\begin{aligned} y_1^*(8 - (2u^* + v^*)) &= 1 \cdot 0 = 0, \\ y_2^*(8 - (u^* + 2v^*)) &= 1 \cdot 0 = 0, \\ u^*(2y_1^* + y_2^* - 3) &= \frac{8}{3} \cdot 0 = 0, \\ v^*(y_1^* + 2y_2^* - 3) &= \frac{8}{3} \cdot 0 = 0. \end{aligned}$$

(d) Equality form and basis. After introducing nonnegative slacks,

$$2u + v + s_1 = 8, \quad u + 2v + s_2 = 8, \quad u, v, s_1, s_2 \geq 0.$$

Equivalently,

$$\underbrace{\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{bmatrix}}_{\tilde{B}} \begin{bmatrix} u \\ v \\ s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 8 \end{bmatrix}.$$

Choose the first two columns as the basis and set the nonbasic variables $s_1 = s_2 = 0$. The basic matrix is

$$B_{\mathcal{B}} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad B_{\mathcal{B}}^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Hence

$$\begin{bmatrix} u \\ v \end{bmatrix} = B_{\mathcal{B}}^{-1} d = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ 8 \end{bmatrix} = \begin{bmatrix} 8/3 \\ 8/3 \end{bmatrix} \geq 0.$$

Thus this basis gives a basic feasible solution, and it is precisely the optimal vertex found above.

Solution to Exercise 3.3

Let $A \in \mathbb{R}^{m \times n}$, $x, c, s \in \mathbb{R}^n$, and $b, y \in \mathbb{R}^m$. Take any primal–dual feasible triple (x, y, s) . Dual feasibility gives $c = A^\top y + s$, while primal feasibility gives $Ax = b$. Therefore, using only the defining adjoint identity,

$$\begin{aligned}\langle c, x \rangle - \langle b, y \rangle &= \langle A^\top y + s, x \rangle - \langle b, y \rangle \\ &= \langle y, Ax \rangle + \langle s, x \rangle - \langle y, b \rangle \\ &= \langle s, x \rangle.\end{aligned}$$

The last quantity is nonnegative because the feasibility assumptions include $x \in K$ and $s \in K^*$, where

$$K^* = \{s : \langle s, z \rangle \geq 0 \text{ for every } z \in K\}.$$

Hence

$$\langle c, x \rangle - \langle b, y \rangle = \langle s, x \rangle \geq 0,$$

which is weak duality. If a feasible triple has zero gap, then its primal and dual objective values coincide. Weak duality forces x to be primal optimal and (y, s) to be dual optimal. When both optima are attained, zero optimal gap is equivalently the conic complementarity condition $\langle s^*, x^* \rangle = 0$ and certifies equality of the two attained optimal values.

Solution to Exercise 3.4

For clarity, define

$$f_i(w, b) = 1 - y_i(wx_i + b).$$

The Lagrangian and dual objective in the course convention are

$$L(w, b, \alpha) = \frac{1}{2}w^2 + \sum_{i=1}^3 \alpha_i [1 - y_i(wx_i + b)],$$

and, whenever $\alpha \geq 0$ and $\sum_i \alpha_i y_i = 0$,

$$q(\alpha) = \sum_{i=1}^3 \alpha_i - \frac{1}{2} \left(\sum_{i=1}^3 \alpha_i y_i x_i \right)^2.$$

(a) Primal feasibility and margin points. At $(w^*, b^*) = (2/3, 1/3)$, the three signed margins are

$$\begin{aligned}y_1(w^*x_1 + b^*) &= (-1) \left(\frac{2}{3}(-2) + \frac{1}{3} \right) = 1, \\ y_2(w^*x_2 + b^*) &= (+1) \left(\frac{2}{3}(1) + \frac{1}{3} \right) = 1, \\ y_3(w^*x_3 + b^*) &= (+1) \left(\frac{2}{3}(3) + \frac{1}{3} \right) = \frac{7}{3}.\end{aligned}$$

All are at least 1, so the point is primal feasible. The first two data points have signed margin exactly 1 and lie on the margin planes; the third lies strictly beyond the positive margin plane.

(b) All KKT conditions. Primal feasibility was just checked. Dual feasibility holds because $\alpha_i^* \geq 0$ for every i . The two stationarity equations are

$$\frac{\partial L}{\partial w} = w - \sum_i \alpha_i y_i x_i = 0, \quad \frac{\partial L}{\partial b} = - \sum_i \alpha_i y_i = 0.$$

For the proposed multipliers,

$$\sum_{i=1}^3 \alpha_i^* y_i = \frac{2}{9}(-1) + \frac{2}{9}(+1) + 0 = 0,$$

and

$$\sum_{i=1}^3 \alpha_i^* y_i x_i = \frac{2}{9}(-1)(-2) + \frac{2}{9}(+1)(1) + 0 = \frac{4}{9} + \frac{2}{9} = \frac{2}{3} = w^*.$$

Thus stationarity holds in both w and b . Finally,

$$(f_1(w^*, b^*), f_2(w^*, b^*), f_3(w^*, b^*)) = \left(0, 0, -\frac{4}{3}\right),$$

so complementary slackness holds componentwise:

$$\alpha_i^* f_i(w^*, b^*) = \left(\frac{2}{9} \cdot 0, \frac{2}{9} \cdot 0, 0 \cdot \left(-\frac{4}{3}\right)\right) = (0, 0, 0).$$

(c) Matching objective values. The primal value is

$$\frac{1}{2}(w^*)^2 = \frac{1}{2}\left(\frac{2}{3}\right)^2 = \frac{2}{9}.$$

The dual value is

$$\begin{aligned} q(\alpha^*) &= \left(\frac{2}{9} + \frac{2}{9} + 0\right) - \frac{1}{2}\left(\frac{2}{3}\right)^2 \\ &= \frac{4}{9} - \frac{2}{9} = \frac{2}{9}. \end{aligned}$$

The primal and dual points are feasible and have equal values. Weak duality therefore certifies that both are optimal. In particular, the first two points are support vectors; the third has zero multiplier and does not determine the maximum-margin boundary.

Solution to Exercise 3.5

Negating the concave utility gives the equivalent convex minimization problem

$$\begin{aligned} &\underset{p_1, p_2}{\text{minimize}} && f_0(p) = -\log(1 + p_1) - \log(1 + 4p_2) \\ &\text{subject to} && -p_1 \leq 0, \quad -p_2 \leq 0, \quad p_1 + p_2 - 1 \leq 0. \end{aligned}$$

Its Lagrangian is

$$L(p, \mu, \nu) = -\log(1 + p_1) - \log(1 + 4p_2) - \mu_1 p_1 - \mu_2 p_2 + \nu(p_1 + p_2 - 1).$$

The KKT conditions are as follows:

$$\begin{aligned} \text{primal feasibility:} & && p_1 \geq 0, \quad p_2 \geq 0, \quad p_1 + p_2 \leq 1, \\ \text{dual feasibility:} & && \mu_1 \geq 0, \quad \mu_2 \geq 0, \quad \nu \geq 0, \\ \text{complementary slackness:} & && \mu_1 p_1 = 0, \quad \mu_2 p_2 = 0, \quad \nu(p_1 + p_2 - 1) = 0, \\ \text{stationarity:} & && -\frac{1}{1 + p_1} - \mu_1 + \nu = 0, \\ & && -\frac{4}{1 + 4p_2} - \mu_2 + \nu = 0. \end{aligned}$$

At the proposed point, $p_1^*, p_2^* > 0$ and $p_1^* + p_2^* = 1$. Hence primal feasibility and complementary slackness hold with $\mu_1^* = \mu_2^* = 0$. Moreover,

$$\frac{1}{1 + p_1^*} = \frac{1}{1 + 1/8} = \frac{8}{9}, \quad \frac{4}{1 + 4p_2^*} = \frac{4}{1 + 4(7/8)} = \frac{4}{9/2} = \frac{8}{9}.$$

Thus both stationarity equations hold with $\nu^* = 8/9$, which is also dual feasible. Since f_0 is convex (indeed, strictly convex on its domain) and a Slater point exists, for example $(p_1, p_2) = (1/4, 1/4)$, these KKT conditions certify the unique optimum.

For the shadow-price interpretation, replace the budget 1 by P and let

$$V(P) = \max_{p_1, p_2 \geq 0, p_1 + p_2 \leq P} [\log(1 + p_1) + \log(1 + 4p_2)].$$

The minimization constraint is $p_1 + p_2 - P \leq 0$. The envelope theorem gives

$$\frac{d[-V(P)]}{dP} = -\nu^*,$$

or equivalently

$$V'(1) = \nu^* = \frac{8}{9},$$

at this regular point. Therefore an additional small budget ΔP increases optimal utility by $(8/9)\Delta P + o(\Delta P)$. This is precisely the marginal value of budget. The same number appears as the common marginal utility of both positively allocated channels.

Solution to Exercise 3.6

For a minimization problem, primal feasibility gives an upper bound and dual feasibility gives a lower bound. Hence

$$12.1 \leq p^* \leq 12.4.$$

If x_p denotes the returned primal point, then

$$0 \leq f_0(x_p) - p^* \leq 12.4 - 12.1 = 0.3.$$

Thus its absolute suboptimality is certified to be at most 0.3. Neither point is certified optimal: the nonzero gap leaves open the possibility that p^* lies strictly between the two bounds (or equals just one bound). Equality of feasible primal and dual values would certify both points, but a gap of 0.3 does not.

Suggested reading. The separation, duality, and KKT developments follow [BV04, Chapters 2 and 5].

Historical note

Farkas' 1902 work gave a foundational alternative for systems of linear inequalities [Far02]. The constrained stationarity conditions now called KKT appeared in Karush's 1939 thesis and in the 1951 paper of Kuhn and Tucker [Kar39, KT51]. Together these developments shifted optimization from merely finding a candidate to also producing a short, independently checkable certificate.

Chapter 4

From Newton and Barriers to Semidefinite Optimization

Why this chapter

Week 3 established what a constrained-optimization certificate must contain. This chapter first asks how a smooth solver can approach that certificate: Newton's method follows a local quadratic model, while a log barrier keeps the iterates in the strict interior. It then makes the representation switch from vectors to positive-semidefinite matrices.

The generic conic pair is specialized to $K = \mathbb{S}_+^n$. The new matrix-specific work is PSD geometry, trace inner products, self-duality, Gram representations, and matrix certificates. This creates the bridge from convex optimization to the SDP language used for Max-Cut in Week 5 and Lovász theta in Week 6; quantum-state discrimination remains optional.

Learning objectives

By the end of this chapter, you should be able to:

1. derive a Newton direction from a local quadratic model and explain why damping may be needed, then explain how a log barrier produces strictly feasible central-path points and an approximate KKT gap;
2. verify the equivalence of eigenvalue, quadratic-form, and Gram-factor views of a positive-semidefinite matrix;
3. formulate an SDP primal–dual pair and derive its trace-form duality gap with the correct bound direction;
4. construct and numerically verify a 2×2 primal–dual trace-gap certificate.

4.1 The conceptual map

smooth objective $\longrightarrow$ local quadratic model $\longrightarrow$ Newton step,
inequality constraints $\longrightarrow$ log barrier $\longrightarrow$ central path,
vectors $\longrightarrow$ Gram matrix $\longrightarrow$ PSD cone $\longrightarrow$ SDP certificate.

The first two rows connect algorithms to the certificate language of Week 3. The last row makes

the central representation switch

many vectors and their pairwise angles $\longleftrightarrow$ one positive-semidefinite matrix.

Week 5 will exploit this switch to replace binary signs by unit vectors.

4.2 Newton's method: follow a local quadratic model

Chapter 2 obtained first-order methods by minimizing regularized linear models. For a twice-differentiable objective $F : \mathbb{R}^d \rightarrow \mathbb{R}$, Newton's method instead uses the second-order model

$$Q_x(p) = F(x) + \nabla F(x)^\top p + \frac{1}{2} p^\top \nabla^2 F(x) p.$$

If $H(x) := \nabla^2 F(x) \succ 0$, its unique minimizing direction satisfies

$$H(x)p_N = -\nabla F(x). \quad (4.1)$$

Moreover,

$$\nabla F(x)^\top p_N = -p_N^\top H(x)p_N < 0$$

whenever $p_N \neq 0$, so the direction is locally descending.

Intuition: predict where the gradient becomes zero. A first-order expansion of the gradient gives

$$\nabla F(x+p) \approx \nabla F(x) + H(x)p.$$

Newton's step chooses $p = p_N$ so that this local prediction is zero. Thus the method can be read either as minimizing the quadratic model Q_x , or as solving the linearized stationarity equation $\nabla F(x+p) = 0$.

The Hessian also changes the units of motion. Along an eigenvector of $H(x)$ with eigenvalue $\lambda_i > 0$, the step component is $p_i = -g_i/\lambda_i$: high-curvature directions receive shorter moves and low-curvature directions receive longer ones. Near a minimizer where the Hessian is nonsingular and varies smoothly, the quadratic model becomes very accurate; this is the source of Newton's local quadratic convergence under the standard regularity assumptions. Farther away, the model may be unreliable, which is why the next algorithm adds damping rather than automatically taking a full step.

Algorithm: damped Newton step

Fix a tolerance $\varepsilon \geq 0$, a backtracking factor $\beta \in (0, 1)$, and $c \in (0, 1/2)$. At an iterate x_k where $\nabla^2 F(x_k) \succ 0$:

1. stop if $\|\nabla F(x_k)\|_2 \leq \varepsilon$;
2. solve $\nabla^2 F(x_k)p_k = -\nabla F(x_k)$;
3. start at $\alpha_k = 1$ and repeatedly multiply by β until the trial point is in the domain and satisfies the Armijo condition

$$F(x_k + \alpha_k p_k) \leq F(x_k) + c\alpha_k \langle \nabla F(x_k), p_k \rangle;$$

4. set $x_{k+1} = x_k + \alpha_k p_k$.

A full step is appropriate near a well-behaved minimizer; damping protects the method when the quadratic model is reliable only locally.

Worked example: a quadratic is solved in one Newton step

For

$$F(x) = \frac{1}{2}x^\top Hx - q^\top x, \quad H \succ 0,$$

we have $\nabla F(x) = Hx - q$ and $\nabla^2 F(x) = H$. Hence

$$Hp = -(Hx - q), \quad x + p = H^{-1}q.$$

The full step reaches the exact minimizer because the local quadratic model is the function itself.

Common trap

Do not form H^{-1} in numerical code; solve the linear system in [equation \(4.1\)](#). If the Hessian is singular or indefinite, the standard Newton direction may be undefined or fail to descend, so a modified method is required.

4.3 From a log barrier to the central path

Consider the convex problem

$$\min_x f_0(x) \quad \text{subject to} \quad f_i(x) \leq 0, \quad i = 1, \dots, m.$$

Assume the functions are differentiable on an open set containing a nonempty strictly feasible region and that the optimum p^* is finite. Assume also that, for every value of t considered, the barrier objective below attains a minimizer in that region; compact relevant sublevel sets are one sufficient route. The log barrier is

$$\phi(x) = -\sum_{i=1}^m \log(-f_i(x)), \quad f_i(x) < 0. \quad (4.2)$$

For $t > 0$, choose an exact minimizer

$$x^*(t) \in \arg \min_x [tf_0(x) + \phi(x)] \quad (4.3)$$

as a point on the central path. Increasing t gives the original objective more weight. The barrier objective is finite only on the strict interior; a line search must keep every *accepted* Newton iterate in that domain.

At an exact central-path point, define

$$\lambda_i(t) = \frac{1}{-tf_i(x^*(t))} > 0. \quad (4.4)$$

Dividing the stationarity condition for [equation \(4.3\)](#) by t gives KKT stationarity for these multipliers, while

$$-\lambda_i(t)f_i(x^*(t)) = \frac{1}{t}.$$

Thus the barrier replaces exact complementary slackness by a controlled error.

Key result: the exact central-path gap

Under these assumptions, convexity makes $x^*(t)$ a minimizer of the Lagrangian for $\lambda(t)$. Therefore

$$0 \leq f_0(x^*(t)) - p^* \leq \frac{m}{t}. \quad (4.5)$$

The quantity m/t controls the barrier contribution to the error. A practical inexact solve must additionally check feasibility and stationarity residuals.

Algorithm: barrier method with Newton inner steps

Start from a strictly feasible point. Approximately minimize the barrier objective in equation (4.3) with damped Newton steps, increase t , and warm-start the next subproblem from the current point. Stop only after the central-path gap and the numerical residuals meet the requested tolerance.

Worked example: the Week 3 halfspace anchor on the central path

For equation (3.1), where $a = (1, 1)^\top$, $b = 1$, and $c = (2, 1)^\top$,

$$F_t(x) = \frac{t}{2} \|x - c\|_2^2 - \log(b - a^\top x), \quad (4.6)$$

$$\nabla F_t(x) = t(x - c) + \frac{a}{b - a^\top x}, \quad (4.7)$$

$$\nabla^2 F_t(x) = tI + \frac{aa^\top}{(b - a^\top x)^2} \succ 0. \quad (4.8)$$

Writing $x^*(t) = c - \alpha(t)a$ and imposing stationarity gives

$$\alpha(t) = \frac{1 + \sqrt{1 + 2/t}}{2} > 1. \quad (4.9)$$

Hence every finite central-path point satisfies $a^\top x^*(t) < b$, and $x^*(t) \rightarrow c - a = (1, 0)^\top$ as $t \rightarrow \infty$.

The scalar barrier is the bridge to SDP. After the PSD cone is introduced, its strict interior will be $X \succ 0$ and the analogous barrier will be $-\log \det X$.

4.4 Symmetric matrices and the PSD order

Definition 4.1 (Symmetric matrix space).

$$\mathbb{S}^n := \left\{ X \in \mathbb{R}^{n \times n} \mid X = X^\top \right\}.$$

It is a vector space of dimension $n(n+1)/2$. Its standard inner product is the trace inner product

$$\langle A, B \rangle := \text{Tr}(A^\top B) = \text{Tr}(AB) = \sum_{i,j=1}^n A_{ij} B_{ij}, \quad A, B \in \mathbb{S}^n. \quad (4.10)$$

The equality $\text{Tr}(A^\top B) = \sum_{i,j} A_{ij} B_{ij}$ shows that $\mathbb{S}^n$ is an ordinary Euclidean space whose coordinates happen to be arranged as a matrix.

Definition 4.2 (Positive semidefinite and positive definite). For $X \in \mathbb{S}^n$,

$$X \succeq 0 \iff z^\top X z \geq 0 \text{ for every } z \in \mathbb{R}^n.$$

If $z^\top X z > 0$ for every nonzero z , then $X \succ 0$. We write

$$\mathbb{S}_+^n = \{X \in \mathbb{S}^n \mid X \succeq 0\}, \quad \mathbb{S}_{++}^n = \{X \in \mathbb{S}^n \mid X \succ 0\}.$$

The relation $A \succeq B$ means $A - B \succeq 0$.

This is not the componentwise vector order used in Chapter 3. For symmetric matrices, $A \succeq B$ is the PSD (Loewner) order and compares quadratic forms in every direction; it is not an entrywise comparison.

Matrix order is a *partial* order. Two matrices need not be comparable. For example,

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

is neither PSD nor negative semidefinite.

Theorem 4.3 (Equivalent views of positive semidefiniteness). *For $X \in \mathbb{S}^n$, the following are equivalent.*

- (i) $X \succeq 0$.
- (ii) Every eigenvalue of X is nonnegative.
- (iii) There is a real matrix V such that $X = VV^\top$.
- (iv) X is a Gram matrix: there are vectors $v_1, \dots, v_n$ with $X_{ij} = \langle v_i, v_j \rangle$.

Proof. By the spectral theorem,

$$X = Q\Lambda Q^\top, \quad \Lambda = \text{Diag}(\lambda_1, \dots, \lambda_n),$$

with Q orthogonal. For $z = Qy$,

$$z^\top X z = y^\top \Lambda y = \sum_i \lambda_i y_i^2.$$

This is nonnegative for every y exactly when every $\lambda_i \geq 0$, proving (i) $\Leftrightarrow$ (ii).

If the eigenvalues are nonnegative, define

$$V = Q\Lambda^{1/2}.$$

Then $VV^\top = Q\Lambda Q^\top = X$, proving (ii) $\Rightarrow$ (iii). Conversely, $X = VV^\top$ implies

$$z^\top X z = \|V^\top z\|_2^2 \geq 0,$$

so (iii) $\Rightarrow$ (i).

Finally, if the i -th row of V is $v_i^\top$, then $(VV^\top)_{ij} = v_i^\top v_j$. Thus factor and Gram views are the same statement. $\square$

Worked example: the complete 2×2 PSD test

For

$$X = \begin{bmatrix} a & b \\ b & c \end{bmatrix},$$

the matrix is PSD if and only if

$$a \geq 0, \quad c \geq 0, \quad ac - b^2 \geq 0. \quad (4.11)$$

Necessity follows by testing the coordinate vectors and requiring the product of the two eigenvalues, $\det X = ac - b^2$, to be nonnegative. Conversely, these conditions make both eigenvalues nonnegative because their sum $a + c$ and product $ac - b^2$ are nonnegative. For a correlation-shaped matrix,

$$X(\rho) = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix},$$

the condition reduces to $|\rho| \leq 1$. Its eigenvalues are $1 + \rho$ and $1 - \rho$.

Common trap

Checking only the diagonal entries is not enough. The matrix

$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

has positive diagonal entries but eigenvalues 3 and -1 . A PSD test must control all directions z , not only coordinate directions.

4.5 PSD-cone geometry and linear matrix inequalities

The PSD cone supplies both the order used by an SDP and the geometry of its feasible regions. We first record the cone and dual-cone facts needed for certificates, then use affine slices of the cone to express linear matrix inequalities.

Proposition 4.4 (Cone geometry). $\mathbb{S}_+^n$ is a closed convex cone, and

$$\text{int}(\mathbb{S}_+^n) = \mathbb{S}_{++}^n.$$

Proof. If $X, Y \succeq 0$ and $\alpha, \beta \geq 0$, then for every z ,

$$z^\top (\alpha X + \beta Y) z = \alpha z^\top X z + \beta z^\top Y z \geq 0,$$

so the set is a convex cone. If $X_k \succeq 0$ and $X_k \rightarrow X$, taking limits in $z^\top X_k z \geq 0$ gives $X \succeq 0$, so the cone is closed. A symmetric matrix lies in the interior exactly when its smallest eigenvalue is strictly positive. $\square$

4.5.1 PSD matrices have nonnegative trace products

Lemma 4.5 (Nonnegative trace product). If $A, B \succeq 0$, then

$$\langle A, B \rangle = \text{Tr}(AB) \geq 0.$$

Proof. Let $A^{1/2}$ be the PSD square root of A . Then

$$A^{1/2}BA^{1/2} \succeq 0$$

because

$$z^\top A^{1/2}BA^{1/2}z = (A^{1/2}z)^\top B(A^{1/2}z) \geq 0.$$

Its eigenvalues, and hence its trace, are nonnegative. Cyclicity of trace gives

$$\text{Tr}(A^{1/2}BA^{1/2}) = \text{Tr}(AB).$$

□

This lemma is the matrix analogue of the scalar fact $ab \geq 0$ for $a, b \geq 0$. It powers SDP weak duality.

Definition 4.6 (Dual cone). For a cone K in an inner-product space,

$$K^* = \{Y \mid \langle Y, X \rangle \geq 0 \text{ for every } X \in K\}.$$

Theorem 4.7 (Self-duality of the PSD cone). *With the trace inner product,*

$$(\mathbb{S}_+^n)^* = \mathbb{S}_+^n.$$

Proof. If $Y \succeq 0$, then [lemma 4.5](#) gives $\langle Y, X \rangle \geq 0$ for every $X \succeq 0$. Hence $\mathbb{S}_+^n \subseteq (\mathbb{S}_+^n)^*$.

Conversely, suppose $Y \in (\mathbb{S}_+^n)^*$. For every z , the rank-one matrix $zz^\top \succeq 0$, so

$$z^\top Yz = \text{Tr}(Yzz^\top) = \langle Y, zz^\top \rangle \geq 0.$$

The quadratic-form definition therefore gives $Y \succeq 0$. □

Key result: why self-duality matters

In scalar optimization, a nonnegative multiplier certifies a scalar inequality. In SDP, a PSD matrix certifies a PSD inequality. Because the PSD cone is self-dual, the primal variable, dual slack, and complementarity condition all live in the same matrix geometry.

4.5.2 Linear matrix inequalities and spectrahedra

Definition 4.8 (Linear matrix inequality). Given $F_0, F_1, \dots, F_d \in \mathbb{S}^n$, a linear matrix inequality (LMI) is

$$F(x) := F_0 + \sum_{j=1}^d x_j F_j \succeq 0, \quad x \in \mathbb{R}^d. \quad (4.12)$$

The set of x satisfying an LMI is called a spectrahedron.

An LMI represents infinitely many scalar inequalities:

$$z^\top F(x)z \geq 0 \quad \text{for every } z \in \mathbb{R}^n.$$

Nevertheless, the feasible set is convex because $F(x)$ is affine and the PSD cone is convex.

Worked example: a Euclidean ball as an LMI

For $x = (x_1, x_2)^\top$,

$$\begin{bmatrix} 1 - x_2 & x_1 \\ x_1 & 1 + x_2 \end{bmatrix} \succeq 0$$

is equivalent to

$$x_1^2 + x_2^2 \leq 1.$$

Indeed, the determinant condition is

$$(1 - x_2)(1 + x_2) - x_1^2 = 1 - x_1^2 - x_2^2 \geq 0,$$

while points in the unit disk automatically satisfy $1 \pm x_2 \geq 0$. An apparently curved feasible region is therefore an affine slice of the PSD cone.

Figure 4.1 uses the Schur-complement criterion: for symmetric $Q \succ 0$,

$$\begin{bmatrix} Q & a \\ a^\top & t \end{bmatrix} \succeq 0 \iff t - a^\top Q^{-1}a \geq 0.$$

The strict condition on Q makes its inverse legitimate.

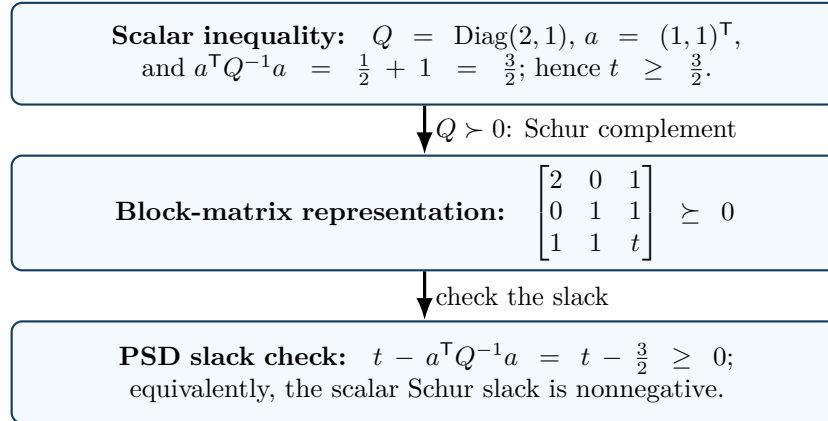


Figure 4.1: A scalar quadratic threshold can be encoded as an LMI. The equivalence $t \geq a^\top Q^{-1}a \iff \begin{bmatrix} Q & a \\ a^\top & t \end{bmatrix} \succeq 0$ requires $Q \succ 0$; without that assumption, this Schur-complement step is not valid as written.

4.6 The SDP primal–dual pair and a complete certificate

We adopt one convention throughout Weeks 4–6.

Definition 4.9 (Standard primal SDP). Given $C, A_1, \dots, A_m \in \mathbb{S}^n$ and $b \in \mathbb{R}^m$, define

$$\mathcal{A}(X) = \begin{bmatrix} \langle A_1, X \rangle \\ \vdots \\ \langle A_m, X \rangle \end{bmatrix}, \quad \mathcal{A}^*(y) = \sum_{i=1}^m y_i A_i.$$

The standard primal SDP is

$$\begin{aligned} \text{(P)} \quad & \underset{X \in \mathbb{S}^n}{\text{maximize}} && \langle C, X \rangle \\ & \text{subject to} && \mathcal{A}(X) = b, \\ & && X \succeq 0. \end{aligned} \tag{4.13}$$

The adjoint identity

$$y^\top \mathcal{A}(X) = \langle \mathcal{A}^*(y), X \rangle$$

is the matrix version of moving a linear map from one side of an inner product to the other.

Definition 4.10 (Standard dual SDP). The dual of equation (4.13) is

$$\begin{aligned} \text{(D)} \quad & \underset{y \in \mathbb{R}^m}{\text{minimize}} && b^\top y \\ & \text{subject to} && S := \mathcal{A}^*(y) - C \succeq 0. \end{aligned} \tag{4.14}$$

The matrix S is the dual slack.

4.6.1 Checking the SDP signs from the conic pair

There is no new duality theorem here. Regard $\mathbb{S}^n$ as the finite-dimensional inner-product space from equation (4.10), set $K = K^* = \mathbb{S}_+^n$, and apply the conic gap identity equation (3.24). The chapter uses a maximization primal, so its feasible matrix value is a lower candidate and the dual value is an upper bound. The dual slack must therefore be

$$S = \mathcal{A}^*(y) - C \succeq 0.$$

This sign check specializes Week 3; it does not re-prove generic conic duality.

Theorem 4.11 (SDP weak duality). *For every primal-feasible X and dual-feasible y ,*

$$\langle C, X \rangle \leq b^\top y.$$

Proof. Let $S = \mathcal{A}^*(y) - C \succeq 0$. Then

$$\begin{aligned} b^\top y - \langle C, X \rangle &= y^\top \mathcal{A}(X) - \langle C, X \rangle \\ &= \langle \mathcal{A}^*(y) - C, X \rangle \\ &= \langle S, X \rangle \geq 0, \end{aligned}$$

where the last inequality uses $S, X \succeq 0$. □

Proof note. The displayed calculation is the PSD instance of equation (3.24); use it to check the signs, not as a second general duality derivation.

Key result: the SDP gap is a trace product

For a feasible primal–dual pair,

$$b^\top y - \langle C, X \rangle = \langle S, X \rangle. \tag{4.15}$$

Thus the upper bound $b^\top y$ meets the lower candidate $\langle C, X \rangle$ exactly when the two PSD matrices are complementary.

4.6.2 Anchor problem: a complete 2×2 certificate

Consider

$$\begin{aligned} \text{(P}_2\text{)} \quad & \underset{X \in \mathbb{S}^2}{\text{maximize}} && 2X_{12} \\ & \text{subject to} && X_{11} = 1, \quad X_{22} = 1, \\ & && X \succeq 0. \end{aligned} \tag{4.16}$$

Writing

$$C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

makes $\langle C, X \rangle = 2X_{12}$. The constraints fix

$$X = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}.$$

By equation (4.11), feasibility is exactly $|\rho| \leq 1$. Hence

$$X^* = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad p^* = 2.$$

The constraint matrices are

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad b = (1, 1)^\top.$$

Therefore the dual is

$$\begin{aligned} (D_2) \quad & \underset{y_1, y_2 \in \mathbb{R}}{\text{minimize}} && y_1 + y_2 \\ & \text{subject to} && S(y) = \begin{bmatrix} y_1 & -1 \\ -1 & y_2 \end{bmatrix} \succeq 0. \end{aligned} \tag{4.17}$$

The choice $y^* = (1, 1)^\top$ is dual feasible:

$$S^* = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \succeq 0, \quad b^\top y^* = 2.$$

Weak duality now gives the complete certificate

$$2 = \langle C, X^* \rangle \leq p^* \leq d^* \leq b^\top y^* = 2.$$

Every inequality is equality.

Complementarity is visible as matrix multiplication:

$$S^* X^* = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = 0.$$

The range of X^* is spanned by $(1, 1)^\top$, while the range of S^* is spanned by $(1, -1)^\top$.

4.7 Spectral optimization: the largest eigenvalue as an SDP

Let $C \in \mathbb{S}^n$. The largest eigenvalue can be written as the optimal value of a small but fundamental SDP:

$$\begin{aligned} & \underset{X \in \mathbb{S}^n}{\text{maximize}} && \langle C, X \rangle \\ & \text{subject to} && \text{Tr}(X) = 1, \quad X \succeq 0. \end{aligned} \tag{4.18}$$

The feasible matrix X distributes one unit of PSD *mass* among directions. To see the bound, use

$$C \preceq \lambda_{\max}(C)I.$$

For every feasible X ,

$$\langle C, X \rangle \leq \lambda_{\max}(C) \langle I, X \rangle = \lambda_{\max}(C).$$

The inequality follows from $\lambda_{\max}(C)I - C \succeq 0$, $X \succeq 0$, and the nonnegative trace product in lemma 4.5. If u is a unit top eigenvector, then $X = uu^\top$ is feasible and

$$\langle C, uu^\top \rangle = u^\top C u = \lambda_{\max}(C).$$

The upper bound is attained.

This argument is already a primal–dual certificate. Specializing equation (4.14) with $A_1 = I$ and $b = 1$ gives

$$\begin{aligned} & \underset{t \in \mathbb{R}}{\text{minimize}} && t \\ & \text{subject to} && tI - C \succeq 0. \end{aligned} \tag{4.19}$$

The smallest feasible t is exactly $\lambda_{\max}(C)$. At optimality, the slack $S^* = \lambda_{\max}(C)I - C$ annihilates every top eigenvector, so $S^*uu^\top = 0$.

Worked example: a 2×2 spectral certificate

Let

$$C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Its eigenvalues are 3 and 1, with unit top eigenvector $u = (1, 1)^\top / \sqrt{2}$. A primal candidate is

$$X^* = uu^\top = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \text{Tr}(X^*) = 1.$$

Choose the dual value $t^* = 3$. Its slack is

$$S^* = 3I - C = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \succeq 0.$$

The two objectives match:

$$\langle C, X^* \rangle = 3 = t^*,$$

and complementarity is visible directly:

$$S^*X^* = 0.$$

We have checked feasibility, matching values, and orthogonal ranges—the full SDP certificate.

This formulation is also the optimization core of the leading direction in principal component analysis: with C a covariance matrix, u is the direction of largest variance. The SDP view says that even if we allow a mixture of directions through X , all mass can be placed on a best eigendirection.

4.8 SDP Slater and KKT as Week 3 specializations

Theorem 4.12 (Slater conditions for the standard SDP). *1. Suppose p^* is finite. If there exists $\bar{X}$ with*

$$\bar{X} \succ 0, \quad \mathcal{A}(\bar{X}) = b,$$

then the dual optimum is attained and $p^ = d^*$.*

2. Suppose d^* is finite. If there exists $\bar{y}$ with

$$\mathcal{A}^*(\bar{y}) - C \succ 0,$$

then the primal optimum is attained and $p^* = d^*$.

For the anchor problem, $\bar{X} = I \succ 0$ is strictly primal feasible. The choice $\bar{y} = (2, 2)^\top$ gives

$$\mathcal{A}^*(\bar{y}) - C = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \succ 0,$$

so both sides satisfy Slater.

Theorem 4.13 (KKT certificate for an SDP). *Suppose strong duality holds and the optima are attained. A primal–dual triple (X^*, y^*, S^*) is optimal if and only if*

$$\begin{aligned} \mathcal{A}(X^*) &= b, & X^* &\succeq 0, & \text{primal feasibility,} \\ S^* &= \mathcal{A}^*(y^*) - C \succeq 0, & & \text{dual feasibility,} \\ \langle S^*, X^* \rangle &= 0, & & \text{complementarity.} \end{aligned}$$

For PSD matrices, complementarity is equivalent to

$$S^* X^* = 0.$$

Proof. Feasibility and equation (4.15) show that the primal–dual gap is $\langle S^*, X^* \rangle$, so zero complementarity is equivalent to matching objective values. To obtain the matrix form, observe that

$$\langle S, X \rangle = \|S^{1/2} X^{1/2}\|_F^2.$$

Here $\|A\|_F^2 := \sum_{i,j} A_{ij}^2 = \text{Tr}(A^\top A)$ is the squared Frobenius norm. If the trace product is zero, then $S^{1/2} X^{1/2} = 0$, which implies $SX = 0$. The converse is immediate by taking the trace. $\square$

4.9 From scalar barriers to the log-determinant barrier

The scalar barrier above diverges when any inequality approaches its boundary. For the PSD cone, the strict interior is $X \succ 0$, and the analogous barrier is

$$\phi(X) = -\log \det X.$$

For $H \in \mathbb{S}^n$, define the directional derivative by

$$D\phi(X)[H] := \left. \frac{d}{ds} \phi(X + sH) \right|_{s=0}.$$

Write $D^2\phi(X)[H, H]$ for the corresponding second derivative. Then

$$D\phi(X)[H] = -\text{Tr}(X^{-1}H), \tag{4.20}$$

$$D^2\phi(X)[H, H] = \text{Tr}(X^{-1}HX^{-1}H) \geq 0. \tag{4.21}$$

If $X_k \succ 0$ converges to a finite singular matrix $\bar{X} \succeq 0$, then $\det X_k \rightarrow 0$ and $-\log \det X_k \rightarrow +\infty$. Primal–dual interior-point methods follow matrix-valued central paths and solve structured Newton systems. The scalar and matrix cases share the same pattern:

strict interior $\longrightarrow$ barrier subproblem $\longrightarrow$ Newton system $\longrightarrow$ approximate KKT point.

Limitations. Dense $n \times n$ PSD variables contain $O(n^2)$ scalar entries, and generic Newton systems can be expensive. Practical large-scale SDPs exploit sparsity, block structure, low-rank factorization, or first-order approximations. Convexity guarantees global meaning; it does not make every instance small.

4.10 Quantum representation bridge: binary state discrimination

This application introduces only the matrix language needed for the optimization problem.

Further exploration: from real symmetric to complex Hermitian matrices

The preceding SDP geometry was written over real symmetric matrices. Over $\mathbb{C}$, replace transpose $(\cdot)^\top$ by conjugate transpose $(\cdot)^\dagger$. A matrix X is Hermitian when $X = X^\dagger$, and $X \succeq 0$ means $z^\dagger X z \geq 0$ for every complex vector z . Hermitian matrices have real eigenvalues and unitary spectral decompositions. The real trace inner product extends to $\langle A, B \rangle = \text{Re Tr}(A^\dagger B)$; for the Hermitian products used below, $\text{Tr}(AB)$ is real. Gram factorizations, PSD order, spectral proofs, and primal–dual trace certificates carry over with these replacements.

Definition 4.14 (Density matrix and measurement effect). A finite-dimensional quantum state is a complex Hermitian matrix ρ satisfying

$$\rho \succeq 0, \quad \text{Tr}(\rho) = 1.$$

A two-outcome measurement can be represented by a Hermitian matrix M with

$$0 \preceq M \preceq I.$$

The two outcome operators are M and $I - M$. For state ρ , the probability of the first outcome is $\text{Tr}(M\rho)$.

Suppose ρ_0 and ρ_1 occur with equal prior probability. We guess state 0 after outcome M and state 1 after outcome $I - M$. The success probability is

$$p_{\text{succ}}(M) = \frac{1}{2} \text{Tr}(M\rho_0) + \frac{1}{2} \text{Tr}((I - M)\rho_1) \quad (4.22)$$

$$= \frac{1}{2} + \frac{1}{2} \text{Tr}(M(\rho_0 - \rho_1)). \quad (4.23)$$

Thus binary discrimination is the SDP

$$\begin{aligned} & \underset{M}{\text{maximize}} && \frac{1}{2} + \frac{1}{2} \text{Tr}(M\Delta) \\ & \text{subject to} && M \succeq 0, \quad I - M \succeq 0, \end{aligned} \quad \Delta := \rho_0 - \rho_1. \quad (4.24)$$

Theorem 4.15 (Holevo–Helstrom formula, equal priors). For two density matrices with equal prior probability,

$$p_{\text{succ}}^* = \frac{1}{2} + \frac{1}{4} \|\rho_0 - \rho_1\|_1, \quad (4.25)$$

where $\|H\|_1 = \sum_i |\lambda_i(H)|$ for Hermitian H . An optimal M is the projector onto the positive-eigenvalue subspace of $\Delta = \rho_0 - \rho_1$.

Proof. Diagonalize

$$\Delta = \sum_i \lambda_i u_i u_i^\dagger.$$

For $0 \preceq M \preceq I$, each number

$$m_i = u_i^\dagger M u_i$$

lies in $[0, 1]$. Therefore

$$\text{Tr}(M\Delta) = \sum_i \lambda_i m_i$$

is maximized by choosing $m_i = 1$ for $\lambda_i > 0$ and $m_i = 0$ for $\lambda_i < 0$. The optimum is $\sum_{\lambda_i > 0} \lambda_i$. Because $\text{Tr}(\Delta) = 0$,

$$\sum_{\lambda_i > 0} \lambda_i = \frac{1}{2} \sum_i |\lambda_i| = \frac{1}{2} \|\Delta\|_1.$$

Substitution into equation (4.23) proves the formula. $\square$

Worked example: distinguishing $|0\rangle$ from $|+\rangle$

Let

$$\rho_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \rho_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Then

$$\Delta = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

has eigenvalues $\pm 1/\sqrt{2}$, so

$$\|\Delta\|_1 = \sqrt{2}, \quad p_{\text{succ}}^* = \frac{1}{2} + \frac{\sqrt{2}}{4} \approx 0.853553.$$

The states are neither identical, which would give success $1/2$, nor orthogonal, which would permit success 1 .

Connection to AI practice: Gram matrices, kernels, and global bounds

The same PSD object has several AI interpretations. A Gram matrix records embedding inner products; a kernel matrix must be PSD; a covariance matrix encodes second moments; and a dual PSD slack can certify a global bound. SDP relaxations also appear in certified robustness and combinatorial subproblems. The reusable skill is to recognize when pairwise interactions can be lifted into a matrix and when the resulting bound is worth the computational cost.

4.11 Misconception checks

Classroom activity: four representations of one matrix

For

$$X = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix},$$

work in groups to:

1. compute its eigenvalues;
2. prove PSD using a quadratic form;
3. find three planar unit vectors having X as their Gram matrix;
4. identify its rank and nullspace;
5. predict why this matrix will be useful for a triangle graph.

Checkpoint. The eigenvalues are $3/2, 3/2, 0$, and three unit vectors separated by 120° produce this Gram matrix.

Summary and connections

- With a positive-definite Hessian, Newton's step minimizes a local quadratic model; damping controls the step length.
- Log barriers enforce strict feasibility. For m scalar inequalities in the convex setting, exact central-path points have gap m/t . The PSD barrier is $-\log \det X$ on $X \succ 0$.
- PSD matrices admit quadratic-form, eigenvalue, and Gram descriptions. Their cone is closed, convex, and self-dual under the trace inner product.
- An SDP combines a linear objective, affine constraints, and a PSD constraint. Conic duality supplies bounds; Slater conditions give strong duality, and complementarity means orthogonal primal and dual ranges.
- The largest-eigenvalue SDP has matching certificates: a rank-one top-eigenvector projector and $\lambda_{\max}(C)I - C \succeq 0$.

4.12 Chapter practice set

Exercise 4.1 (PSD recognition). For each matrix, determine whether it is PSD using both eigenvalues and a quadratic-form or determinant argument:

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}.$$

Exercise 4.2 (Trace product). Let $A = uu^\top$ and $B = vv^\top$.

- (a) Prove $\langle A, B \rangle = (u^\top v)^2$.
- (b) Interpret $\langle A, B \rangle = 0$ geometrically.
- (c) Explain how this rank-one calculation foreshadows PSD-cone self-duality.

Exercise 4.3 (LMI conversion). Let $x \in \mathbb{R}^n$ and let $I = I_n$. Show that

$$\|x\|_2 \leq t \iff \begin{bmatrix} tI & x \\ x^\top & t \end{bmatrix} \succeq 0$$

for $t \geq 0$. You may use the Schur complement.

Exercise 4.4 (Repeated top eigenvalue). Let $C = \text{Diag}(3, 3, 1)$ and consider

$$\max_{X \succeq 0} \langle C, X \rangle \quad \text{subject to} \quad \text{Tr}(X) = 1.$$

- (a) Derive the scalar dual and construct an optimal dual slack.
- (b) Characterize *every* primal optimizer.
- (c) Exhibit one rank-one and one rank-two optimizer. Explain why this does not contradict the rank-one construction in the worked example.

Exercise 4.5 (Check Slater's condition). Let $X \in \mathbb{S}^n$, with $n \geq 2$ for the second set. For each PSD feasibility region below, determine whether primal Slater holds:

$$\{X \succeq 0 \mid \text{Tr}(X) = 1\}, \quad \{X \succeq 0 \mid X_{11} = 0, X_{22} = 1\}.$$

Does failure of the displayed Slater condition prove a duality gap? Explain.

Exercise 4.6 (Quantum discrimination). Assume equal prior probabilities and the measurement convention $0 \preceq M \preceq I$, where outcome M guesses the first state. Verify every step of the $|0\rangle$ versus $|+\rangle$ example, including the eigenvalues and the feasibility of the positive-space projector M . Then repeat for two orthogonal basis states.

Exercise 4.7 (Newton and barrier bridge). For

$$F(x_1, x_2) = e^{x_1+x_2} + \frac{1}{2}(x_1^2 + 2x_2^2),$$

derive the gradient and Hessian, then write—without inverting the matrix—the 2×2 Newton system at a general point. For

$$\min_x \frac{1}{2} \|x\|_2^2 \quad \text{subject to} \quad a^\top x \leq b,$$

derive the gradient and Hessian of

$$\frac{t}{2} \|x\|_2^2 - \log(b - a^\top x).$$

Show that the Hessian is positive definite for $t > 0$, explain why a line search must preserve $a^\top x < b$. Separately, for a barrier problem with $m = 40$ scalar inequalities, find a t for which the ideal central-path gap satisfies $m/t \leq 10^{-3}$.

4.13 Complete solutions

Practice guide

Attempt the complete chapter practice set before consulting this section. Each numbered box supplies a checkable derivation, certificate, or computational verification for the corresponding exercise.

Solution conventions. Throughout this section,

$$\langle A, B \rangle := \text{Tr}(A^\top B)$$

is the trace inner product on real symmetric matrices, and $X \succeq 0$ means that X is positive semidefinite (PSD). For Hermitian matrices in Exercise 4.6, $A^\top$ is replaced by $A^\dagger$.

Solution to Exercise 4.1

For A_1 ,

$$\det(A_1 - \lambda I) = (2 - \lambda)^2 - 1 = (\lambda - 1)(\lambda - 3),$$

so its eigenvalues are 1 and 3. Both are positive. Independently, for $z = (a, b)^\top$,

$$z^\top A_1 z = 2a^2 + 2ab + 2b^2 = (a + b)^2 + a^2 + b^2,$$

which is positive for every $z \neq 0$. Hence $A_1 \succ 0$.

For A_2 , its trace and determinant are 5 and 0, respectively, so its eigenvalues are 5 and 0. Moreover,

$$z^\top A_2 z = (a + 2b)^2 \geq 0, \quad A_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}.$$

Thus $A_2 \succeq 0$, but it is singular and has rank one.

For A_3 , the vectors $(1, 1)^\top$ and $(1, -1)^\top$ are eigenvectors with eigenvalues 3 and -1 , respectively. The negative eigenvalue rules out positive semidefiniteness. The quadratic-form check gives the same conclusion:

$$\begin{bmatrix} 1 & -1 \end{bmatrix} A_3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -2 < 0.$$

Equivalently, $\det A_3 = -3 < 0$, so A_3 is indefinite.

Solution to Exercise 4.2

By cyclicity of the trace,

$$\begin{aligned} \langle A, B \rangle &= \text{Tr}(u u^\top v v^\top) = \text{Tr}(u (u^\top v) v^\top) \\ &= (u^\top v) \text{Tr}(u v^\top) = (u^\top v)(v^\top u) = (u^\top v)^2. \end{aligned}$$

Consequently, $\langle A, B \rangle = 0$ if and only if $u^\top v = 0$. For nonzero u and v , this says that the one-dimensional ranges of A and B are orthogonal. The statement also includes the degenerate case in which $u = 0$ or $v = 0$.

To see the connection with self-duality, use spectral decompositions

$$X = \sum_i \lambda_i u_i u_i^\top, \quad Y = \sum_j \mu_j v_j v_j^\top, \quad \lambda_i, \mu_j \geq 0.$$

The rank-one identity yields

$$\langle X, Y \rangle = \sum_{i,j} \lambda_i \mu_j (u_i^\top v_j)^2 \geq 0.$$

Thus every PSD Y belongs to the dual cone of the PSD cone. Conversely, if a symmetric matrix Y has nonnegative inner product with every PSD matrix, then testing only the rank-one PSD matrices $uu^\top$ gives

$$u^\top Y u = \langle Y, uu^\top \rangle \geq 0 \quad \text{for every } u.$$

Hence $Y \succeq 0$. This proves $(\mathbb{S}_+^n)^* = \mathbb{S}_+^n$ and shows why rank-one matrices are enough to reveal self-duality.

Solution to Exercise 4.3

First suppose $t > 0$. The leading block tI is positive definite, so the Schur-complement criterion gives

$$\begin{aligned} K(t, x) \succeq 0 &\iff t - x^\top (tI)^{-1} x \geq 0 \\ &\iff t - \frac{\|x\|_2^2}{t} \geq 0 \\ &\iff \|x\|_2^2 \leq t^2 \iff \|x\|_2 \leq t, \end{aligned}$$

where the last equivalence uses $t > 0$.

It remains to handle the singular endpoint $t = 0$, where the inverse-based Schur complement cannot be used. If $K(0, x) \succeq 0$, then every principal 2×2 submatrix

$$\begin{bmatrix} 0 & x_i \\ x_i & 0 \end{bmatrix}$$

must be PSD. Its determinant is $-x_i^2$, so $x_i = 0$ for every i . Thus $x = 0$. Conversely, $x = 0$ makes $K(0, 0)$ the zero matrix, which is PSD. Since $\|x\|_2 \leq 0$ also holds exactly when $x = 0$, the equivalence includes $t = 0$.

Solution to Exercise 4.4

Introduce a scalar multiplier t for $\text{Tr}(X) = 1$. The Lagrangian is

$$L(X, t) = \langle C, X \rangle + t(1 - \text{Tr}(X)) = t + \langle C - tI, X \rangle.$$

The supremum over $X \succeq 0$ is finite precisely when $C - tI \preceq 0$. Hence the dual is

$$\begin{aligned} &\text{minimize} && t \\ &\text{subject to} && S(t) := tI - C \succeq 0. \end{aligned}$$

Here

$$S(t) = \text{Diag}(t - 3, t - 3, t - 1),$$

so dual feasibility is equivalent to $t \geq 3$. Therefore

$$t^* = 3, \quad S^* = 3I - C = \text{Diag}(0, 0, 2).$$

The primal and dual Slater conditions both hold: $I/3 \succ 0$ is primal strictly feasible, and any $t > 3$ gives $tI - C \succ 0$. In particular, strong duality applies.

For a primal-feasible X ,

$$\langle C, X \rangle = 3X_{11} + 3X_{22} + X_{33} = 3 \operatorname{Tr}(X) - 2X_{33} = 3 - 2X_{33} \leq 3,$$

because $X_{33} \geq 0$. Equality holds exactly when $X_{33} = 0$. If a PSD matrix has $X_{33} = 0$, then its 2×2 principal minors on indices $\{i, 3\}$ satisfy

$$0 \leq X_{ii}X_{33} - X_{i3}^2 = -X_{i3}^2, \quad i = 1, 2,$$

so $X_{13} = X_{23} = 0$. Thus X is optimal if and only if it has the form

$$X^* = \begin{bmatrix} A & 0 \\ 0 & 0 \end{bmatrix}, \quad A \in \mathbb{S}_+^2, \quad \operatorname{Tr}(A) = 1.$$

For every such X^* , the matching certificate is also visible from $S^*X^* = 0$.
Examples of ranks one and two are

$$X_1 = \operatorname{Diag}(1, 0, 0), \quad X_2 = \operatorname{Diag}\left(\frac{1}{2}, \frac{1}{2}, 0\right).$$

There is no contradiction with the rank-one spectral construction. That construction guarantees the existence of a rank-one optimizer uu^T for a top eigenvector u ; it does not assert that every optimizer has rank one. Since the top eigenvalue 3 has a two-dimensional eigenspace, any PSD unit-trace matrix supported on that eigenspace is optimal, including rank-two mixtures.

Solution to Exercise 4.5

Primal Slater requires a matrix satisfying all equality constraints and lying in the interior of the ambient PSD cone, that is, a matrix $X \succ 0$. For $\mathcal{F}_1$,

$$X = \frac{1}{n}I_n \succ 0, \quad \operatorname{Tr}(X) = 1,$$

so primal Slater holds.

For $\mathcal{F}_2$, any positive-definite matrix would satisfy

$$X_{11} = e_1^T X e_1 > 0,$$

contradicting $X_{11} = 0$. Hence primal Slater fails. The region is nevertheless nonempty: for example, $\operatorname{Diag}(0, 1, 0, \dots, 0) \in \mathcal{F}_2$. In fact, if $X \succeq 0$ and $X_{11} = 0$, then

$$0 = e_1^T X e_1 = \|X^{1/2}e_1\|_2^2,$$

so $Xe_1 = 0$. Every feasible matrix therefore lies on a proper face of the PSD cone, which explains geometrically why no strictly feasible point exists.

Failure of primal Slater does not prove a duality gap. Slater's condition is a sufficient constraint qualification, not a necessary condition for strong duality. For a completed SDP satisfying the usual finiteness hypotheses, dual Slater or a reduction to the minimal face may still establish strong duality. Moreover, the displayed sets contain no objective, so a primal-dual gap is not defined until an objective is supplied. As a concrete no-gap example, attach the zero objective to $\mathcal{F}_2$:

$$\begin{aligned} & \text{maximize} && 0 \\ & \text{subject to} && X_{11} = 0, \quad X_{22} = 1, \quad X \succeq 0. \end{aligned}$$

Its primal value is 0. With $E_{ii} = e_i e_i^\top$, its dual is

$$\text{minimize } y_2 \quad \text{subject to} \quad y_1 E_{11} + y_2 E_{22} \succeq 0.$$

The choice $y_1 = y_2 = 0$ is feasible and has value 0, so this problem has no duality gap despite failure of primal Slater.

Solution to Exercise 4.6

For two states ρ_0, ρ_1 with equal prior probabilities, a two-outcome effect M obeys $0 \preceq M \preceq I$. If outcome M means *guess state* 0, then

$$\begin{aligned} p_{\text{succ}}(M) &= \frac{1}{2} \text{Tr}(M\rho_0) + \frac{1}{2} \text{Tr}((I - M)\rho_1) \\ &= \frac{1}{2} + \frac{1}{2} \text{Tr}(M\Delta), \quad \Delta := \rho_0 - \rho_1, \end{aligned}$$

because $\text{Tr}(\rho_1) = 1$. The optimal effect is the projector onto the positive-eigenvalue subspace of Δ .

For

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

the density matrices and their difference are

$$\rho_0 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \rho_1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \Delta = \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}.$$

The characteristic polynomial is

$$\det(\Delta - \lambda I) = \lambda^2 - \frac{1}{2},$$

so

$$\lambda_+ = \frac{1}{\sqrt{2}}, \quad \lambda_- = -\frac{1}{\sqrt{2}}.$$

For example, a positive-eigenvalue eigenvector is proportional to $(1, 1 - \sqrt{2})^\top$. Since $\Delta^2 = I/2$, the positive-space projector can be written without choosing a normalization:

$$\begin{aligned} M = P_+ &= \frac{1}{2}(I + \sqrt{2}\Delta) \\ &= \frac{1}{2} \begin{bmatrix} 1 + 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & 1 - 1/\sqrt{2} \end{bmatrix}. \end{aligned}$$

It is real symmetric, and $\Delta^2 = I/2$ gives $M^2 = M$. Therefore its eigenvalues are 0 and 1. The matrix $I - M$ is the complementary orthogonal projector and also has eigenvalues 0 and 1. Hence

$$M \succeq 0, \quad I - M \succeq 0,$$

which is exactly the required feasibility condition $0 \preceq M \preceq I$.

Because M selects the positive eigenvalue,

$$\text{Tr}(M\Delta) = \lambda_+ = \frac{1}{\sqrt{2}}.$$

Thus

$$p_{\text{succ}}^* = \frac{1}{2} + \frac{1}{2\sqrt{2}} = \frac{1}{2} + \frac{\sqrt{2}}{4} \approx 0.853553.$$

The same value follows directly from the two conditional success probabilities:

$$\langle 0|M|0\rangle = \langle +|(I-M)|+\rangle = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}}\right).$$

Finally,

$$\|\rho_0 - \rho_1\|_1 = |\lambda_+| + |\lambda_-| = \sqrt{2},$$

so the Holevo–Helstrom formula $p_{\text{succ}}^* = \frac{1}{2} + \frac{1}{4}\|\rho_0 - \rho_1\|_1$ agrees with the explicit optimization. For two orthogonal basis states, take

$$\rho_0 = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \rho_1 = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\Delta = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad M = P_+ = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad I - M = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Both measurement effects are PSD, the eigenvalues of Δ are 1 and -1 , and $\|\Delta\|_1 = 2$. Therefore

$$p_{\text{succ}}^* = \frac{1}{2} + \frac{1}{4}(2) = 1.$$

Indeed, the computational-basis measurement identifies the two orthogonal states with certainty.

Solution to Exercise 4.7

Set $r = e^{x_1+x_2} > 0$. Direct differentiation gives

$$\nabla F(x) = \begin{bmatrix} r + x_1 \\ r + 2x_2 \end{bmatrix}, \quad \nabla^2 F(x) = \begin{bmatrix} r + 1 & r \\ r & r + 2 \end{bmatrix}.$$

Thus the Newton direction $p = (p_1, p_2)^\top$ is defined by

$$\begin{bmatrix} r + 1 & r \\ r & r + 2 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = - \begin{bmatrix} r + x_1 \\ r + 2x_2 \end{bmatrix}.$$

The matrix should be solved as a linear system, not explicitly inverted.

For the barrier calculation, let $\delta(x) = b - a^\top x > 0$. The chain rule gives

$$\nabla \Phi_t(x) = tx + \frac{a}{\delta(x)}, \quad \nabla^2 \Phi_t(x) = tI + \frac{aa^\top}{\delta(x)^2}.$$

For every nonzero z ,

$$z^\top \nabla^2 \Phi_t(x) z = t \|z\|_2^2 + \frac{(a^\top z)^2}{\delta(x)^2} > 0,$$

so the Hessian is positive definite when $t > 0$. A trial point with $a^\top x \geq b$ leaves the logarithm's domain; backtracking must therefore keep every accepted point strictly feasible.

Finally,

$$\frac{m}{t} \leq 10^{-3} \iff t \geq \frac{40}{10^{-3}} = 4 \times 10^4.$$

Hence $t = 40000$ is the smallest threshold value for the stated ideal bound.

Suggested reading. The Newton, barrier, SDP, and interior-point viewpoints follow [BV04, NW06, Nes18, VB96]. The binary state-discrimination formulation follows [Wat18, Chapter 3], and the optional modeling layer uses [DB16].

Historical note

Newton iteration began as a method for solving equations; the optimization version applies the same idea to the stationarity equation $\nabla F(x) = 0$. Karmarkar's 1984 polynomial-time algorithm for linear programming made interior-point methods a central modern paradigm [Kar84]; by the 1990s, primal–dual interior-point methods had also become a standard way to solve semidefinite programs [VB96].

Chapter 5

Max-Cut, SDP Relaxation, and Randomized Rounding

Why this chapter

Week 4 supplied PSD representations and primal–dual bounds. This chapter asks what to do when the desired rank-one matrix encodes a combinatorial decision and is therefore hard to optimize over directly. Weighted Max-Cut is NP-hard: the goal is not to disguise an exact exponential search as a convex problem, but to replace it by a tractable relaxation with an explicit approximation guarantee.

The representation switch is the elegant step. Binary signs are lifted to the rank-one matrix $X = xx^T$; dropping only the rank constraint leaves an SDP. Under the usual finite-precision assumptions, this convex relaxation can be solved to prescribed accuracy in polynomial time. For the maximization problem, its value is an upper bound on the exact discrete optimum. Randomized rounding returns a feasible cut, hence a lower candidate, and the Goemans–Williamson analysis guarantees in expectation at least the Goemans–Williamson constant $\alpha_{\text{GW}} \approx 0.878$ times the SDP value for nonnegative edge weights. The central pattern is

discrete formulation $\longrightarrow$ rank-one lifting $\longrightarrow$ convex relaxation
 $\longrightarrow$ randomized rounding $\longrightarrow$ provable approximation.

This pattern reaches beyond Max-Cut: continuous geometry supplies an estimate and a certificate, while rounding reconstructs a valid structured decision.

For AI, the edge weights may be learned pairwise scores. Max-Cut begins where prediction ends: it converts many conflicting local preferences into one globally consistent binary partition. A complete analysis therefore keeps four error sources separate:

1. statistical error in the learned scores;
2. the gap introduced by the convex relaxation;
3. randomness and loss introduced by rounding;
4. numerical error in solving the SDP.

Learning objectives

By the end of this chapter, you should be able to:

1. formulate nonnegative pairwise disagreement as a Max-Cut quadratic optimization problem;
2. derive the rank-one PSD lift and its SDP relaxation by dropping rank;
3. compute one random-hyperplane rounding outcome from a Gram representation;
4. distinguish and order a realized cut, its expected rounded value, the exact optimum, and the SDP upper bound.

5.1 The approximation process

$$\begin{array}{ccccccc}
 x_i \in \{-1, +1\} & \implies & X = xx^\top & \implies & X \succeq 0, X_{ii} = 1 & \implies & X_{ij} = \langle v_i, v_j \rangle \\
 \text{cut} & & \text{rank-one matrix} & & \text{SDP relaxation} & & \text{unit vectors} \\
 & & & & & & \downarrow \text{random hyperplane} \\
 & & & & & & \hat{x}_i = \text{sign}(\langle v_i, g \rangle).
 \end{array}$$

There are two logically different losses:

1. **Relaxation loss:** dropping rank one may increase the maximization value.
2. **Rounding loss:** converting vectors back to signs may decrease the value.

The SDP value measures the first gap; the Goemans–Williamson analysis controls the second in expectation.

Connection to AI practice: from pairwise predictions to one consistent decision

Imagine that a model compares many pairs of objects. For each pair $\{i, j\}$, it outputs a score $\hat{w}_{ij} \geq 0$: a larger score means stronger preference for placing the two objects on opposite sides. The pairwise wishes may conflict. For example, three objects cannot all be opposite to one another using only two labels.

The system must therefore choose one label $x_i \in \{-1, +1\}$ for every object and optimize the pairwise wishes jointly:

$$\text{observations} \longrightarrow \text{learned scores } \hat{w}_{ij} \longrightarrow \text{one valid binary partition } x.$$

Weighted Max-Cut is the cleanest model of this “learn scores, then make a globally consistent discrete decision” step. It is a useful prototype for pairwise structured inference, but it does not imply that every clustering, ranking, or assignment problem is literally Max-Cut.

Once the learned weights are fixed, optimization supplies an auditable bracket:

$$\text{realized cut} \leq \text{OPT}(G) \leq \text{SDP}(G).$$

The rounding theorem relates the left and right sides in expectation. It does not certify that the learned weights reflect the deployment population; that is a separate statistical question.

5.2 Weighted Max-Cut

Definition 5.1 (Weighted undirected graph). A weighted undirected graph is $G = (V, E, w)$, where

$$V = \{1, \dots, n\}, \quad E \subseteq \{\{i, j\} \mid i \neq j\}, \quad w_{ij} \geq 0 \quad \text{for } \{i, j\} \in E.$$

Every unordered edge in sums below is counted exactly once.

Definition 5.2 (Cut and cut value). A cut is a partition $(S, V \setminus S)$. Its value is

$$\text{Cut}(S) = \sum_{\substack{\{i,j\} \in E \\ i \in S, j \notin S}} w_{ij}.$$

Max-Cut asks for

$$\text{OPT}(G) = \max_{S \subseteq V} \text{Cut}(S).$$

Encode the partition using

$$x_i = \begin{cases} +1, & i \in S, \\ -1, & i \notin S. \end{cases}$$

For one edge,

$$\frac{1 - x_i x_j}{2} = \begin{cases} 1, & \text{the edge crosses the cut,} \\ 0, & \text{the endpoints are on the same side.} \end{cases}$$

Therefore

$$\text{Cut}(x) = \frac{1}{2} \sum_{\{i,j\} \in E} w_{ij} (1 - x_i x_j), \quad x \in \{-1, +1\}^n. \quad (5.1)$$

5.2.1 The graph Laplacian

Let W be the symmetric weighted adjacency matrix and let

$$D = \text{Diag}(d_1, \dots, d_n), \quad d_i = \sum_{j: \{i,j\} \in E} w_{ij}.$$

The graph Laplacian is

$$L = D - W.$$

For every $z \in \mathbb{R}^n$,

$$z^\top L z = \sum_{\{i,j\} \in E} w_{ij} (z_i - z_j)^2. \quad (5.2)$$

Consequently $L \succeq 0$, and for a sign vector,

$$\text{Cut}(x) = \frac{1}{4} x^\top L x. \quad (5.3)$$

Worked example: the triangle K_3

The unit-weight triangle has

$$L = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

The all-equal sign patterns cut no edges. Every other sign pattern isolates one vertex and cuts two edges. Hence

$$\text{OPT}(K_3) = 2.$$

For $x = (1, 1, -1)^\top$,

$$x^\top Lx = 8, \quad \frac{1}{4}x^\top Lx = 2,$$

confirming the coefficient in equation (5.3).

Common trap

Factors of two depend on whether an undirected edge is counted once or both ordered pairs (i, j) and (j, i) are summed. This chapter counts each edge once. The invariant check is simple: one crossed unit-weight edge must contribute exactly 1 to $\text{Cut}(x)$.

5.2.2 When the edge weights are learned

Suppose $w_e \geq 0$ is the true pairwise value of edge e , while a model produces a nonnegative prediction $\hat{w}_e$. Write $\text{Cut}_w(S)$ and $\text{Cut}_{\hat{w}}(S)$ for the value of the same cut under the two sets of weights, and let

$$\text{OPT}_w := \max_{S \subseteq V} \text{Cut}_w(S), \quad \text{OPT}_{\hat{w}} := \max_{S \subseteq V} \text{Cut}_{\hat{w}}(S).$$

Define the total score error

$$\Delta_w := \sum_{e \in E} |w_e - \hat{w}_e|. \quad (5.4)$$

Proposition 5.3 (Stability of a cut under learned edge scores). *For every cut S ,*

$$|\text{Cut}_w(S) - \text{Cut}_{\hat{w}}(S)| \leq \Delta_w. \quad (5.5)$$

Accessible linear reading: for every fixed cut S , the absolute difference between its true-weight value and predicted-weight value is at most Δ_w .

Moreover, let $\hat{S}$ be the random cut returned by an algorithm that, for some $\alpha \in [0, 1]$, satisfies

$$\mathbb{E}[\text{Cut}_{\hat{w}}(\hat{S})] \geq \alpha \text{OPT}_{\hat{w}}.$$

Then its performance under the true weights obeys

$$\mathbb{E}[\text{Cut}_w(\hat{S})] \geq \alpha \text{OPT}_w - (1 + \alpha)\Delta_w. \quad (5.6)$$

Proof. For a fixed cut, let $c_e(S) \in \{0, 1\}$ indicate whether edge e crosses the cut. Then

$$|\text{Cut}_w(S) - \text{Cut}_{\hat{w}}(S)| = \left| \sum_{e \in E} (w_e - \hat{w}_e) c_e(S) \right| \leq \sum_{e \in E} |w_e - \hat{w}_e| = \Delta_w.$$

Apply this inequality first to the returned cut and then to an optimal cut for the true weights. The second application gives $\text{OPT}_{\hat{w}} \geq \text{OPT}_w - \Delta_w$. Therefore

$$\begin{aligned} \mathbb{E}[\text{Cut}_w(\hat{S})] &\geq \mathbb{E}[\text{Cut}_{\hat{w}}(\hat{S})] - \Delta_w \\ &\geq \alpha \text{OPT}_{\hat{w}} - \Delta_w \\ &\geq \alpha \text{OPT}_w - (1 + \alpha)\Delta_w. \end{aligned} \quad \square$$

Worked example: reading the score-error bound

Suppose a three-edge graph has true weights $(4, 2, 1)$ and predicted weights $(3.5, 2.4, 0.9)$. Then

$$\Delta_w = 0.5 + 0.4 + 0.1 = 1.$$

Every fixed cut changes value by at most 1. For example, a cut crossing the first and third edges has true value 5, predicted value 4.4, and difference $0.6 \leq 1$. The bound is deliberately simple: it separates score-estimation error from whatever relaxation and rounding error the optimizer introduces.

Mastery check

The learning–optimization error decomposition has two independently controllable terms. Rearranging equation (5.6) gives

$$\text{OPT}_w - \mathbb{E}[\text{Cut}_w(\hat{S})] \leq \underbrace{(1 - \alpha) \text{OPT}_w}_{\text{optimization / rounding}} + \underbrace{(1 + \alpha) \Delta_w}_{\text{score estimation}}.$$

Use this as a diagnostic, not merely as a bound:

- if $\Delta_w = 0$, only the relaxation-and-rounding loss remains;
- if $\alpha = 1$, an exact optimizer can still lose up to $2\Delta_w$ because it optimizes the learned objective;
- improving the SDP solver does not reduce Δ_w , while collecting better pairwise data does not change the approximation factor α .

Before interpreting a poor cut, identify which term the experiment actually measures.

Key result: separating learning and optimization error

The learned scores determine the objective; the optimizer determines the decision. A strong optimizer cannot repair badly estimated scores, while perfect scores do not remove the computational difficulty of the discrete problem. Equation (5.6) keeps the two sources of error visible in one inequality.

5.3 Rank-one lifting

For $x \in \{-1, +1\}^n$, define

$$X = xx^\top.$$

Then

$$X \succeq 0, \quad X_{ii} = 1, \quad \text{rank}(X) = 1,$$

and

$$x^\top Lx = \langle L, X \rangle.$$

Thus Max-Cut is exactly

$$\begin{aligned} & \underset{X \in \mathbb{S}^n}{\text{maximize}} && \frac{1}{4} \langle L, X \rangle \\ & \text{subject to} && X_{ii} = 1, \quad i = 1, \dots, n, \\ & && X \succeq 0, \\ & && \text{rank}(X) = 1. \end{aligned} \tag{5.7}$$

Proposition 5.4 (Why the lifting is exact). *The feasible sign vectors of Max-Cut correspond exactly, up to the global sign symmetry $x \leftrightarrow -x$, to the feasible matrices of equation (5.7).*

Proof. Every sign vector produces a feasible $X = xx^\top$. Conversely, if $X \succeq 0$ and $\text{rank}(X) = 1$, then $X = vv^\top$ for some v . The diagonal constraints give $v_i^2 = 1$, so $v_i \in \{-1, +1\}$. Replacing v by $-v$ leaves X unchanged and swaps the two sides of the same cut. $\square$

The only nonconvex ingredient in equation (5.7) is the rank constraint.

5.4 The SDP relaxation

Drop $\text{rank}(X) = 1$ and keep every convex constraint:

$$\begin{aligned} \text{SDP}(G) &:= \underset{X \in \mathbb{S}^n}{\text{maximize}} && \frac{1}{4} \langle L, X \rangle \\ &\text{subject to} && X_{ii} = 1, \quad i = 1, \dots, n, \\ &&& X \succeq 0. \end{aligned} \tag{5.8}$$

For a feasible matrix X , write $v(X) := \frac{1}{4} \langle L, X \rangle$ for its objective value. Thus $v(X) \leq \text{SDP}(G)$, with equality only for an optimal matrix. The symbol $\text{SDP}(G)$ always denotes the optimum, not a solver's current objective.

Proposition 5.5 (Relaxation bound).

$$\text{OPT}(G) \leq \text{SDP}(G).$$

Proof. Every feasible sign vector maps to a feasible SDP matrix with exactly the same objective value. The SDP maximizes over this set and possibly additional higher-rank matrices, so its optimum cannot be smaller. $\square$

5.4.1 The dual certificate

Applying the Week 4 standard dual with $C = L/4$ gives

$$\begin{aligned} &\underset{y \in \mathbb{R}^n}{\text{minimize}} && \sum_{i=1}^n y_i \\ &\text{subject to} && \text{Diag}(y) - \frac{1}{4}L \succeq 0. \end{aligned} \tag{5.9}$$

Write $U(y) := \sum_i y_i$. Every dual-feasible y certifies

$$v(X) \leq \text{SDP}(G) \leq U(y)$$

for every primal-feasible X . The computable gap $U(y) - v(X)$ bounds the suboptimality of the matrix X . The relaxation is computationally useful not only because it produces a matrix X , but also because its dual can certify the relaxed value.

5.5 Gram-vector representation

Because a feasible $X \succeq 0$ with $X_{ii} = 1$ is a Gram matrix, there exist unit vectors $v_1, \dots, v_n$ such that

$$X_{ij} = v_i^\top v_j.$$

The SDP is equivalently the vector program

$$\underset{\|v_i\|_2=1}{\text{maximize}} \quad \frac{1}{2} \sum_{\{i,j\} \in E} w_{ij} (1 - v_i^\top v_j). \quad (5.10)$$

If v_i and v_j make angle

$$\theta_{ij} = \arccos(v_i^\top v_j),$$

then the SDP contribution of edge $\{i, j\}$ is

$$w_{ij} \frac{1 - \cos \theta_{ij}}{2}.$$

Binary signs permit only two antipodal unit vectors. The SDP permits the vertices to spread over a higher-dimensional sphere.

Key result: what the relaxation really changed

The SDP did not replace a graph by an arbitrary matrix. It replaced each binary sign $x_i \in \{-1, +1\}$ by a unit vector v_i , while preserving every pairwise interaction through $X_{ij} = v_i^\top v_j$. Rank one means all vectors lie on a single line; dropping rank allows a spherical embedding.

Key result: the triangle makes the relaxation gap visible

For K_3 , place three unit vectors 120° apart. Their Gram matrix has diagonal 1, off-diagonal entries $-1/2$, and eigenvalues $3/2, 3/2, 0$, so it is SDP feasible. Each of the three edges contributes

$$\frac{1 - (-1/2)}{2} = \frac{3}{4}.$$

Therefore

$$\text{OPT}(K_3) = 2 < \text{SDP}(K_3) = \frac{9}{4}.$$

The equality is certified, not inferred from the feasible Gram matrix alone: since $L = 3I - J$, the dual choice $y_i = 3/4$ has slack

$$\text{Diag}(y) - \frac{1}{4}L = \frac{1}{4}J \succeq 0, \quad \sum_i y_i = \frac{9}{4}.$$

The primal value and dual upper bound match. The strict inequality shows exactly what dropping rank can change. The next example uses K_4 for a different purpose: a symmetric Gram geometry with a matching dual certificate and nontrivial randomized rounding.

5.6 Anchor example: K_4 and a tetrahedron

The unit-weight complete graph K_4 has six edges. Any 2-versus-2 partition cuts $2 \cdot 2 = 4$ edges, and no cut can exceed four. Hence

$$\text{OPT}(K_4) = 4.$$

Consider

$$X^* = \frac{4}{3}I - \frac{1}{3}\mathbf{1}\mathbf{1}^\top. \quad (5.11)$$

It has diagonal entries 1, off-diagonal entries $-1/3$, eigenvalue 0 along $\mathbf{1}$, and eigenvalue $4/3$ on the three-dimensional orthogonal complement. Thus $X^* \succeq 0$ and is SDP feasible.

Every edge contributes

$$\frac{1 - X_{ij}^*}{2} = \frac{1 + 1/3}{2} = \frac{2}{3}.$$

With six edges,

$$\frac{1}{4} \langle L, X^* \rangle = 6 \cdot \frac{2}{3} = 4.$$

This feasible matrix proves $\text{SDP}(K_4) \geq 4$. To prove optimality, we still need an upper bound of four; the next subsection supplies it with a dual certificate.

The following unit vectors form a Gram factor:

$$\begin{aligned} v_1 &= \frac{1}{\sqrt{3}}(1, 1, 1)^\top, & v_2 &= \frac{1}{\sqrt{3}}(1, -1, -1)^\top, \\ v_3 &= \frac{1}{\sqrt{3}}(-1, 1, -1)^\top, & v_4 &= \frac{1}{\sqrt{3}}(-1, -1, 1)^\top. \end{aligned}$$

They point to the vertices of a regular tetrahedron and satisfy $v_i^\top v_j = -1/3$ for $i \neq j$.

5.6.1 A matching dual certificate

For K_4 ,

$$L = 4I - \mathbf{1}\mathbf{1}^\top.$$

Choose $y_i^* = 1$ for every vertex. The dual slack is

$$S^* = I - \frac{1}{4}L = \frac{1}{4}\mathbf{1}\mathbf{1}^\top \succeq 0,$$

and the dual objective is $\sum_i y_i^* = 4$. Moreover,

$$S^* X^* = 0$$

because $X^* \mathbf{1} = 0$. The tetrahedral primal matrix and rank-one dual slack therefore form a complete SDP certificate. Together with the cut of value four, the matching bounds prove

$$\text{OPT}(K_4) = \text{SDP}(K_4) = 4.$$

Common trap

Although the K_4 relaxation has zero value gap, X^* has rank three and is not itself a cut. Value exactness does not imply that every SDP optimizer is rank one. Rounding is still required to output a feasible partition.

5.7 Random-hyperplane rounding

From this point onward, condition on the graph and on the chosen SDP matrix X . The only new randomness is the rounding vector g . $\mathbb{P}(\cdot)$ and $\mathbb{E}[\cdot]$ therefore refer to repeated draws of this

vector unless another source of randomness is named explicitly. In particular, the expectation guarantee below is not an average over graphs or over solver errors.

Factor an SDP solution as

$$X = VV^\top,$$

where row i of V is $v_i^\top$. Draw

$$g \sim \mathcal{N}(0, I_r)$$

and define

$$\hat{x}_i = \text{sign}(v_i^\top g). \quad (5.12)$$

We define $\text{sign}(0) = +1$, so the algorithm is a total function. For a nonzero unit vector v_i and a nondegenerate Gaussian g , the event $v_i^\top g = 0$ has probability zero, so this convention does not alter the distributional guarantee. The hyperplane

$$\{z \mid g^\top z = 0\}$$

passes through the origin. The sign records which side contains v_i . Gaussian rotational invariance makes its orientation uniform; the length of g is irrelevant.

Algorithm: Goemans–Williamson rounding

Input: weighted graph G , feasible or optimal SDP matrix X .

1. Compute a Gram factor $X = VV^\top$.
2. Draw $g \sim \mathcal{N}(0, I_r)$.
3. Set $\hat{x}_i = \text{sign}(v_i^\top g)$.
4. Return the cut $S = \{i \mid \hat{x}_i = +1\}$.

In practice, repeat steps 2–4 and keep the best sampled cut. The theorem below concerns one sample in expectation; repetition improves the chance of observing a strong realization but does not change the theorem’s constant.

Lemma 5.6 (Random-hyperplane separation probability). *For two unit vectors with angle $\theta_{ij} \in [0, \pi]$,*

$$\mathbb{P}(\hat{x}_i \neq \hat{x}_j) = \frac{\theta_{ij}}{\pi}. \quad (5.13)$$

Proof. If $\theta_{ij} = 0$, then $v_i = v_j$ and the signs agree almost surely. If $\theta_{ij} = \pi$, then $v_j = -v_i$ and the signs differ almost surely. Thus the formula holds at both endpoints. Now assume $0 < \theta_{ij} < \pi$.

Only the projection of g onto the two-dimensional plane spanned by v_i and v_j affects their signs. Rotational invariance makes the direction of this projection uniform over angle 2π .

The signs differ exactly when the normal direction g lies in either of two angular sectors of width θ_{ij} . Their total angular measure is $2\theta_{ij}$, so

$$\mathbb{P}(\hat{x}_i \neq \hat{x}_j) = \frac{2\theta_{ij}}{2\pi} = \frac{\theta_{ij}}{\pi}.$$

□

Classroom activity: three-angle checkpoint for hyperplane rounding

Before using equation (5.13) in a proof, test it at three angles that can be pictured without calculation. The same check also compares the SDP contribution of one unit-weight edge with its probability of being cut.

Angle θ	Vector geometry	SDP edge value $\frac{1-\cos\theta}{2}$	Rounding probability $\frac{\theta}{\pi}$
0	identical	0	0
$\pi/2$	orthogonal	1/2	1/2
π	antipodal	1	1

Interpretation. Identical vectors always receive the same sign; antipodal vectors always receive opposite signs; and a uniformly oriented hyperplane separates orthogonal vectors half of the time. If a derived formula fails any of these checks, its angle range or normalization is wrong.

Therefore the expected rounded cut value is

$$\mathbb{E}[\text{Cut}(\hat{x})] = \sum_{\{i,j\} \in E} w_{ij} \frac{\theta_{ij}}{\pi}. \quad (5.14)$$

5.8 The Goemans–Williamson guarantee

The SDP values an edge through $1 - \cos \theta$, while rounding values it through θ . The approximation theorem follows from comparing these two scalar functions uniformly.

Definition 5.7 (Goemans–Williamson constant).

$$\alpha_{\text{GW}} := \min_{0 < \theta \leq \pi} \frac{\theta/\pi}{(1 - \cos \theta)/2} = \min_{-1 \leq t < 1} \frac{2 \arccos(t)}{\pi(1 - t)} \approx 0.878567. \quad (5.15)$$

The minimizing angle solves

$$1 - \cos \theta = \theta \sin \theta,$$

equivalently $\tan(\theta/2) = \theta$, and has $\cos \theta \approx -0.689$. The numerical value is not guessed from graph experiments; it is the worst scalar ratio over every possible pairwise angle.

Theorem 5.8 (Goemans–Williamson). *Let X^* solve the Max-Cut SDP and let $\hat{x}$ be produced by random-hyperplane rounding. Then*

$$\mathbb{E}[\text{Cut}(\hat{x})] \geq \alpha_{\text{GW}} \text{SDP}(G) \geq \alpha_{\text{GW}} \text{OPT}(G). \quad (5.16)$$

Moreover,

$$\alpha_{\text{GW}} \text{SDP}(G) \leq \text{OPT}(G) \leq \text{SDP}(G). \quad (5.17)$$

Proof note. Follow the scalar edgewise comparison and apply the expectation guarantee. Reproducing the minimization that produces the numerical constant is not required.

Proof. Let $\theta_{ij} = \arccos(X_{ij}^*)$. By the definition of α_{GW} , for every edge,

$$\frac{\theta_{ij}}{\pi} \geq \alpha_{\text{GW}} \frac{1 - \cos \theta_{ij}}{2}.$$

Multiplying by $w_{ij} \geq 0$, summing, and using equations (5.10) and (5.14) gives

$$\mathbb{E}[\text{Cut}(\hat{x})] \geq \alpha_{\text{GW}} \text{SDP}(G).$$

Because the SDP is a relaxation, $\text{SDP}(G) \geq \text{OPT}(G)$, which gives the second inequality in equation (5.16).

Every realization of $\hat{x}$ is a valid cut and therefore has value at most $\text{OPT}(G)$. Taking expectations gives

$$\text{OPT}(G) \geq \mathbb{E}[\text{Cut}(\hat{x})] \geq \alpha_{\text{GW}} \text{SDP}(G).$$

Together with proposition 5.5, this proves the sandwich. $\square$

Key result: read the theorem in the correct order

For rounding an optimal SDP matrix,

$$\alpha_{\text{GW}} \text{SDP}(G) \leq \mathbb{E}[\text{Cut}(\hat{x})] \leq \text{OPT}(G) \leq \text{SDP}(G).$$

Each realized cut satisfies $\text{Cut}(\hat{x}) \leq \text{OPT}(G)$, but may lie above or below the expectation. The guarantee is a lower bound on that expectation, not an equality or a guarantee for every trial.

Remark 5.9 (Rounding a feasible, not necessarily optimal, matrix). The same edgewise proof applies to any feasible X and gives

$$\mathbb{E}[\text{Cut}(\hat{x})] \geq \alpha_{\text{GW}} v(X).$$

If $v(X) \geq \text{SDP}(G) - \varepsilon$, this becomes $\mathbb{E}[\text{Cut}(\hat{x})] \geq \alpha_{\text{GW}}(\text{SDP}(G) - \varepsilon)$. An arbitrary feasible matrix need not give an optimum-relative guarantee: on K_4 , $X = J$ has identical Gram vectors and always rounds to a cut of value zero, although $\text{SDP}(K_4) = 4$.

5.8.1 What the theorem does and does not say

- It gives an *expected* guarantee for the randomized output.
- It gives a worst-case guarantee over all nonnegative weighted graphs.
- A single sample may fall below the expectation.
- The empirical ratio on one graph is not α_{GW} .
- The theorem does not say the relaxation gap is always $1/\alpha_{\text{GW}}$; many graphs, including K_4 , have no value gap.

5.9 Rounding the tetrahedron

In the K_4 embedding, every pair has angle

$$\theta = \arccos(-1/3) \approx 1.910633.$$

Thus

$$\mathbb{E}[\text{Cut}(\hat{x})] = 6 \frac{\arccos(-1/3)}{\pi} \approx 3.649.$$

Because $v_1 + \dots + v_4 = 0$, a nondegenerate hyperplane cannot put all four vectors on the same side. The rounded cut is therefore either:

- a 1-versus-3 cut of value 3, or
- a 2-versus-2 cut of value 4.

Let q be the probability of value 4. Then

$$3(1 - q) + 4q = 6 \frac{\arccos(-1/3)}{\pi},$$

so

$$q = 6 \frac{\arccos(-1/3)}{\pi} - 3 \approx 0.649.$$

One random hyperplane finds an exact maximum cut about 65% of the time; repeated independent rounding rapidly increases the chance that the best sample has value four.

Classroom activity: relax, embed, and round a triangle

For K_3 , consider the feasible Gram matrix with diagonal 1 and all off-diagonal entries $-1/2$.

1. Verify its eigenvalues $3/2, 3/2, 0$.
2. Draw three unit vectors separated by 120° .
3. Compute the SDP value and compare it with $\text{OPT}(K_3) = 2$.
4. Show that a generic hyperplane always produces a cut of value 2.
5. Explain why the SDP has a relaxation gap even though rounding is exact on every sample.

Checkpoint. The SDP value is $3 \cdot 3/4 = 9/4$, while every rounded cut has value 2.

Mastery check

You are ready to leave this chapter if you can:

1. derive $\text{Cut}(x) = x^\top Lx/4$ and its Gram-matrix relaxation;
2. derive the separation probability θ/π and use the edgewise comparison to prove the Goemans–Williamson bound;
3. distinguish an exact optimum, an SDP bound, an expected rounded value, and a sampled feasible value.

Summary and connections

- Lifting $\text{Cut}(x) = x^\top Lx/4$ to $X = xx^\top$ and dropping rank one gives an SDP upper bound on Max-Cut.
- A random hyperplane separates two unit vectors with probability equal to their angle divided by π .
- For nonnegative edge weights and an optimal SDP solution, rounding gives $\mathbb{E}[\text{Cut}(x)] \geq \alpha_{\text{GW}} \text{OPT}$, where $\alpha_{\text{GW}} \approx 0.878567$.
- A sampled cut gives a feasible lower bound; a dual feasible point gives an upper bound. An expectation guarantee is not a per-sample guarantee.

5.10 Chapter practice set

Exercise 5.1 (Binary formulation). For the consecutively labeled path P_4 with edges $\{\{1, 2\}, \{2, 3\}, \{3, 4\}\}$, write its Laplacian, enumerate distinct cuts up to global sign symmetry, and verify

$$\max_{x \in \{-1, +1\}^4} \frac{1}{4} x^\top L x = 3.$$

Exercise 5.2 (Triangle relaxation). For K_3 :

- (a) verify feasibility of the 120° Gram matrix;
- (b) compute its SDP value $9/4$;
- (c) construct a matching dual certificate;
- (d) calculate the relaxation ratio OPT / SDP .

Exercise 5.3 (Dual of the Max-Cut SDP). Starting from the Week 4 standard form, derive equation (5.9). For any X with $\text{diag}(X) = \mathbf{1}$ and any $y \in \mathbb{R}^n$, show directly that

$$\sum_i y_i - \frac{1}{4} \langle L, X \rangle = \langle \text{Diag}(y) - L/4, X \rangle.$$

Exercise 5.4 (Rounding probability). Give a complete planar proof of lemma 5.6 for $\theta = 0, \pi/2, \pi$, then generalize to arbitrary $\theta \in [0, \pi]$.

Exercise 5.5 (The scalar approximation ratio).

$$r(\theta) = \frac{2\theta}{\pi(1 - \cos \theta)}, \quad 0 < \theta \leq \pi.$$

Differentiate r and show that every interior stationary point satisfies

$$\tan(\theta/2) = \theta.$$

Using $\theta_\star \approx 2.3311$, verify that $r(\theta_\star) \approx 0.8786$.

Exercise 5.6 (Expectation versus best-of- R). For the regular-tetrahedral optimal Gram embedding of K_4 , suppose each independent rounding trial returns a maximum cut with probability $q \approx 0.649$.

- (a) What is the probability that at least one of R trials is optimal?
- (b) Find the smallest R making this probability at least 0.99.
- (c) Explain why this calculation is instance-specific and not a new worst-case approximation theorem.

Exercise 5.7 (Transfer: weighted graph). Let K_4 have edge $\{1, 2\}$ of weight 3 and all other edges of weight 1. Enumerate the exact optimum. Formulate its SDP and dual. Show that the heavy pair can be chosen antipodal in an optimal Gram embedding, describe any remaining nonuniqueness, and test the prediction computationally.

5.11 Complete solutions

Conventions and a factor check. Let $G = (V, E, w)$ be an undirected graph with nonnegative edge weights, and count every unordered edge $\{i, j\}$ exactly once. Its weighted adjacency matrix, degree matrix, and Laplacian are

$$W_{ij} = w_{ij}, \quad D_{ii} = \sum_{j: \{i, j\} \in E} w_{ij}, \quad L = D - W.$$

For a sign vector $x \in \{-1, +1\}^n$,

$$x^\top Lx = \sum_{\{i, j\} \in E} w_{ij} (x_i - x_j)^2 = 2 \sum_{\{i, j\} \in E} w_{ij} (1 - x_i x_j) = 4 \text{Cut}(x).$$

This verifies the factor

$$\text{Cut}(x) = \frac{1}{4} x^\top Lx.$$

After lifting $X = xx^\top$ and dropping the rank-one constraint, the primal and dual Max-Cut SDPs are

$$\begin{aligned} \text{SDP}(G) &= \max_{X \in \mathbb{S}^n} \left\{ \frac{1}{4} \langle L, X \rangle : X_{ii} = 1 \ (1 \leq i \leq n), \ X \succeq 0 \right\}, \\ &= \min_{y \in \mathbb{R}^n} \left\{ \sum_{i=1}^n y_i : \text{Diag}(y) - \frac{1}{4} L \succeq 0 \right\}. \end{aligned}$$

Here $\langle A, B \rangle = \text{Tr}(A^\top B)$, and $\text{Diag}(y)$ is the diagonal matrix whose diagonal is y . The equality of the primal and dual optimal values follows, for example, from Slater's condition because $X = I$ is strictly primal feasible. If $X_{ij} = v_i^\top v_j = \cos \theta_{ij}$ is a Gram representation by unit vectors, then

$$\frac{1}{4} \langle L, X \rangle = \sum_{\{i, j\} \in E} w_{ij} \frac{1 - \cos \theta_{ij}}{2}.$$

A random hyperplane separates v_i and v_j with probability θ_{ij}/π . For a positive-weight edge with $0 < \theta \leq \pi$, the edgewise ratio between expected rounded value and SDP value is exactly

$$\frac{\theta/\pi}{(1 - \cos \theta)/2} = \frac{2\theta}{\pi(1 - \cos \theta)}.$$

These identities fix all factors of two and four used below.

Solution to Exercise 5.1

Label the path consecutively, so that

$$V = \{1, 2, 3, 4\}, \quad E = \{\{1, 2\}, \{2, 3\}, \{3, 4\}\},$$

with unit edge weights. Then

$$L = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

The vectors x and $-x$ represent the same cut, so fixing $x_1 = +1$ gives one representative of each of the $2^{4-1} = 8$ cuts. Direct enumeration gives

(x_2, x_3, x_4)	crossed edges	$\frac{1}{4}x^\top Lx$
$(+, +, +)$	$\emptyset$	0
$(+, +, -)$	$\{3, 4\}$	1
$(+, -, +)$	$\{2, 3\}, \{3, 4\}$	2
$(+, -, -)$	$\{2, 3\}$	1
$(-, +, +)$	$\{1, 2\}, \{2, 3\}$	2
$(-, +, -)$	$\{1, 2\}, \{2, 3\}, \{3, 4\}$	3
$(-, -, +)$	$\{1, 2\}, \{3, 4\}$	2
$(-, -, -)$	$\{1, 2\}$	1

Thus the alternating vectors $(1, -1, 1, -1)^\top$ and its global negative cut all three edges. No cut can have value larger than the total weight 3, so

$$\max_{x \in \{-1, +1\}^4} \frac{1}{4}x^\top Lx = 3.$$

Solution to Exercise 5.2

The Gram matrix of three coplanar unit vectors separated pairwise by 120° is

$$X_{120} = \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{bmatrix} = \frac{3}{2}I - \frac{1}{2}\mathbf{1}\mathbf{1}^\top.$$

Its eigenvalue on $\text{span}\{\mathbf{1}\}$ is 0, and its two eigenvalues on $\mathbf{1}^\perp$ are $3/2$. Hence $X_{120} \succeq 0$; its diagonal entries are all one, so it is SDP feasible. Each of the three edges contributes

$$\frac{1 - X_{ij}}{2} = \frac{1 - (-1/2)}{2} = \frac{3}{4},$$

and therefore

$$\frac{1}{4}\langle L, X_{120} \rangle = 3 \cdot \frac{3}{4} = \frac{9}{4}.$$

For K_3 , $L = 3I - \mathbf{1}\mathbf{1}^\top$. Choose $y_i = 3/4$ for $i = 1, 2, 3$. Its dual slack is

$$S = \text{Diag}(y) - \frac{1}{4}L = \frac{3}{4}I - \frac{1}{4}(3I - \mathbf{1}\mathbf{1}^\top) = \frac{1}{4}\mathbf{1}\mathbf{1}^\top \succeq 0,$$

and the dual objective is $\sum_i y_i = 9/4$. The feasible primal and dual objectives agree, so weak duality certifies

$$\text{SDP}(K_3) = \frac{9}{4}.$$

Every nonconstant sign assignment on a triangle cuts exactly two edges, whereas a constant assignment cuts none. Thus $\text{OPT}(K_3) = 2$, and the relaxation ratio is

$$\frac{\text{OPT}(K_3)}{\text{SDP}(K_3)} = \frac{2}{9/4} = \frac{8}{9}.$$

Solution to Exercise 5.3

Write the primal in the standard form

$$\max \langle C, X \rangle \quad \text{subject to} \quad \langle E_{ii}, X \rangle = 1 \quad (1 \leq i \leq n), \quad X \succeq 0,$$

where $C = L/4$ and $E_{ii} = e_i e_i^\top$. With unrestricted equality multipliers $y_i \in \mathbb{R}$, the Lagrangian is

$$\begin{aligned} \mathcal{L}(X, y) &= \langle C, X \rangle + \sum_{i=1}^n y_i (1 - \langle E_{ii}, X \rangle) \\ &= \sum_{i=1}^n y_i + \langle C - \text{Diag}(y), X \rangle. \end{aligned}$$

For $X \succeq 0$, the supremum of this expression over X is finite if and only if

$$\text{Diag}(y) - C = \text{Diag}(y) - \frac{1}{4}L \succeq 0.$$

When this condition holds, the supremum is $\sum_i y_i$. Hence the dual is

$$\min_{y \in \mathbb{R}^n} \sum_{i=1}^n y_i \quad \text{subject to} \quad \text{Diag}(y) - \frac{1}{4}L \succeq 0.$$

For any symmetric X with $X_{ii} = 1$, irrespective of whether it is PSD,

$$\sum_i y_i = \sum_i y_i X_{ii} = \langle \text{Diag}(y), X \rangle.$$

Subtracting $\langle L, X \rangle/4$ gives the claimed identity. To conclude that this gap is nonnegative, additionally require $X \succeq 0$ and dual feasibility: $S = \text{Diag}(y) - L/4 \succeq 0$. Then

$$\langle S, X \rangle = \text{Tr}(S^{1/2} X S^{1/2}) \geq 0,$$

which is weak duality. The diagonal constraints alone prove the identity; positive semidefiniteness of both X and S proves nonnegativity.

Solution to Exercise 5.4

Let u, v be unit vectors with angle $\theta = \arccos(u^\top v)$, draw a rotationally invariant Gaussian g , and set

$$\hat{x}_u = \text{sign}(u^\top g), \quad \hat{x}_v = \text{sign}(v^\top g).$$

The zero inner-product events have probability zero.

For the three checkpoint angles:

- If $\theta = 0$, then $u = v$, so the two signs agree almost surely and the separation probability is 0.
- If $\theta = \pi$, then $v = -u$, so the two signs disagree almost surely and the separation probability is 1.
- If $\theta = \pi/2$, choose planar coordinates $u = e_1$ and $v = e_2$. The signs differ exactly when the projected normal lies in the second or fourth quadrant. These two quadrants have total angle π out of 2π , so the probability is $1/2$.

Now suppose $0 < \theta < \pi$ and let $P = \text{span}\{u, v\}$. Only the projection of g onto P affects the two signs. By rotational invariance, the direction of this nonzero projection is uniform on the circle. For a fixed vector, the directions producing a positive inner product form a semicircle. The two relevant semicircles have centers separated by angle θ ; their symmetric difference consists of two arcs, each of length θ . Hence the set of normal directions that give opposite signs has angular measure 2θ , and

$$\mathbb{P}(\hat{x}_u \neq \hat{x}_v) = \frac{2\theta}{2\pi} = \frac{\theta}{\pi}.$$

Together with the separately treated endpoints, this proves the formula on all of $[0, \pi]$.

Solution to Exercise 5.5

Differentiation gives

$$r'(\theta) = \frac{2}{\pi} \frac{1 - \cos \theta - \theta \sin \theta}{(1 - \cos \theta)^2}.$$

Hence an interior stationary point satisfies $1 - \cos \theta = \theta \sin \theta$. For $0 < \theta < \pi$, the half-angle identities and division by $2 \sin(\theta/2) \cos(\theta/2) > 0$ give

$$\tan(\theta/2) = \theta.$$

At the supplied value,

$$r(2.3311) = \frac{2(2.3311)}{\pi(1 - \cos(2.3311))} \approx 0.8786.$$

Thus the scalar constant used in the chapter is $\alpha_{\text{GW}} \approx 0.8786$.

Solution to Exercise 5.6

The probability that all R independent trials fail is $(1 - q)^R$. Therefore

$$\mathbb{P}\{\text{at least one optimal trial}\} = 1 - (1 - q)^R.$$

The inequality $1 - (1 - q)^R \geq 0.99$ is equivalent to

$$R \geq \frac{\log(0.01)}{\log(1 - q)}.$$

For the tetrahedral K_4 embedding, every pairwise angle is $\theta = \arccos(-1/3)$, so the expected number of cut edges is $6\theta/\pi$. Also $v_1 + v_2 + v_3 + v_4 = 0$, which rules out a generic zero-versus-four rounded cut; hence the realized cut value is either 3 or 4. If q denotes the probability of value 4, then $3(1 - q) + 4q = 6\theta/\pi$. Therefore the exact one-trial probability is

$$q = \frac{6 \arccos(-1/3)}{\pi} - 3 = 0.6490406878 \dots$$

It gives

$$\frac{\log(0.01)}{\log(1 - q)} = 4.398086 \dots, \quad R_{\min} = 5.$$

The adjacent integer checks are

$$1 - (1 - q)^4 = 0.98482855 \dots < 0.99, \quad 1 - (1 - q)^5 = 0.99467544 \dots > 0.99.$$

This is an instance-specific success probability for the regular tetrahedral optimal Gram embedding of K_4 . It is not uniform over all graphs, and it is not even intrinsic to every optimal SDP representation of K_4 : a rank-one optimal Gram matrix already encodes a maximum cut and rounds optimally with probability one. A worst-case approximation theorem must instead compare the expected cut value with $\text{OPT}(G)$ uniformly over every admissible graph. Repeating this particular rounding experiment does not establish such a uniform comparison.

Solution to Exercise 5.7

By relabeling vertices, take the heavy edge to be $\{1, 2\}$. Thus

$$W = \begin{bmatrix} 0 & 3 & 1 & 1 \\ 3 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 5 & -3 & -1 & -1 \\ -3 & 5 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}.$$

Fixing $x_1 = +1$ again removes global sign duplication. The complete enumeration is

(x_2, x_3, x_4)	$\text{Cut}_w(x)$
$(+, +, +)$	0
$(+, +, -)$	3
$(+, -, +)$	3
$(+, -, -)$	4
$(-, +, +)$	5
$(-, +, -)$	6
$(-, -, +)$	6
$(-, -, -)$	5

Therefore

$$\text{OPT}(G) = 6,$$

attained, up to global sign, by $(1, -1, 1, -1)^\top$ and $(1, -1, -1, 1)^\top$.

The primal and dual SDPs are, explicitly,

$$\begin{aligned} \max_{X \in \mathbb{S}^4} \quad & \frac{1}{4} \langle L, X \rangle = \frac{1}{2} \sum_{1 \leq i < j \leq 4} w_{ij} (1 - X_{ij}) \\ \text{subject to} \quad & X_{11} = X_{22} = X_{33} = X_{44} = 1, \quad X \succeq 0, \end{aligned}$$

and

$$\begin{aligned} \min_{y \in \mathbb{R}^4} \quad & y_1 + y_2 + y_3 + y_4 \\ \text{subject to} \quad & \text{Diag}(y) - \frac{1}{4}L \succeq 0. \end{aligned}$$

An explicit primal certificate is

$$X^* = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

It is the Gram matrix of

$$v_1 = e_1, \quad v_2 = -e_1, \quad v_3 = e_2, \quad v_4 = -e_2,$$

and has eigenvalues 0, 0, 2, 2. Hence it is feasible. The heavy edge contributes 3, edge {3, 4} contributes 1, and the four remaining edges contribute 1/2 each, giving

$$\frac{1}{4}\langle L, X^\star \rangle = 3 + 1 + 4 \cdot \frac{1}{2} = 6.$$

For a matching dual certificate, choose

$$y^\star = (2, 2, 1, 1)^\top.$$

The slack matrix is

$$S^\star = \text{Diag}(y^\star) - \frac{1}{4}L = \begin{bmatrix} \frac{3}{4} & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{3}{4} & \frac{3}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix}.$$

For every $z \in \mathbb{R}^4$, let $a = z_1 + z_2$ and $b = z_3 + z_4$. Then

$$z^\top S^\star z = \frac{1}{4}(3a^2 + 2ab + b^2) = \frac{1}{4}(2a^2 + (a + b)^2) \geq 0.$$

Thus $S^\star \succeq 0$, while the dual objective is $\sum_i y_i^\star = 6$. The primal and dual values match, so the exact certificate is

$$\boxed{\text{OPT}(G) = \text{SDP}(G) = 6}.$$

It also follows directly that $S^\star X^\star = 0$.

The optimal geometry can be characterized completely. If X is any optimal primal solution, equality of the certified primal and dual values implies $\langle S^\star, X \rangle = 0$. Since both matrices are PSD, this forces $S^\star X = 0$, so

$$\text{range}(X) \subseteq \ker S^\star = \text{span}\{(1, -1, 0, 0)^\top, (0, 0, 1, -1)^\top\}.$$

Imposing $X_{ii} = 1$ shows that every optimal Gram matrix has the form

$$X(t) = \begin{bmatrix} 1 & -1 & t & -t \\ -1 & 1 & -t & t \\ t & -t & 1 & -1 \\ -t & t & -1 & 1 \end{bmatrix}, \quad -1 \leq t \leq 1.$$

Its eigenvalues are 0, 0, $2(1 - t)$, $2(1 + t)$, so conversely every $X(t)$ in this parameter range is feasible and has value 6. One Gram realization is

$$v_1 = e_1, \quad v_2 = -e_1, \quad v_3 = te_1 + \sqrt{1 - t^2}e_2, \quad v_4 = -v_3.$$

Thus every optimal embedding makes the heavy pair antipodal. It also makes v_3 and v_4 antipodal, while the relative angle between the two axes is free. The symmetric choice $t = 0$ makes the axes orthogonal; the endpoint choices $t = \pm 1$ are the rank-one maximum cuts. Random-hyperplane rounding of any member of this family satisfies $\hat{x}_2 = -\hat{x}_1$ and $\hat{x}_4 = -\hat{x}_3$ almost surely, so it always returns one of the two maximum cuts.

The following solver-independent NumPy check was executed. It enumerates all cuts and verifies both PSD certificates, equality of objectives, and complementary slackness.

```
1 import itertools
2 import numpy as np
3
4 W = np.array([[0,3,1,1],
```

```

5         [3,0,1,1],
6         [1,1,0,1],
7         [1,1,1,0]], dtype=float)
8 L = np.diag(W.sum(axis=1)) - W
9
10 def cut_value(x):
11     return sum(W[i,j] for i in range(4) for j in range(i+1,4)
12               if x[i] != x[j])
13
14 xs = [np.array((1,)+tail)
15       for tail in itertools.product([1,-1], repeat=3)]
16 vals = [cut_value(x) for x in xs]
17
18 X = np.array([[ 1,-1, 0, 0],
19              [-1, 1, 0, 0],
20              [ 0, 0, 1,-1],
21              [ 0, 0,-1, 1]], dtype=float)
22 y = np.array([2,2,1,1], dtype=float)
23 S = np.diag(y) - L/4
24
25 def clean_eigs(A):
26     e = np.linalg.eigvalsh(A)
27     e[np.abs(e) < 1e-10] = 0
28     return np.round(e, 6)
29
30 print("cut values =", [int(v) for v in vals])
31 print("exact OPT =", int(max(vals)))
32 print("eig(X*) =", clean_eigs(X))
33 print("eig(S*) =", clean_eigs(S))
34 print("primal, dual =", np.sum(L*X)/4, y.sum())
35 print("||S*X*||_F =", np.linalg.norm(S@X))

```

The observed output is

```

1 cut values = [0, 3, 3, 4, 5, 6, 6, 5]
2 exact OPT = 6
3 eig(X*) = [0. 0. 2. 2.]
4 eig(S*) = [0.      0.      0.292893 1.707107]
5 primal, dual = 6.0 6.0
6 ||S*X*||_F = 0.0

```

The computation therefore confirms, rather than merely suggests, the analytic prediction about the heavy edge and the optimum.

Suggested reading. The relaxation and rounding analysis follows the original Goemans–Williamson paper [GW95]. The SDP formulation and duality conventions build on [VB96, BV04].

Historical note

Goemans and Williamson’s 1995 paper combined an SDP relaxation with random hyperplane rounding and proved the 0.87856-type expected guarantee studied in this chapter [GW95]. It became a landmark example of a broader approximation-algorithm strategy: optimize over a tractable continuous geometry, then round while controlling the loss.

Chapter 6

Shannon Capacity and the Lovász Theta Function

Why this chapter

Week 4 built the PSD language and Week 5 used it to approximate one NP-hard discrete objective: solve a convex relaxation, then round back to a feasible cut. Week 6 uses the same language for a different purpose. Here the output is an efficiently computable upper certificate for an asymptotic quantity, not a rounded code and not a general exact solution of an NP-hard problem.

The central question is:

How many messages can a noisy channel transmit with exactly zero probability of confusion?

The answer for one channel use is the independence number, whose computation is NP-hard in general. Repeated uses are harder still: they create exponentially growing strong graph powers, and Shannon capacity depends on their limiting growth rate. The Lovász theta function replaces this intractable sequence by one semidefinite program:

$$\text{channel} \longrightarrow \text{confusability graph} \longrightarrow \text{strong powers} \longrightarrow \text{Shannon capacity} \leq \vartheta(G).$$

The elegance is that product behavior lets one convex certificate control every block length. In general, $\vartheta(G)$ is an upper bound rather than the exact independence number or capacity. For the pentagon C_5 , a matching code construction and SDP certificate close the gap and prove $\Theta(C_5) = \vartheta(C_5) = \sqrt{5}$.

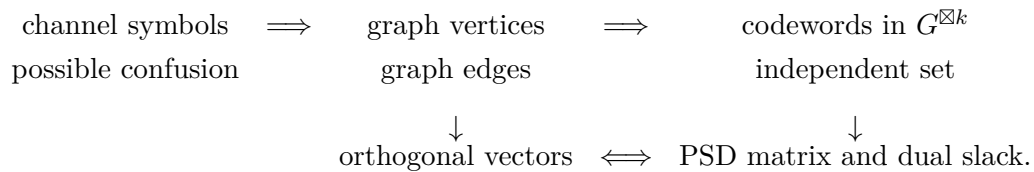
For AI, this is a reliability and representation-limit question. Once a noise, quantization, or perturbation model specifies which inputs can produce the same observation, no optimizer can recover distinctions erased by that observation process. The graph records possible confusions, a code constructs a reliably distinguishable subset, and theta certifies a limit on every such subset. This is a worst-case zero-error statement, not a statistical generalization theorem or a claim about average benchmark accuracy.

Learning objectives

By the end of this chapter, you should be able to:

1. formulate a confusability graph from a channel or fixed ambiguity model and connect one-shot zero-error codes to independent sets;
2. derive the strong-product model for repeated uses and define Shannon capacity;
3. formulate Lovász theta as an SDP and verify $\alpha(G) \leq \vartheta(G)$;
4. compute matching C_5 primal–dual solutions and, using multiplicativity, conclude $\Theta(C_5) = \sqrt{5}$.

6.1 The chapter's representation map



Object or notation	Meaning under this chapter's convention
edge $uv \in E(G)$	symbols u and v may be confused
independent set	pairwise distinguishable one-shot code
$G^{\boxtimes k}$	confusability graph for k channel uses
$\alpha(G)$	largest one-shot zero-error code
$\Theta(G)$	asymptotic zero-error growth rate
$\vartheta(G)$	efficiently checkable SDP upper bound
$\overline{G}$	complement graph; flips adjacency conventions

This chapter always uses edges for confusability; every imported theta formula must be translated to this convention before use.

Four objects will repeatedly describe the same restriction:

1. **Graph:** an edge records a pair that might be confused.
2. **Code:** every pair of selected words must be distinguishable.
3. **Handle vectors:** nonadjacent vertices receive orthogonal unit vectors.
4. **Primal matrix:** a different Gram representation uses zeros on edges and a trace normalization; duality connects the two forms.

Definition 6.1 (Graph notation used in this chapter). Throughout this chapter, graphs have at least one vertex. A finite simple graph $G = (V, E)$ has no loops or repeated edges. Vertices $i \neq j$ are adjacent when $\{i, j\} \in E$. An independent set $S \subseteq V$ contains no adjacent pair, and

$$\alpha(G) := \max \{|S| \mid S \subseteq V \text{ is independent}\}$$

is the independence number. The complement $\overline{G}$ has the same vertex set and reverses adjacency between distinct vertices. These conventions matter because some theta formulas are written for $\overline{G}$ rather than G .

Connection to AI practice: AI lens: when two inputs can look the same

Consider a finite collection of classes, commands, or prototypes. For each input i , let $B_i \subseteq \mathcal{Y}$ be the set of observations that the specified sensor, noise, or perturbation model might produce. If

$$B_i \cap B_j \neq \emptyset,$$

then some observation is compatible with both inputs. A decoder seeing that observation cannot guarantee whether i or j was intended, so we connect the two inputs by a confusability edge.

A set of inputs is guaranteed distinguishable exactly when its possible-output sets are pairwise disjoint—equivalently, when it is an independent set of the confusability graph. The sequence is

$$\begin{aligned} \text{noise or perturbation model} &\longrightarrow \text{possible-output sets } \{B_i\}, \\ \{B_i\} &\longrightarrow \text{confusability graph}, \\ \text{confusability graph} &\longrightarrow \text{zero-error code and capacity bound}. \end{aligned}$$

The conceptual shift for AI is that optimization can certify an information limit, not only choose model parameters. A better training algorithm cannot separate two inputs that the declared observation model makes identical; to improve the guarantee one must change the representation, sensing process, or allowable ambiguity model. Theta turns that limit into a checkable convex certificate.

This is an intentionally strict abstraction. It asks what is guaranteed for every allowed observation after the ambiguity model has been fixed. It does not by itself describe average test error, training from finite data, or distribution shift.

6.2 From a noisy channel to an independent set

6.2.1 Discrete memoryless channels

Definition 6.2 (Discrete memoryless channel). A discrete memoryless channel is specified by finite input and output alphabets $\mathcal{X}, \mathcal{Y}$ and transition probabilities

$$W(y | x) \geq 0, \quad \sum_{y \in \mathcal{Y}} W(y | x) = 1 \quad \text{for every } x \in \mathcal{X}.$$

For k independent channel uses,

$$W^{\otimes k}(y_1, \dots, y_k | x_1, \dots, x_k) = \prod_{\ell=1}^k W(y_\ell | x_\ell).$$

Ordinary channel capacity allows a decoding error probability that tends to zero with block length. Zero-error communication imposes a different and strictly stronger requirement: no output sequence may be compatible with two different codewords.

Definition 6.3 (Confusability graph). The confusability graph of W is the simple undirected graph

$$G_W = (\mathcal{X}, E),$$

where distinct inputs $x, x' \in \mathcal{X}$ are adjacent when there exists an output $y \in \mathcal{Y}$ such that

$$W(y | x) > 0 \quad \text{and} \quad W(y | x') > 0.$$

An edge therefore means “the receiver might see the same output from either input.”

This is precisely the set model above with

$$B_x = \{y \in \mathcal{Y} \mid W(y \mid x) > 0\}.$$

The channel notation adds probabilities, but its zero-error graph retains only which observations are possible.

Only the support pattern of W , not the positive probability values, matters for zero-error communication.

Proposition 6.4 (One-shot zero-error codes). *A set $\mathcal{C} \subseteq \mathcal{X}$ is a one-shot zero-error code if and only if it is an independent set of G_W . Consequently, the maximum number of one-shot messages is*

$$M_1(W) = \alpha(G_W).$$

Proof. If two code symbols are adjacent, some output can arise from either symbol, so no decoder can always distinguish them. Conversely, if no selected pair is adjacent, the sets of possible outputs generated by the selected symbols are pairwise disjoint. The decoder can therefore identify the transmitted symbol without error. $\square$

Worked example: a five-symbol cyclic channel

Let the inputs and outputs both be $\{0, 1, 2, 3, 4\}$, with arithmetic modulo 5, and suppose input i can produce only outputs i and $i + 1$, both with positive probability. Inputs i and $i + 1$ share output $i + 1$; similarly, i and $i - 1$ share output i . No other pair shares an output.

Thus the confusability graph is the pentagon C_5 . Its largest independent sets have size two, for example $\{0, 2\}$, so

$$M_1 = \alpha(C_5) = 2.$$

The precise probabilities assigned to the two possible outputs do not change this zero-error conclusion as long as both remain positive.

Common trap

An edge means *confusable*, not distinguishable. This convention makes zero-error codes independent sets. Some sources instead draw a distinguishability graph; every edge/nonedge statement must then be complemented.

6.3 Repeated uses and the strong graph product

One use of a channel can hide a coding gain that appears only when symbols are combined into longer words. The graph operation must reproduce coordinatewise confusability.

Definition 6.5 (Strong graph product). For simple graphs $G = (V, E)$ and $H = (U, F)$, the strong product $G \boxtimes H$ has vertex set $V \times U$. Two *distinct* vertices (v, u) and (v', u') are adjacent exactly when

$$(v = v' \text{ or } \{v, v'\} \in E) \quad \text{and} \quad (u = u' \text{ or } \{u, u'\} \in F).$$

The k -fold strong power is

$$G^{\boxtimes k} = \underbrace{G \boxtimes \cdots \boxtimes G}_{k \text{ factors}}.$$

Worked example: decoding three pairs in $C_5 \boxtimes C_5$

Reuse the labels 0, 1, 2, 3, 4 from the cyclic-channel example, so two symbols are adjacent in C_5 when they differ by ± 1 modulo 5. Check the strong-product condition coordinate by coordinate:

Pair of product vertices	First coordinate	Second coordinate	Product conclusion
(0, 0), (1, 1)	adjacent	adjacent	adjacent
(0, 0), (0, 1)	equal	adjacent	adjacent
(0, 0), (1, 2)	adjacent	nonadjacent	nonadjacent

The definition uses an *and* across coordinates. Product vertices are confusable only when every coordinate is equal or confusable. Conversely, one coordinate that is neither equal nor adjacent is enough to distinguish the two words. The vertices must also be distinct: because the graph is simple, (0, 0) is not adjacent to itself.

Two length- k input words are confusable precisely when, in every coordinate, their symbols are either equal or confusable. Therefore the confusability graph for k channel uses is $G_W^{\boxtimes k}$, and

$$M_k(W) = \alpha(G_W^{\boxtimes k}).$$

Definition 6.6 (Zero-error rate and Shannon capacity). Using logarithms to base 2, the zero-error capacity in bits per channel use is

$$C_0(G) = \sup_{k \geq 1} \frac{1}{k} \log_2 \alpha(G^{\boxtimes k}).$$

It is convenient to exponentiate this rate and define the Shannon capacity

$$\Theta(G) = 2^{C_0(G)} = \sup_{k \geq 1} \alpha(G^{\boxtimes k})^{1/k}. \quad (6.1)$$

Thus $C_0(G) = \log_2 \Theta(G)$.

Theorem 6.7 (Fekete's lemma, superadditive form). *If a real sequence $(a_k)_{k \geq 1}$ satisfies $a_{k+\ell} \geq a_k + a_\ell$ for all $k, \ell \geq 1$, then*

$$\lim_{k \rightarrow \infty} \frac{a_k}{k} = \sup_{k \geq 1} \frac{a_k}{k},$$

with the common value allowed to be $+\infty$. Thus a superadditive sequence has a well-defined asymptotic growth rate even when the individual terms do not form a monotone sequence.

Proposition 6.8 (Why the asymptotic limit exists). *For every finite nonempty graph G ,*

$$\Theta(G) = \lim_{k \rightarrow \infty} \alpha(G^{\boxtimes k})^{1/k}.$$

Proof. If S and T are independent sets in $G^{\boxtimes k}$ and $G^{\boxtimes \ell}$, respectively, then $S \times T$ is independent in $G^{\boxtimes (k+\ell)}$. Hence

$$\alpha(G^{\boxtimes (k+\ell)}) \geq \alpha(G^{\boxtimes k}) \alpha(G^{\boxtimes \ell}).$$

The sequence

$$a_k = \log \alpha(G^{\boxtimes k})$$

is therefore superadditive. Fekete's lemma gives

$$\lim_{k \rightarrow \infty} \frac{a_k}{k} = \sup_{k \geq 1} \frac{a_k}{k}.$$

Exponentiating proves the result. □

Worked example: a five-word code in $C_5^{\boxtimes 2}$

Label the vertices of C_5 by $\mathbb{Z}_5$, with i adjacent to $i \pm 1$. Consider

$$\mathcal{C}_2 = \{(i, 2i \bmod 5) \mid i \in \mathbb{Z}_5\}.$$

Take two distinct codewords indexed by i, j , and let $d = i - j \not\equiv 0 \pmod{5}$.

- If $d \equiv \pm 1$, then the first coordinates are confusable, but the second-coordinate difference is $2d \equiv \pm 2$, which is nonadjacent in C_5 .
- If $d \equiv \pm 2$, then the first coordinates are already nonadjacent.

In every pair, at least one coordinate distinguishes the two words, so $\mathcal{C}_2$ is independent and

$$\alpha(C_5^{\boxtimes 2}) \geq 5.$$

Therefore

$$\Theta(C_5) \geq 5^{1/2} = \sqrt{5} > \alpha(C_5) = 2.$$

Two channel uses are genuinely better than repeating the best one-shot code, which would provide only $2^2 = 4$ words.

Key result: why direct computation is not a satisfactory strategy

The definition of $\Theta(G)$ asks for maximum independent sets in an unbounded sequence of exponentially growing graph powers. Even if the first few powers are solved, no finite computation alone proves that a later power will not improve the normalized rate. We need an upper bound that is:

1. at least $\alpha(G)$;
2. multiplicative under $\boxtimes$;
3. computationally tractable.

The Lovász theta function is designed to satisfy exactly these three needs.

6.4 Lovász theta: fix the convention first

The literature contains several equivalent theta formulations, often with a graph complement inserted into one of them. In this chapter, G is always the *confusability graph*, and we adopt the convention

$$\alpha(G) \leq \vartheta(G), \quad \vartheta(\overline{K_n}) = n, \quad \vartheta(K_n) = 1. \quad (6.2)$$

6.4.1 Orthonormal representations

Definition 6.9 (Orthonormal representation). An orthonormal representation of a graph $G = (V, E)$ is a collection of unit vectors $\{u_i : i \in V\}$ in some Euclidean space such that

$$u_i^\top u_j = 0 \quad \text{whenever } i \neq j \text{ and } \{i, j\} \notin E. \quad (6.3)$$

Thus *nonadjacent* vertices, which are distinguishable symbols, receive orthogonal vectors.

Theorem 6.10 (Handle formulation of Lovász theta). *Let c be a unit vector called the handle. Then*

$$\vartheta(G) = \min_{\substack{\|c\|_2=1 \\ \{u_i\} \text{ an orthonormal} \\ \text{representation of } G}} \max_{i \in V} \frac{1}{(c^\top u_i)^2}, \quad (6.4)$$

where a zero denominator is interpreted as $+\infty$.

Proof note. For this quoted theorem, know the convention, run the empty/complete-graph checks, and apply the formula. A full proof is not required.

For the empty graph, the u_i must be pairwise orthogonal; a balanced handle gives value n . For the complete graph, all u_i may equal c , giving value 1. These two checks confirm [equation \(6.2\)](#).

Proposition 6.11 (Theta dominates the independence number). *For every finite graph G ,*

$$\alpha(G) \leq \vartheta(G).$$

Proof. Let S be an independent set of size $r \geq 1$. In every orthonormal representation, the vectors $\{u_i : i \in S\}$ are pairwise orthogonal. Bessel's inequality gives

$$\sum_{i \in S} (c^\top u_i)^2 \leq \|c\|_2^2 = 1.$$

At least one $i \in S$ must therefore satisfy

$$(c^\top u_i)^2 \leq \frac{1}{r}.$$

Hence $\max_i (c^\top u_i)^{-2} \geq r$. Minimizing over the representation and handle gives $\vartheta(G) \geq r$, and then maximize over independent sets. $\square$

6.5 The semidefinite program

Let $G = (V, E)$ have n vertices. Write $J = \mathbf{1}\mathbf{1}^\top \in \mathbb{S}^n$, where $\mathbf{1}$ is the all-ones vector.

Theorem 6.12 (Primal theta SDP). *The handle definition in [equation \(6.4\)](#) is equivalent to*

$$\begin{aligned} \vartheta(G) = \underset{X \in \mathbb{S}^n}{\text{maximize}} \quad & \langle J, X \rangle \\ \text{subject to} \quad & \text{Tr}(X) = 1, \\ & X_{ij} = 0, \quad \{i, j\} \in E, \\ & X \succeq 0. \end{aligned} \quad (6.5)$$

Proof note. Proof roadmap only; verify feasible Gram matrices, graph-convention constraints, and objective values. The full handle-SDP equivalence is optional.

Main idea. Both formulations optimize over Gram geometry, but use different vector representations. The handle formulation uses unit vectors orthogonal on nonedges and controls their worst overlap with one direction. The primal SDP uses a trace-one Gram matrix with zeros on edges and maximizes total correlation $\langle J, X \rangle = \mathbf{1}^\top X \mathbf{1}$.

These zero patterns cannot be interchanged by nonzero diagonal scaling of the same Gram matrix. The equivalence uses SDP duality and the relation between representations of a graph and its complement [Lov79]; it is quoted here rather than proved in full. For this course, the operational checks are:

1. a feasible $X \succeq 0$ is a Gram matrix;
2. edge constraints enforce the selected graph convention;
3. trace one removes arbitrary scaling;
4. the empty- and complete-graph values match [equation \(6.2\)](#).

Proposition 6.13 (Independent sets are primal certificates). *If $S \subseteq V$ is an independent set of size $r \geq 1$, define*

$$X_S = \frac{1}{r} \mathbf{1}_S \mathbf{1}_S^\top,$$

where $\mathbf{1}_S \in \mathbb{R}^n$ is the indicator vector of S . Then X_S is feasible in [equation \(6.5\)](#) and

$$\langle J, X_S \rangle = r.$$

Proof. The matrix is rank-one PSD and

$$\text{Tr}(X_S) = \frac{1}{r} \|\mathbf{1}_S\|_2^2 = 1.$$

If $\{i, j\} \in E$, an independent set cannot contain both endpoints, so $(X_S)_{ij} = 0$. Finally,

$$\langle J, X_S \rangle = \frac{1}{r} (\mathbf{1}^\top \mathbf{1}_S)^2 = \frac{r^2}{r} = r.$$

□

This supplies a second, purely matrix-based proof of $\alpha(G) \leq \vartheta(G)$.

6.5.1 Deriving the dual

For an edge $e = \{i, j\}$, define

$$H_e = e_i e_j^\top + e_j e_i^\top.$$

Then

$$\langle H_e, X \rangle = 2X_{ij}.$$

Let $t \in \mathbb{R}$ be the multiplier for $\text{Tr}(X) = 1$, and let $y_e \in \mathbb{R}$ be the unrestricted multiplier for $\langle H_e, X \rangle = 0$. The Lagrangian can be written

$$\begin{aligned} \mathcal{L}(X, t, y) &= \langle J, X \rangle + t(1 - \langle I, X \rangle) - \sum_{e \in E} y_e \langle H_e, X \rangle \\ &= t + \left\langle J - tI - \sum_{e \in E} y_e H_e, X \right\rangle. \end{aligned}$$

The supremum over $X \succeq 0$ is finite exactly when

$$tI + \sum_{e \in E} y_e H_e - J \succeq 0.$$

Therefore the dual is

$$\begin{aligned} &\underset{t \in \mathbb{R}, y \in \mathbb{R}^{|E|}}{\text{minimize}} && t \\ &\text{subject to} && S(t, y) := tI + \sum_{e \in E} y_e H_e - J \succeq 0. \end{aligned} \tag{6.6}$$

Proposition 6.14 (The theta SDP satisfies strong duality). *For every finite graph, the primal in equation (6.5) attains its optimum, the dual in equation (6.6) attains its optimum, and their optimal values are equal.*

Proof. The primal feasible set is nonempty because I/n has trace one, is positive definite, and has zero off-diagonal entries on every edge. It is closed. It is also bounded: if $X \succeq 0$ and $\text{Tr}(X) = 1$, its eigenvalues are nonnegative and sum to one, so

$$\|X\|_F^2 = \sum_i \lambda_i(X)^2 \leq \left(\sum_i \lambda_i(X) \right)^2 = 1.$$

Thus the feasible set is compact and the linear primal objective attains its maximum. Moreover, $I/n \succ 0$ lies in the interior of the PSD cone while satisfying every affine equality; equality constraints themselves do not need a “strict” version. The SDP Slater theorem from Week 4 therefore gives zero duality gap and dual attainment. $\square$

Proposition 6.15 (Certificate identity). *For every primal-feasible X and dual-feasible (t, y) ,*

$$t - \langle J, X \rangle = \langle S(t, y), X \rangle \geq 0. \quad (6.7)$$

Proof. Use $\text{Tr}(X) = 1$ and the edge equalities:

$$\begin{aligned} \langle S(t, y), X \rangle &= t \langle I, X \rangle + \sum_{e \in E} y_e \langle H_e, X \rangle - \langle J, X \rangle \\ &= t - \langle J, X \rangle. \end{aligned}$$

The last quantity is nonnegative because

$$\langle S, X \rangle = \text{Tr}(S^{1/2} X S^{1/2}) \geq 0;$$

the matrix inside the trace is PSD. $\square$

Thus a primal matrix gives a lower bound on $\vartheta(G)$, while a dual PSD slack gives an upper bound. Matching values prove optimality without trusting a solver’s status message.

6.6 Why theta bounds Shannon capacity

Theorem 6.16 (Multiplicativity). *For finite graphs G, H ,*

$$\vartheta(G \boxtimes H) = \vartheta(G)\vartheta(H). \quad (6.8)$$

Proof note. Know the statement, assumptions, graph convention, and where it enters the capacity bound. The complete tensor-product proof is enrichment and is not a closed-note derivation target.

Proof structure. The two inequalities use the two representations of theta.

Primal direction. If X_G, X_H are primal feasible, then

$$X_G \otimes X_H \succeq 0, \quad \text{Tr}(X_G \otimes X_H) = 1.$$

An edge of $G \boxtimes H$ contains an edge in at least one coordinate, so the corresponding entry of $X_G \otimes X_H$ is zero. Moreover,

$$\langle J_G \otimes J_H, X_G \otimes X_H \rangle = \langle J_G, X_G \rangle \langle J_H, X_H \rangle.$$

Taking optimal matrices gives

$$\vartheta(G \boxtimes H) \geq \vartheta(G)\vartheta(H).$$

Handle direction. Take orthonormal representations $\{u_i\}, \{v_j\}$ with handles c, d . Assign

$$(i, j) \mapsto u_i \otimes v_j, \quad \text{handle } c \otimes d.$$

If two product vertices are nonadjacent, at least one coordinate is a distinct nonedge, so the corresponding tensor vectors are orthogonal. Also

$$((c \otimes d)^\top (u_i \otimes v_j))^2 = (c^\top u_i)^2 (d^\top v_j)^2.$$

For any two chosen representations, the product objective is the product of their two objectives. Taking infima (equivalently, using arbitrarily near-optimal choices) gives the reverse inequality. $\square$

Theorem 6.17 (Lovász upper bound on Shannon capacity). *For every finite graph G ,*

$$\boxed{\alpha(G) \leq \Theta(G) \leq \vartheta(G)}. \quad (6.9)$$

Proof. The first inequality follows by using Cartesian products of a one-shot independent set:

$$\alpha(G^{\boxtimes k}) \geq \alpha(G)^k.$$

For the second inequality, apply [proposition 6.11](#) and [theorem 6.16](#):

$$\alpha(G^{\boxtimes k}) \leq \vartheta(G^{\boxtimes k}) = \vartheta(G)^k.$$

Taking k -th roots and the supremum over k proves the result. $\square$

Key result: the role of multiplicativity

A one-shot upper bound $U(G) \geq \alpha(G)$ does not automatically bound $\Theta(G)$. It must remain controlled on every graph power. Theta succeeds because

$$\alpha(G^{\boxtimes k}) \leq \vartheta(G^{\boxtimes k}) = \vartheta(G)^k.$$

Multiplicativity converts one SDP on G into an upper bound for infinitely many coding problems.

6.7 Anchor example: exact capacity of the pentagon

Let $A \in \mathbb{S}^5$ be the adjacency matrix of C_5 , with vertices $\mathbb{Z}_5$ and edges $\{i, i+1\}$. Define

$$J = \mathbf{1}\mathbf{1}^\top, \quad B = J - I - A, \quad \phi = \frac{1 + \sqrt{5}}{2}.$$

The matrix B is the adjacency matrix of the complement of C_5 , which is another five-cycle.

6.7.1 The primal certificate

Consider

$$X^* = \frac{1}{5}I + \frac{1}{5\phi}B. \quad (6.10)$$

Its trace is one. Because B has zeros on the edges of C_5 , $X_{ij}^* = 0$ for every $\{i, j\} \in E(C_5)$.

The eigenvalues of A are

$$2, \quad \frac{1}{\phi}, \quad \frac{1}{\phi}, \quad -\phi, \quad -\phi. \quad (6.11)$$

Along $\mathbf{1}$, B has eigenvalue 2. On the orthogonal complement, $J = 0$, so $B = -I - A$. Therefore the B -eigenvalues corresponding to equation (6.11) are

$$2, \quad -\phi, \quad -\phi, \quad \frac{1}{\phi}, \quad \frac{1}{\phi}.$$

The eigenvalues of X^* are consequently nonnegative:

$$\frac{1}{5} + \frac{2}{5\phi}, \quad 0, \quad 0, \quad \frac{1}{5} + \frac{1}{5\phi^2}, \quad \frac{1}{5} + \frac{1}{5\phi^2}.$$

Thus $X^* \succeq 0$. Its objective value is

$$\langle J, X^* \rangle = 1 + \frac{2}{\phi} = \sqrt{5}. \quad (6.12)$$

This proves $\vartheta(C_5) \geq \sqrt{5}$.

6.7.2 The dual certificate

By rotational symmetry, assign the same multiplier to every edge:

$$t^* = \sqrt{5}, \quad y^* = \frac{\sqrt{5}}{\phi} = \frac{5 - \sqrt{5}}{2}.$$

Because $\sum_{e \in E(C_5)} H_e = A$, the dual slack is

$$S^* = \sqrt{5}I + y^*A - J. \quad (6.13)$$

On the three A -eigenspaces, the eigenvalues of S^* are:

subspace	$\lambda(A)$	$\lambda(S^*)$
$\text{span}\{\mathbf{1}\}$	2	$\sqrt{5} + 2y^* - 5 = 0$
$\mathbf{1}^\perp$, first pair	$1/\phi$	$\sqrt{5} + y^*/\phi > 0$
$\mathbf{1}^\perp$, second pair	$-\phi$	$\sqrt{5} - \phi y^* = 0$.

Hence $S^* \succeq 0$, so dual feasibility gives

$$\vartheta(C_5) \leq t^* = \sqrt{5}.$$

Together with equation (6.12),

$$\boxed{\vartheta(C_5) = \sqrt{5}.} \quad (6.14)$$

The positive eigenspaces of X^* are exactly zero eigenspaces of S^* , so

$$S^*X^* = 0.$$

This is complementary slackness in a form that can be checked directly.

6.7.3 Representation switch: the pentagon umbrella

The same value has a three-dimensional geometric certificate. For $k \in \mathbb{Z}_5$, let

$$u_k = \begin{bmatrix} \cos \beta \\ \sin \beta \cos(2\pi k/5) \\ \sin \beta \sin(2\pi k/5) \end{bmatrix}, \quad c = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Nonadjacent vertices of C_5 differ by 2 modulo 5. Their inner product is

$$u_k^\top u_{k+2} = \cos^2 \beta + \sin^2 \beta \cos(4\pi/5).$$

Choose β so this equals zero. Solving gives

$$\cos^2 \beta = \frac{1}{\sqrt{5}}.$$

The vectors form an orthonormal representation and every handle overlap is

$$(c^\top u_k)^2 = \cos^2 \beta = \frac{1}{\sqrt{5}}.$$

The handle objective is therefore $\sqrt{5}$, matching the SDP certificates.

Closing the capacity proof

The code in [section 6.3](#) and the theta bound give

$$\sqrt{5} \leq \Theta(C_5) \leq \vartheta(C_5) = \sqrt{5}.$$

Therefore

$$\boxed{\Theta(C_5) = \sqrt{5}, \quad C_0(C_5) = \frac{1}{2} \log_2 5.} \quad (6.15)$$

6.8 Chapter synthesis

Mastery check

You are ready to leave this chapter if you can reconstruct the chain

$$\alpha(G) \leq \alpha(G^{\boxtimes k})^{1/k} \leq \vartheta(G^{\boxtimes k})^{1/k} = \vartheta(G)$$

and can answer all of the following:

1. Why do zero-error codes correspond to independent sets in strong graph powers?
2. Which theta entries are zero, and what do primal and dual feasible points certify?
3. How do multiplicativity, the five-word code, and the SDP certificate determine $\Theta(C_5)$?

Summary and connections

- Confusability edges join inputs sharing a possible output. Zero-error codes are independent sets; repeated uses give strong graph powers.
- Shannon capacity is the asymptotic independent-set growth factor $\Theta(G) = \sup_k \alpha(G^{\boxtimes k})^{1/k}$.
- Theta bounds the independence number and is multiplicative under strong products, giving $\Theta(G) \leq \vartheta(G)$.
- For C_5 , a five-word two-use code and matching primal–dual SDP certificates meet at $\Theta(C_5) = \vartheta(C_5) = \sqrt{5}$.
- A zero-error guarantee for a fixed ambiguity graph does not establish statistical generalization or average-case accuracy.

6.9 Chapter practice set

Exercise 6.1 (Recall: graph convention). For each of the channels below, draw the confusability graph and compute the maximum one-shot zero-error code size.

- Every input has a disjoint set of possible outputs.
- Every pair of inputs shares at least one possible output.
- Input $i \in \mathbb{Z}_4$ can produce outputs i and $i + 1$.
- Four inputs have possible-output sets

$$B_1 = \{a, b\}, \quad B_2 = \{b, c\}, \quad B_3 = \{d\}, \quad B_4 = \{c, d\}.$$

Check your answers against $\vartheta(\overline{K_n}) = n$ and $\vartheta(K_n) = 1$.

Exercise 6.2 (Derive: supermultiplicativity). Let S and T be independent sets of $G^{\boxtimes k}$ and $G^{\boxtimes \ell}$. Prove directly from the strong-product adjacency definition that $S \times T$ is independent. Then derive [proposition 6.8](#).

Exercise 6.3 (Compute: the pentagon code). List all five codewords $(i, 2i \bmod 5)$. Create a 5×5 table whose (i, j) entry records a coordinate that distinguishes codewords i and j . Explain why this table is a complete zero-error proof.

Exercise 6.4 (Derive: the theta dual). Starting from [equation \(6.5\)](#), reproduce the Lagrangian derivation of [equation \(6.6\)](#). Explain why the edge multipliers are unrestricted in sign and why the PSD slack direction is $S \succeq 0$.

Exercise 6.5 (Compute: C_5 spectra). Use

$$\lambda_k(A) = 2 \cos(2\pi k/5), \quad k = 0, \dots, 4,$$

to derive [equation \(6.11\)](#). Then compute the spectra of B, X^*, S^* without numerical software.

Exercise 6.6 (Interpret: complementary slackness). Show algebraically that $S^*X^* = 0$. Identify which eigenspaces support the primal matrix and which eigenspace carries the positive dual slack. Explain why $\langle S^*, X^* \rangle = 0$ proves zero duality gap.

Exercise 6.7 (Transfer: complete and empty graphs). Construct explicit primal and dual theta certificates for K_n and $\overline{K_n}$. Confirm

$$\vartheta(K_n) = 1, \quad \vartheta(\overline{K_n}) = n.$$

Then interpret both values as zero-error communication limits.

6.10 Complete solutions

Conventions and notation. Throughout, graphs are finite and simple, and an edge of a confusability graph means that the two corresponding channel inputs can be confused. Thus zero-error codes are independent sets. For distinct vertices $x = (x_1, \dots, x_k)$ and $x' = (x'_1, \dots, x'_k)$, the strong-power adjacency rule is

$$x \sim_{G^{\boxtimes k}} x' \iff \text{for every } r \in \{1, \dots, k\}, \quad x_r = x'_r \text{ or } \{x_r, x'_r\} \in E(G). \quad (6.16)$$

In particular, one coordinate containing a distinct nonedge is a certificate that two words are distinguishable.

We use the chapter's theta convention

$$\vartheta(G) = \max \{ \langle J, X \rangle : \text{Tr}(X) = 1, X_{ij} = 0 \text{ } (\{i, j\} \in E(G)), X \succeq 0 \}, \quad (6.17)$$

where $J = \mathbf{1}\mathbf{1}^\top$ and $\langle U, V \rangle = \text{Tr}(U^\top V)$. If $e = \{i, j\}$ is an unordered edge, set

$$H_e = e_i e_j^\top + e_j e_i^\top, \quad \langle H_e, X \rangle = 2X_{ij}. \quad (6.18)$$

The associated dual is

$$\vartheta(G) = \min_{t \in \mathbb{R}, y \in \mathbb{R}^{E(G)}} \left\{ t : S(t, y) := tI + \sum_{e \in E(G)} y_e H_e - J \succeq 0 \right\}. \quad (6.19)$$

Consequently, this convention gives $\vartheta(K_n) = 1$ and $\vartheta(\overline{K}_n) = n$.

Solution to Exercise 6.1

Two distinct inputs i, j are adjacent precisely when $B_i \cap B_j \neq \emptyset$.

(a) All intersections are empty, so the graph is $\overline{K}_n$. Every vertex may be used, and hence

$$M_1 = \alpha(\overline{K}_n) = n.$$

This agrees with $\vartheta(\overline{K}_n) = n$.

(b) Every distinct pair is adjacent, so the graph is K_n . An independent set contains at most one vertex, and therefore

$$M_1 = \alpha(K_n) = 1,$$

in agreement with $\vartheta(K_n) = 1$.

(c) Here $B_i = \{i, i+1\}$ with indices modulo 4. The only intersections between distinct sets occur for indices differing by ± 1 . Thus the graph is the cycle

$$C_4 : \quad 0 - 1 - 2 - 3 - 0.$$

The sets $\{0, 2\}$ and $\{1, 3\}$ are independent. No independent set can contain three vertices, since any three vertices of C_4 contain two consecutive vertices. Hence $M_1 = \alpha(C_4) = 2$.

(d) The nonempty pairwise intersections are exactly

$$B_1 \cap B_2 = \{b\}, \quad B_2 \cap B_4 = \{c\}, \quad B_4 \cap B_3 = \{d\}.$$

The graph is therefore the path $1 - 2 - 4 - 3$. An independent set takes at most one endpoint from each of the disjoint edges $\{1, 2\}$ and $\{3, 4\}$, so it has size at most two. This is attained, for example, by $\{1, 3\}$ (also by $\{1, 4\}$ or $\{2, 3\}$). Therefore $M_1 = 2$.

Solution to Exercise 6.2

Identify a pair $(s, t) \in S \times T$ with the concatenated word of length $k + \ell$. Consider two distinct pairs (s, t) and (s', t') .

If $s \neq s'$, independence of S says that s and s' are not adjacent in $G^{\boxtimes k}$. By (6.16), some coordinate $r \leq k$ therefore has $s_r \neq s'_r$ and $\{s_r, s'_r\} \notin E(G)$. That coordinate remains a distinct nonedge in the concatenated words, so the two concatenated words are nonadjacent in $G^{\boxtimes(k+\ell)}$. If instead $s = s'$, distinctness of the pairs forces $t \neq t'$, and the same argument applied to T supplies a distinguishing coordinate among the final ℓ coordinates. Thus every two distinct elements of $S \times T$ are nonadjacent, so $S \times T$ is independent.

Taking S and T to be maximum independent sets gives

$$\alpha(G^{\boxtimes(k+\ell)}) \geq \alpha(G^{\boxtimes k})\alpha(G^{\boxtimes \ell}). \quad (6.20)$$

For a finite nonempty graph, let $a_m = \log \alpha(G^{\boxtimes m})$. Equation (6.20) says $a_{k+\ell} \geq a_k + a_\ell$. Moreover, $1 \leq \alpha(G^{\boxtimes m}) \leq |V(G)|^m$, so every a_m is finite and the normalized sequence is bounded above. Fekete's lemma therefore yields

$$\lim_{m \rightarrow \infty} \frac{a_m}{m} = \sup_{m \geq 1} \frac{a_m}{m}.$$

Exponentiating gives the claimed asymptotic formula

$$\lim_{m \rightarrow \infty} \alpha(G^{\boxtimes m})^{1/m} = \sup_{m \geq 1} \alpha(G^{\boxtimes m})^{1/m} = \Theta(G).$$

Solution to Exercise 6.3

Label C_5 by $\mathbb{Z}_5$, with $a \sim b$ when $a - b \equiv \pm 1 \pmod{5}$. The five codewords are

$$c_0 = (0, 0), \quad c_1 = (1, 2), \quad c_2 = (2, 4), \quad c_3 = (3, 1), \quad c_4 = (4, 3).$$

In the following table, an entry 1 means that the first coordinate is a distinct nonedge of C_5 , and an entry 2 means that the second coordinate is a distinct nonedge. The dash marks comparison of a word with itself.

	c_0	c_1	c_2	c_3	c_4
c_0	—	2	1	1	2
c_1	2	—	2	1	1
c_2	1	2	—	2	1
c_3	1	1	2	—	2
c_4	2	1	1	2	—

To verify every entry at once, take $i \neq j$ and put $d = i - j \pmod{5}$. If $d \equiv \pm 2$, then the first coordinates are distinct and nonadjacent, so coordinate 1 distinguishes the words. If $d \equiv \pm 1$, then the second-coordinate difference is $2d \equiv \pm 2$, so coordinate 2 distinguishes them. These are all four possible nonzero residues modulo 5. The table therefore covers all ten unordered pairs of codewords. By the strong-product rule, each pair is nonadjacent in $C_5 \boxtimes C_5$, so the five words form an independent set and prove $\alpha(C_5^{\boxtimes 2}) \geq 5$.

Solution to Exercise 6.4

Write every edge constraint as $\langle H_e, X \rangle = 0$, counting each undirected edge once. Let $t \in \mathbb{R}$ multiply $1 - \langle I, X \rangle = 0$, and choose the sign convention in which $y_e \in \mathbb{R}$ multiplies $-\langle H_e, X \rangle = 0$. The Lagrangian is

$$\begin{aligned} L(X; t, y) &= \langle J, X \rangle + t(1 - \langle I, X \rangle) - \sum_{e \in E(G)} y_e \langle H_e, X \rangle \\ &= t + \left\langle J - tI - \sum_{e \in E(G)} y_e H_e, X \right\rangle. \end{aligned} \quad (6.21)$$

Both t and every y_e are unrestricted because they multiply equality constraints. Replacing y_e by $-y_e$ would give an equivalent sign convention; what matters is using one convention consistently.

Set

$$C(t, y) = J - tI - \sum_e y_e H_e.$$

The dual function takes the supremum of (6.21) over $X \succeq 0$. If some vector v satisfies $v^\top C(t, y) v > 0$, then taking $X = \lambda v v^\top$ and $\lambda \rightarrow \infty$ makes the Lagrangian unbounded above. Conversely, if $C(t, y) \preceq 0$, then $\langle C(t, y), X \rangle \leq 0$ for every $X \succeq 0$, and the supremum is t , attained at $X = 0$. Finiteness is therefore equivalent to

$$S(t, y) := -C(t, y) = tI + \sum_e y_e H_e - J \succeq 0.$$

Minimizing the finite dual function gives exactly (6.19). The primal feasible set is compact: $X \succeq 0$ and $\text{Tr}(X) = 1$ imply $\|X\|_F \leq 1$. Hence the primal optimum is attained. Moreover, $I/n \succ 0$ satisfies all primal affine equalities, so primal Slater gives zero duality gap and attainment of the dual optimum.

The factor of two in (6.18) is essential. If one instead attaches a multiplier μ_e directly to the scalar equality $X_{ij} = 0$, then

$$-\mu_e X_{ij} = -\frac{\mu_e}{2} \langle H_e, X \rangle,$$

so the coefficient in (6.19) is $y_e = \mu_e/2$. Omitting this conversion changes the displayed edge multipliers, although not the optimum. Finally, the direction $S \succeq 0$ rather than $S \preceq 0$ follows from the fact that the primal is a maximization over the PSD cone: its coefficient in the Lagrangian must be nonpositive semidefinite, i.e., $C = -S \preceq 0$.

Solution to Exercise 6.5

Let $\omega = e^{2\pi i/5}$. The Fourier vectors $f_k = (1, \omega^k, \omega^{2k}, \dots, \omega^{4k})^\top$ diagonalize every circulant matrix appearing here, and

$$A f_k = (\omega^k + \omega^{-k}) f_k = 2 \cos(2\pi k/5) f_k.$$

The exact trigonometric values are

$$2 \cos(2\pi/5) = \frac{\sqrt{5} - 1}{2} = \frac{1}{\phi}, \quad 2 \cos(4\pi/5) = -\frac{\sqrt{5} + 1}{2} = -\phi.$$

Thus, with multiplicities shown implicitly,

$$\text{spec}(A) = \left(2, \frac{1}{\phi}, \frac{1}{\phi}, -\phi, -\phi \right). \quad (6.22)$$

For clarity, decompose the real vertex space into the mutually orthogonal A -eigenspaces

$$E_0 = \text{span}\{\mathbf{1}\}, \quad E_+ = \text{span}_{\mathbb{R}}\{\text{Re } f_1, \text{Im } f_1\}, \quad E_- = \text{span}_{\mathbb{R}}\{\text{Re } f_2, \text{Im } f_2\},$$

of dimensions 1, 2, 2. Here J has eigenvalue 5 on E_0 and zero on $E_+ \oplus E_-$. Since $B = J - I - A$, all four spectra can now be read off exactly:

space	dim	$\lambda(A)$	$\lambda(B)$	$\lambda(X^*)$	$\lambda(S^*)$
E_0	1	2	2	$\frac{1}{\sqrt{5}}$	0
E_+	2	$\frac{1}{\phi}$	$-\phi$	0	$\frac{5}{\phi}$
E_-	2	$-\phi$	$\frac{1}{\phi}$	$\frac{5 - \sqrt{5}}{10}$	0

In particular,

$$\begin{aligned} \text{spec}(B) &= \left(2, -\phi, -\phi, \frac{1}{\phi}, \frac{1}{\phi}\right), \\ \text{spec}(X^*) &= \left(\frac{1}{\sqrt{5}}, 0, 0, \frac{5 - \sqrt{5}}{10}, \frac{5 - \sqrt{5}}{10}\right), \\ \text{spec}(S^*) &= \left(0, \frac{5}{\phi}, \frac{5}{\phi}, 0, 0\right). \end{aligned}$$

The identities $\phi^2 = \phi + 1$, $\phi^{-1} = \phi - 1$, $\phi^{-1} + \phi^{-2} = 1$, and $\sqrt{5} = 1 + 2/\phi$ give the simplifications in the table. They also show directly that both X^* and S^* are PSD. Moreover, $\text{Tr}(X^*) = 1$ and

$$\langle J, X^* \rangle = 5\lambda_{E_0}(X^*) = \sqrt{5}.$$

Because B has zero entries on the edges of C_5 , X^* is primal feasible. Since $\sum_{e \in E(C_5)} H_e = A$, the displayed S^* is the dual slack for $t^* = \sqrt{5}$ and the constant edge multiplier $y_e = y^*$; its spectrum proves dual feasibility.

Solution to Exercise 6.6

For the pentagon,

$$A^2 = 2I + B, \quad AB = BA = J - I, \quad AJ = JA = 2J, \quad BJ = JB = 2J. \quad (6.23)$$

The first identity counts two-step walks in C_5 ; the others follow from $B = J - I - A$ and 2-regularity. Put $s = \sqrt{5}$ and $y = s/\phi$. Using (6.23),

$$\begin{aligned} 5S^*X^* &= (sI + yA - J)(I + \phi^{-1}B) \\ &= sI + yA - J + \phi^{-1}(sB + y(J - I) - 2J) \\ &= s(1 - \phi^{-1} - \phi^{-2})I + (y - s/\phi)A + (-1 + s/\phi + s/\phi^2 - 2/\phi)J \\ &= 0. \end{aligned}$$

The last equality uses $y = s/\phi$, $\phi^{-1} + \phi^{-2} = 1$, and $s = 1 + 2/\phi$.

The spectral table in Exercise 6.5 gives the sharper support statement

$$\text{range}(X^*) = E_0 \oplus E_- = \ker S^*, \quad \text{range}(S^*) = E_+ = \ker X^*.$$

Thus the primal matrix has rank three and is supported on $E_0 \oplus E_-$, whereas the positive dual slack is carried by the two-dimensional space E_+ .

For any primal-feasible X and dual-feasible (t, y) , the edge and trace equalities imply the certificate identity

$$t - \langle J, X \rangle = \langle S(t, y), X \rangle \geq 0.$$

Here $S^*X^* = 0$ gives $\langle S^*, X^* \rangle = \text{Tr}(S^*X^*) = 0$. Hence the feasible primal value equals the feasible dual value:

$$\langle J, X^* \rangle = t^* = \sqrt{5}.$$

Weak duality then leaves no room for a different optimum, proving zero duality gap and $\vartheta(C_5) = \sqrt{5}$.

Solution to Exercise 6.7

For $G = K_n$, the primal edge constraints set every off-diagonal entry of X to zero. Therefore every primal-feasible X satisfies

$$\langle J, X \rangle = \text{Tr}(X) = 1.$$

For example, $X = I/n$ is a primal certificate of value 1. For the dual, take $t = 1$ and $y_e = 1$ for every edge. Since each unordered edge is counted once,

$$\sum_{e \in E(K_n)} H_e = J - I, \quad S = I + (J - I) - J = 0 \succeq 0.$$

The matching certificates prove $\vartheta(K_n) = 1$.

For $G = \overline{K}_n$, there are no edge constraints or edge multipliers. The rank-one matrix

$$X = \frac{1}{n}J = \frac{1}{n}\mathbf{1}\mathbf{1}^\top$$

is PSD, has trace one, and has objective $\langle J, X \rangle = n$. For the dual, take $t = n$; then

$$S = nI - J \succeq 0,$$

because its eigenvalue is 0 on $\text{span}\{\mathbf{1}\}$ and n on $\mathbf{1}^\perp$. Thus $\vartheta(\overline{K}_n) = n$.

In communication terms, K_n means every two distinct symbols may be confused. Every strong power is again complete, so only one zero-error message can be sent and $\Theta(K_n) = 1$ (equivalently, $C_0 = 0$ bits per use). The empty graph means that no distinct symbols can be confused. All n^k length- k words are usable, so $\Theta(\overline{K}_n) = n$ and $C_0 = \log_2 n$ bits per use. The theta values are exact in both cases.

Suggested reading.

The zero-error communication framework originates with Shannon [Sha56]. The orthonormal-representation bound and the exact capacity of the pentagon are due to Lovász [Lov79]. For the SDP and dual-certificate viewpoint, see [VB96, BV04].

Historical note

Shannon introduced zero-error channel capacity in 1956 [Sha56]. Lovász's 1979 theta function supplied the missing spectral-convex upper bound for the pentagon and proved $\Theta(C_5) = \sqrt{5}$ [Lov79], a striking example in which one finite-dimensional certificate controls an asymptotic coding problem.

Appendix A

Formative Checks and Exit Tickets

Use one poll and exit ticket per week for a quick understanding check. Choose an option, discuss the reasoning, and state what the exit-ticket result certifies. The answer key is at the end; full practice sets remain in the chapters.

A.1 Week 1: risk, first optimization loop, and attainment

Formative check: one-minute poll

For a fixed parameter θ chosen before an i.i.d. sample is drawn, which statement follows when the loss has finite expectation?

- A. $\hat{R}_S(\theta) = R(\theta)$ for every sample.
- B. $\mathbb{E}[\hat{R}_S(\theta)] = R(\theta)$.
- C. The empirical-risk minimizer has zero generalization gap.
- D. Repeated validation tuning preserves an unbiased validation estimate.

Formative check: two-minute exit ticket

Use the data $(x_i, y_i) = (1, 2), (2, 3), (3, 5)$, model $h_w(x) = wx$, squared loss, and regularizer $0.1w^2$. Write the average regularized ERM objective and take one gradient step from $w_0 = 0$ with $\eta = 0.1$. Explain why the minimizer is unique and attained, and why test-set mean squared error must remain outside the training optimizer's input.

A.2 Week 2: convexity and first-order operators

Formative check: one-minute poll

For which set is the Euclidean projection $\arg \min_{x \in \mathcal{X}} \frac{1}{2} \|x - v\|_2^2$ guaranteed to exist and be unique for every $v \in \mathbb{R}^d$?

- A. Every nonempty convex set.
- B. Every closed set, even if nonconvex.
- C. Only a compact affine subspace.
- D. Every nonempty closed convex set.

Formative check: two-minute exit ticket

For $g(x) = \lambda|x|$, at $v = -0.3$, $\eta = 0.5$, and $\lambda = 1$, compute $\text{prox}_{\eta g}(v)$. State which optimality condition certifies your answer and which closedness/convexity facts make this local subproblem well posed.

A.3 Week 3: separation, Farkas certificates, and conic duality**Formative check: one-minute poll**

Which KKT statement is correct?

- A. For differentiable convex problems KKT is sufficient; a qualification such as Slater supports necessity.
- B. KKT is always necessary and sufficient.
- C. Slater makes a nonconvex problem convex.
- D. Complementary slackness forces both factors to be zero.

Formative check: two-minute exit ticket

For $A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ and $b = (1, 2)^\top$, either produce $x \geq 0$ with $Ax = b$ or a Farkas witness y . Verify every inequality and explain why the two alternatives cannot both hold.

A.4 Week 4: Newton, barriers, and semidefinite certificates**Formative check: one-minute poll**

Why does a damped Newton step use backtracking for a log-barrier objective?

- A. To make the Hessian diagonal.
- B. To remove the need for gradients.
- C. To preserve the strict domain and obtain sufficient decrease.
- D. To make every constraint active.

Formative check: two-minute exit ticket

For $M = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, compute the eigenvalues and decide whether $M \succeq 0$. Verify the same conclusion using a quadratic form. Explain why the SDP pair is the Week 3 conic pair specialized to $K = \mathbb{S}_+^n$.

A.5 Week 5: relaxation and randomized rounding

Formative check: one-minute poll

For a graph with nonnegative edge weights, what does random-hyperplane rounding of an optimal Max-Cut SDP matrix guarantee?

- A. The expected rounded value is at least α_{GW} SDP.
- B. Every rounded sample beats α_{GW} OPT.
- C. The SDP always equals the exact optimum.
- D. One graph's empirical ratio defines α_{GW} .

Formative check: two-minute exit ticket

Place a feasible cut value, the exact optimum, and the SDP optimum in the correct inequality order. State which number is a candidate and which is a bound.

A.6 Week 6: zero-error capacity and theta

Formative check: one-minute poll

Why is $\vartheta(G) \geq \alpha(G)$ alone insufficient to show $\Theta(G) \leq \vartheta(G)$?

- A. $\alpha(G)$ is not computable.
- B. Theta is defined only for complete graphs.
- C. Independent sets are randomized.
- D. Capacity depends on all strong powers, so the bound must remain controlled under products.

Formative check: two-minute exit ticket

Give the five-word two-use zero-error code in $C_5^{\otimes 2}$. Then write and label the two bounds $\sqrt{5} \leq \Theta(C_5) \leq \vartheta(C_5) = \sqrt{5}$.

A.7 Week 7: cumulative certificate discipline

Formative check: one-minute poll

A feasible cut of value 9 has been independently verified. A report claims $\text{OPT} = 8$ and $\text{SDP} = 10$. Which reported value is refuted by the verified cut?

- A. The SDP value 10.
- B. The rounded cut 9.
- C. The claimed optimum 8, because a feasible cut of value 9 exists.
- D. None of the values.

Formative check: two-minute exit ticket

Given one result from Weeks 1–6, write its output as a “number with a noun,” state the bound direction, and list the one assumption whose failure would invalidate the certificate.

A.8 Answer key and diagnostic rationale

Week	Answers	Diagnostic focus
1	B	Unbiasedness is pointwise in a parameter fixed before sampling.
2	D	A nonempty closed convex set has a unique Euclidean projection.
3	A	Convex KKT sufficiency and qualification-supported necessity differ.
4	C	Backtracking protects the strict barrier domain and sufficient decrease.
5	A	Optimal SDP rounding gives an expectation guarantee, not a per-trial bound.
6	D	Product control is needed to pass from one shot to capacity.
7	C	A feasible candidate cannot exceed a correctly stated maximum optimum.

Exit-ticket numerical checkpoints. Week 1: $w_1 = 23/15$ and $w^* = 230/143$. Week 2: the proximal point is 0. Week 3: $y = (1, -1)^\top$ has $A^\top y = (0, 2)^\top$ and $b^\top y = -1$. Week 4: the eigenvalues are 3, -1 , and $(1, -1)M(1, -1)^\top = -2$. Week 6: use $\{(i, 2i \bmod 5) : i = 0, \dots, 4\}$. These checkpoints supplement the required certificate and interpretation.

Appendix B

Normal Cones and the Unified Fermat Rule

This reference sheet supplies one optimality interface for the smooth, nonsmooth, constrained, projection, and proximal problems used in the main chapters. It records the assumptions and reusable formulas without repeating their chapter examples.

B.1 Constraint and normal-cone language

For a nonempty set $C \subseteq \mathbb{R}^d$, its extended-real indicator is

$$\delta_C(x) = \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C. \end{cases} \quad (\text{B.1})$$

Thus minimizing f over C is equivalent to minimizing $f + \delta_C$ over $\mathbb{R}^d$; this rewriting does not by itself prove finiteness or attainment.

For nonempty convex C , the convex-analysis normal cone is

$$N_C(x) = \left\{ v \in \mathbb{R}^d \mid \langle v, z - x \rangle \leq 0 \text{ for every } z \in C \right\}, \quad x \in C, \quad (\text{B.2})$$

and $N_C(x) = \emptyset$ outside C . The identity

$$\partial \delta_C(x) = N_C(x)$$

turns constraint geometry into subgradient language.

For the halfspace $C = \{x \mid a^\top x \leq b\}$, $a \neq 0$,

$$N_C(x) = \begin{cases} \{0\}, & a^\top x < b, \\ \{\lambda a \mid \lambda \geq 0\}, & a^\top x = b, \\ \emptyset, & a^\top x > b. \end{cases} \quad (\text{B.3})$$

The sign follows from the convention $a^\top x \leq b$: a points outward.

B.2 Unified convex Fermat rule

Theorem B.1 (Fermat rule with a normal cone). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ be proper, closed, and convex, and let $C \subseteq \mathbb{R}^d$ be nonempty, closed, and convex. Under the qualification*

$$\text{relint}(\text{dom } f) \cap \text{relint}(C) \neq \emptyset,$$

a point $x^ \in C \cap \text{dom } f$ is optimal exactly when*

$$x^* \in \arg \min_{x \in C} f(x) \iff 0 \in \partial f(x^*) + N_C(x^*). \quad (\text{B.4})$$

The rule combines unconstrained convex Fermat with the qualified sum rule for $\partial(f + \delta_C)$.

The inclusion tests a supplied feasible point; it does not assert that an optimizer exists. Compactness plus continuity and closed coercive objectives are two common, sufficient existence routes.

B.3 Specializations to reuse

Problem	Optimality condition	Extra fact to check
Smooth, unconstrained	$\nabla f(x^*) = 0$	Convexity makes stationarity sufficient.
Nonsmooth, unconstrained	$0 \in \partial f(x^*)$	Use the convex subdifferential.
Smooth over C	$-\nabla f(x^*) \in N_C(x^*)$	Verify feasibility and the outward-normal sign.
Projection onto C	See equation (B.5).	Nonempty closed convex C gives existence and uniqueness.
Proximal point	See equation (B.6).	Proper closed convex r and $\eta > 0$ give a unique minimizer.

For later cross-reference, the projection and proximal rows are

$$p = \Pi_C(y) \iff y - p \in N_C(p), \quad (\text{B.5})$$

$$p = \text{prox}_{\eta r}(v) \iff 0 \in \partial r(p) + \frac{1}{\eta}(p - v). \quad (\text{B.6})$$

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